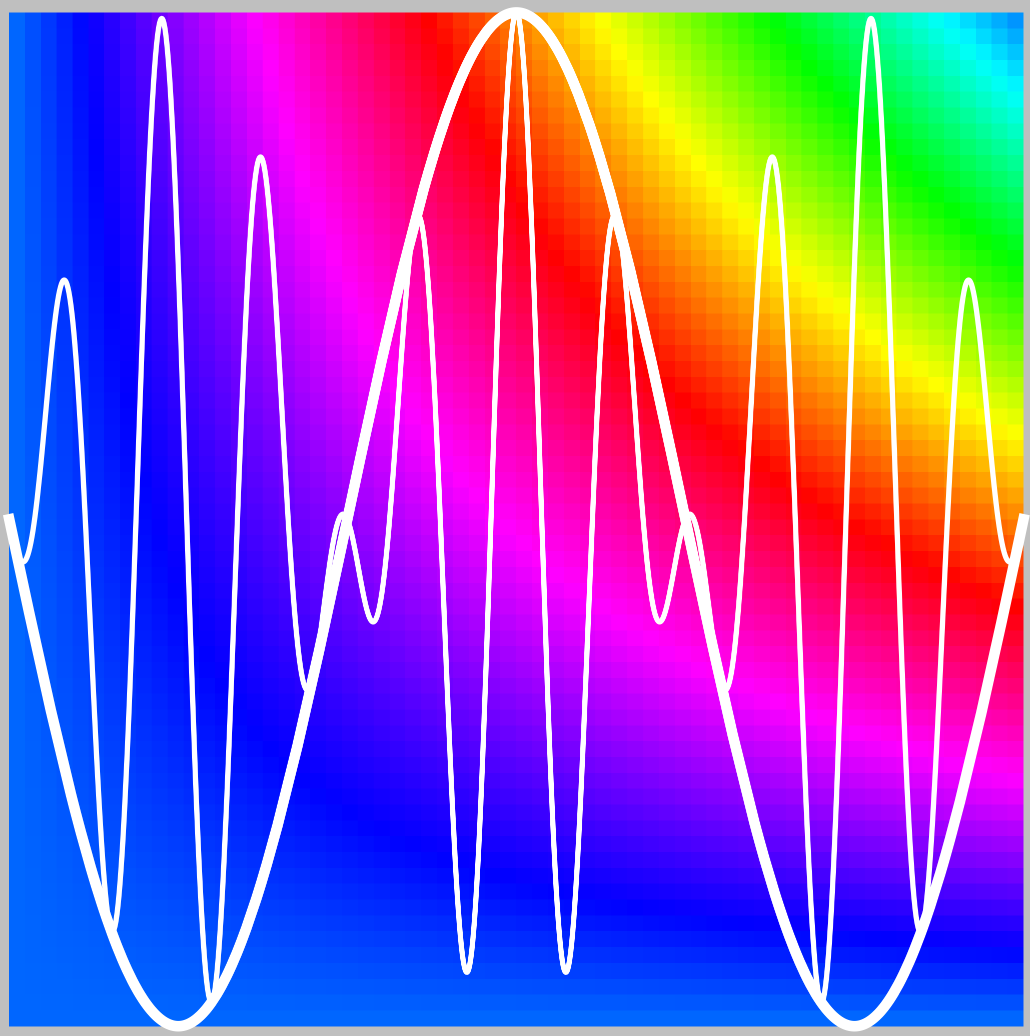
SOLUTIONS MANUAL FOR ADVANCED TOPICS IN APPLIED MATHEMATICS



Sudhakar Nair

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Preface

This is the solutions manual to accompany the text, "Advanced Topics in Applied Mathematics." I have attempted to show all the intermediate steps in the solutions. For the convenience of posting solutions, a new solution appears on a new page. Please let me know if anyone notices any errors. Some figures were created using the Mathematica^{TM} package and I am indebted to Wolfram Research Incorporated.

S. Nair (nair@iit.edu)
Illinois Institute of Technology,
Chicago, Illinois

Chapter 1

GREEN'S FUNCTIONS

1.1 The deflection of a beam is governed by the equation

$$EI\frac{d^4v}{dx^4} = -p(x)$$

where EI is the bending stiffness and $p(x)$ is the distributed loading on the beam. If the beam has a length ℓ, and at both the ends the deflection and slope are zero, obtain expressions for the deflection by direct integration, using the Macaulay brackets when necessary, if
a) $p(x) = p_0$,
b) $p(x) = P_0\delta(x-\xi)$,
c) $p(x) = M_0\delta'(x-\xi)$.
Obtain the Green's function for the deflection equation from the preceding calculations.

Solution

Let us make the substitution $x \to x/\ell$. The derivative of the delta function in Part (c) will bring $1/\ell$ when the non-dimensional x is used. The beam equation becomes

(a)

$$\begin{aligned}
v'''' &= -\frac{p_0\ell^4}{EI}, \quad v''' = -\frac{p_0\ell^4}{EI}[x + C_1], \\
v'' &= -\frac{p_0\ell^4}{EI}[\frac{x^2}{2} + C_1 x + C_2], \\
v' &= -\frac{p_0\ell^4}{EI}[\frac{x^3}{3} + C_1\frac{x^2}{2} + C_2 x + C_3], \\
v &= -\frac{p_0\ell^4}{EI}[\frac{x^4}{24} + C_1\frac{x^3}{6} + C_2\frac{x^2}{2} + C_3 x + C_4].
\end{aligned}$$

Using the boundary conditions

$$v(0)=0, \quad v'(0)=0, \quad C_4=0, \quad C_3=0.$$

Using

$$v(1)=0, \quad v'(1)=0, \frac{1}{2}C_1+C_2=-\frac{1}{6}, \quad \frac{1}{6}C_1+\frac{1}{2}C_2=-\frac{1}{24}.$$

Solving

$$C_1=-\frac{1}{2}, \quad C_2=\frac{1}{12}.$$

$$\begin{aligned} v(x) &= -\frac{p_0\ell^4}{EI}\left[\frac{x^4}{24}-\frac{x^3}{12}+\frac{x^2}{24}\right], \\ &= -\frac{p_0\ell^4}{24EI}(x-1)^2x^2. \end{aligned}$$

(b)

$$\begin{aligned} v'''' &= -\frac{P_0\ell^4}{EI}\delta(x-\xi), \quad v'''=-\frac{P_0\ell^4}{EI}[\langle x-\xi\rangle^0+C_1], \\ v'' &= -\frac{P_0\ell^4}{EI}[\langle x-\xi\rangle^1+C_1x+C_2], \\ v' &= -\frac{P_0\ell^4}{EI}[\frac{1}{2}\langle x-\xi\rangle^2+\frac{1}{2}C_1x^2+C_2x+C_3], \\ v &= -\frac{P_0\ell^4}{EI}[\frac{1}{6}\langle x-\xi\rangle^3+\frac{1}{6}C_1x^3+\frac{1}{2}C_2x^2+C_3x+C_4]. \end{aligned}$$

Using

$$v(0)=0=v'(0), \quad C_4=C_3=0.$$

Using

$$v(1)=0=v'(1), \quad \frac{1}{2}C_1+C_2=-\frac{1}{2}(1-\xi)^2, \quad \frac{1}{6}C_1+\frac{1}{2}C_2=-\frac{1}{6}(1-\xi)^3,$$

Then

$$C_1=-(1+2\xi)(1-\xi)^2, \quad C_2=\xi(1-\xi)^2.$$

$$v(x) = -\frac{P_0\ell^4}{6EI}\left[\langle x-\xi\rangle^3 - (1+2\xi)(1-\xi)^2x^3 + 3\xi(1-\xi)^2x^2\right].$$

(c)

$$\begin{aligned}
v'''' &= -\frac{M_0\ell^3}{EI}\delta'(x-\xi), \quad v''' = -\frac{M_0\ell^3}{EI}[\delta(x-\xi)+C_1],\\
v'' &= -\frac{M_0\ell^3}{EI}[\langle x-\xi\rangle^0 + C_1x + C_2],\\
v' &= -\frac{M_0\ell^3}{EI}[\langle x-\xi\rangle^1 + \frac{1}{2}C_1x^2 + C_2x + C_3],\\
v &= -\frac{M_0\ell^3}{EI}[\frac{1}{2}\langle x-\xi\rangle^2 + \frac{1}{6}C_1x^3 + \frac{1}{2}C_2x^2 + C_3x + C_4].
\end{aligned}$$

Using

$$v(0) = 0 = v'(0), \quad C_3 = C_4 = 0,$$

Using

$$v(1) = 0 = v'(1), \quad \frac{1}{2}C_1 + C_2 = -(1-\xi), \quad \frac{1}{6}C_1 + \frac{1}{2}C_2 = -\frac{1}{2}(1-\xi)^2.$$

Then

$$C_1 = -6\xi(1-\xi), \quad C_2 = (3\xi-1)(1-\xi).$$

$$v(x) = -\frac{M_0\ell^3}{2EI}\left[\langle x-\xi\rangle^2 - 2\xi(1-\xi)x^3 + (3\xi-1)(1-\xi)x^2\right].$$

The Green's function in terms of non-dimensional coordinate x/ℓ corresponds to $P_0 = -1$;

$$g(x,\xi) = \frac{\ell^4}{6EI}\left[\langle x-\xi\rangle^3 - (1-\xi)^2x^2(3\xi + x + 2\xi x)\right].$$

$$g(x,\xi) = \begin{cases}(1-\xi)^2x^2(3\xi - x - 2\xi x), & x<\xi\\ (1-x)^2\xi^2(3x-\xi-2x\xi), & x>\xi\end{cases}$$

1.2 Solve the preceding problem when the beam is simply supported. That is,

$$v(0) = v(\ell) = 0, \quad v''(0) = v''(\ell) = 0.$$

Solution

Let us make the substitution $x \to x/\ell$. The derivative of the delta function will have a $1/\ell$ in front when the non-dimensional x is used. The beam equation becomes

(a)

$$\begin{aligned}
v'''' &= -\frac{p_0\ell^4}{EI}, \quad v''' = -\frac{p_0\ell^4}{EI}[x + C_1], \\
v'' &= -\frac{p_0\ell^4}{EI}[\frac{x^2}{2} + C_1 x + C_2], \\
v' &= -\frac{p_0\ell^4}{EI}[\frac{x^3}{3} + C_1\frac{x^2}{2} + C_2 x + C_3], \\
v &= -\frac{p_0\ell^4}{EI}[\frac{x^4}{24} + C_1\frac{x^3}{6} + C_2\frac{x^2}{2} + C_3 x + C_4].
\end{aligned}$$

Using the boundary conditions

$$v(0) = 0, \quad v''(0) = 0, \quad C_2 = 0, \quad C_4 = 0.$$

Using

$$v(1) = 0, \quad v''(1) = 0, \quad C_1 = -\frac{1}{2}, \quad C_3 = \frac{1}{24}.$$

$$\begin{aligned}
v(x) &= -\frac{p_0\ell^4}{EI}\left[\frac{x^4}{24} - \frac{x^3}{12} + \frac{x}{24}\right], \\
&= -\frac{p_0\ell^4}{24EI}x(x^3 - 2x + 1).
\end{aligned}$$

(b)

$$\begin{aligned}
v'''' &= -\frac{P_0\ell^4}{EI}\delta(x-\xi), \quad v''' = -\frac{P_0\ell^4}{EI}[\langle x-\xi\rangle^0 + C_1], \\
v'' &= -\frac{P_0\ell^4}{EI}[\langle x-\xi\rangle^1 + C_1x + C_2], \\
v' &= -\frac{P_0\ell^4}{EI}[\frac{1}{2}\langle x-\xi\rangle^2 + \frac{1}{2}C_1x^2 + C_2x + C_3], \\
v &= -\frac{P_0\ell^4}{EI}[\frac{1}{6}\langle x-\xi\rangle^3 + \frac{1}{6}C_1x^3 + \frac{1}{2}C_2x^2 + C_3x + C_4].
\end{aligned}$$

Using

$$v(0) = 0 = v''(0), \quad C_4 = C_2 = 0.$$

Using

$$v(1) = 0 = v''(1), \quad C_1 = \xi - 1, \quad C_2 = \frac{1}{6}\xi(1-\xi)(2-\xi),$$

$$v(x) = -\frac{P_0\ell^4}{6EI}\left[\langle x-\xi\rangle^3 - (1-\xi)(1-\xi)^2x^3 + \xi(1-\xi)(2-\xi)x\right].$$

(c)

$$v'''' = -\frac{M_0\ell^3}{EI}\delta'(x-\xi), \quad v''' = -\frac{M_0\ell^3}{EI}[\delta(x-\xi) + C_1],$$

Note: δ' is now the derivative with respect to the non-dimensional variable x.

$$\begin{aligned}
v'' &= -\frac{M_0\ell^3}{EI}[\langle x-\xi\rangle^0 + C_1x + C_2], \\
v' &= -\frac{M_0\ell^3}{EI}[\langle x-\xi\rangle^1 + \frac{1}{2}C_1x^2 + C_2x + C_3], \\
v &= -\frac{M_0\ell^3}{EI}[\frac{1}{2}\langle x-\xi\rangle^2 + \frac{1}{6}C_1x^3 + \frac{1}{2}C_2x^2 + C_3x + C_4].
\end{aligned}$$

Using

$$v(0) = 0 = v''(0), \quad C_2 = C_4 = 0,$$

Using

$$v(1) = 0 = v''(1), \quad C_1 = -1, \quad C_3 = \frac{1}{6}[1 - 3(-\xi)^2].$$

$$v(x) = -\frac{M_0 \ell^3}{6EI} \left\{ [3\langle x - \xi \rangle^2 - x^3 + [1 - 3(1 - \xi)^2] x \right\} .$$

The Green's function in terms of the non-dimensional coordinate x/ℓ corresponds to $P_0 = -1$;

$$g(x, \xi) = \frac{\ell^4}{6EI} \left[\langle x - \xi \rangle^3 - (1 - \xi)^2 x^3 + \xi(1 - \xi)(2 - \xi)x \right] .$$

$$g(x, \xi) = \begin{cases} x(1 - \xi)(2\xi - x^2 - \xi^2), & x < \xi \\ \xi(1 - x)(2x - \xi^2 - x^2), & x > \xi \end{cases}$$

1.3 Obtain the derivative of the function

$$g(x) = |f(x)|,$$

in $a < x < b$, assuming $f(x)$ has a simple zero at the point c inside the interval (a, b). Use the Signum function and/or delta function to express the result.

Solution

Let

$$g(x) = f(x)\ \mathrm{sgn}[f(x)].$$

(a) Assuming $f'(c) > 0$,

$$\mathrm{sgn}[f(x)] = \ \mathrm{sgn}(x - c).$$

(b) Assuming $f'(c) < 0$,

$$\mathrm{sgn}[f(x)] = -\ \mathrm{sgn}(x - c).$$

Then

$$g(x) = f(x)\ \mathrm{sgn}[f'(c)]\ \mathrm{sgn}(x-c), \quad g'(x) = [f'(x)\ \mathrm{sgn}(x-c) + 2f(x)\delta(x-c)]\ \mathrm{sgn}[f'(c)].$$

1.4 Assuming a function $f(x)$ has simple zeros at $x_i, i = 1, 2, \ldots, n$, find an expression for
$$\delta(f(x)).$$

Solution

Near a zero, $x = x_i$,

$$f(x) = f'(x_i)(x - x_i) + \cdots ,$$

For a simple zero, $f'(x_i) \neq 0$.

Using a test function ϕ,

$$\begin{aligned}
\int_{-\infty}^{\infty} \delta(f(x))\phi(x)dx &= \sum_{i=1}^{n} \int_{x_i-\epsilon}^{x_i+\epsilon} \phi(x)\delta[f'(x_i)(x - x_i)]dx \\
&= \sum_{i=1}^{n} \int_{-\epsilon}^{\epsilon} \phi(x_i + \xi)\delta[f'(x_i)\xi]d\xi, \\
&= \sum_{i=1}^{n} \frac{1}{f'(x_i)} \int_{-\epsilon}^{\epsilon} \phi\left[x_i + \frac{\eta}{f'(x_i)}\right] \delta(\eta)d\eta \\
&= \sum_{i=1}^{n} \frac{1}{f'(x_i)}\phi(x_i).
\end{aligned}$$

Then
$$\delta(f) = \sum_{i=1}^{n} \frac{1}{f'(x_i)}\delta(x - x_i).$$

1.5 Convert the equation

$$Lu = \frac{d^2u}{dx^2} + x^2\frac{du}{dx} + 2u = f$$

into the Sturm-Liouville form.

Solution

The Sturm-Liouville form is

$$(pu')' + qu = f^*.$$

Expanding

$$u'' + \frac{p'}{p}u' + \frac{q}{p}u = \frac{f^*}{p}.$$

Comparing with the given equation

$$\frac{p'}{p} = x^2, \quad \frac{q}{p} = 2, \quad \frac{f^*}{p} = f.$$

$$\log p = x^3/3, \quad p = e^{x^3/3}, \quad q = 2e^{x^3/3}, \quad f^* = e^{x^3/3}f.$$

$$(e^{x^3/3}u')' + 2e^{x^3/3}u = e^{x^3/3}f.$$

1.6 Find the adjoint system for

$$xu'' + u' + u = 0, \quad u(1) = 0, \quad u(2) + u'(2) = 0.$$

Solution

$$\langle v, Lu\rangle - \langle u, L^*v\rangle = 0.$$

$$\begin{aligned}
\langle v, Lu\rangle &= \int_1^2 v[xu'' + u' + u]dx \\
&= [xvu' - (xv)'u + vu]\|_1^2 + \int_1^2 u[(xv)'' - v' + v]dx, \\
L^*v &= (xv)'' - v' + v = xv'' + 2v' - v' + 1. \\
L^* &= x\frac{d^2}{dx^2} + \frac{d}{dx} + 1.
\end{aligned}$$

The adjoint boundary conditions are found from the bilinear concomitant,

$$[xvu' - (xv)'u + vu]\|_1^2 = 0.$$

$$x = 1, \quad xv = 0, \quad v(1) = 0,$$

$$x = 2, \quad u' = -u, \quad -2v - v - 2v' + v = 0, \quad v'(2) + v(2) = 0.$$

1.7 Solve the differential system

$$u'' + u' - 2u = x^2, \quad u(0) = 0, \quad u'(1) = 0.$$

Solution

For this constant coefficient equation, we try

$$u = e^{\alpha x},$$

to get

$$\alpha^2 + \alpha - 2 = 0, \alpha = -\frac{1}{2} \pm \frac{3}{2}.$$
$$\alpha_1 = 1, \quad \alpha_2 = -2.$$

For a particular solution, we assume

$$u_p = Cx^2 + Dx + E,$$

which, upon substituting in the differential equation gives

$$2C + 2Cx + D - 2Cx^2 - 2Dx - 2E = x^2.$$

Matching the coefficients of equal powers of x,

$$C = -1/2, \quad D = -1/2, \quad E = -3/4.$$

The general solution is

$$u = Ae^x + Be^{-2x} - \frac{1}{2}x^2 - \frac{1}{2}x - \frac{3}{4}.$$

The boundary conditions give

$$u(0) = 0, \quad A + B - \frac{3}{4} = 0,$$

$$u'(1) = 0, \quad Ae - 2Be^{-2} - \frac{3}{2} = 0.$$

Then

$$A = \frac{3}{2}\frac{e^2 + 1}{e^3 + 2}, \quad B = \frac{3}{4}\frac{e^3 - 2e^2}{e^3 + 2}.$$

1.8 Solve the differential system

$$(x^2u')' - n(n+1)u = 0, \quad u(0) = 0, \quad u(1) = 1.$$

Solution

This equation is of variable coefficients of the Euler type. We try

$$u = x^\alpha.$$

The indicial equation is

$$\alpha(\alpha+1) - n(n+1) = 0, \quad \alpha^2 + \alpha - n(n+1) = 0.$$

$$\alpha = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + n(n+1)},$$

$$\alpha_1 = n, \quad \alpha_2 = -(n+1).$$

So, our solution is

$$u = Ax^n + Bx^{-(n+1)}.$$

(a) $n > 0$

$$u(0) = 0, \quad B = 0, \quad u(1) = 1, \quad A = 1.$$

$$u = x^n.$$

(b) $n = 0$

$$u = A + B/x, \quad u(0) = 0, A = B = 0,$$

and $u(1) = 1$ cannot be satisfied. Thus, there is no solution.

(c) $n = -1$

$$u = A/x + B, \quad u(0) = 0, A = B = 0,$$

again, no solution.

(d) $n < -1$

$$u(0) = 0, \quad A = 0, \quad u(1) = 1, \quad B = 1.$$

$$u = x^{-(n+1)}.$$

1.9 Convert the following system to one with homogeneous boundary conditions:

$$(x^2u')' - n(n+1)u = 0, \quad u(0) = 0, \quad u(1) = 1.$$

Solution

Let

$$u = v + Ax + B.$$

$$u(0) = v(0) + B = 0, \quad v(0) = 0, \quad B = 0.$$

$$u(1) = 1 = v(1) + A, \quad A = 0, V(1) = 0.$$

$$u = v + x, \quad [x^2(v' + 1)]' - n(n+1)[v + x] = 0.$$

$$(x^2v')' - n(n+1)v = [n(n+1) - 2]x, \quad v(0) = v(1) = 0.$$

1.10 Obtain the Green's function for the equation

$$(xu')' - \frac{9}{x}u = f(x), \quad u(0) = 0, \quad u(1) = 0.$$

Using the Green's function explicitly, find the solution when $f(x) = x^n$. Check if there are any values for the integer n for which your solution does not satisfy the boundary conditions.

Solution

Let $u = x^\alpha$.

$$\alpha^2 - 9 = 0, \quad u = x^3, x^{-3}.$$

To satisfy $u(0) = 0$, we choose

$$u_1 = x^3.$$

To satisfy $u(1) = 0$, we choose

$$u_2 = x^3 - x^{-3}.$$

Then

$$g = C \begin{cases} x^3(\xi^3 - \xi^{-3}), & x < \xi \\ \xi^3(x^3 - x^{-3}), & x > \xi \end{cases}$$

$$[[g']]_{x=\xi} = \frac{1}{\xi}, \quad C[\xi^3(3\xi^2 + 3\xi^{-4}) - 3\xi^2(\xi^3 - \xi^{-3})] = \frac{1}{\xi}.$$

$$C = \frac{1}{6}. \tag{1.1}$$

$$g = \frac{1}{6} \begin{cases} x^3(\xi^3 - \xi^{-3}), & x < \xi \\ \xi^3(x^3 - x^{-3}), & x > \xi \end{cases} \quad .$$

For $f = x^n$

$$
\begin{aligned}
u &= \frac{1}{6}\left[\int_0^x (x^3 - x^{-3})\xi^{n+3}d\xi + \int_x^1 x^3(\xi^{n+3} - \xi^{n-3})d\xi\right] \\
&= \frac{1}{6}\left[(x^3 - x^{-3})\frac{x^{n+4}}{n+4} + x^3(\frac{\xi^{n+4}}{n+4} - \frac{\xi^{n-2}}{n-2})\Big|_x^1\right] \\
&= \frac{1}{6}\left[\frac{1}{n+4}(x^{n+7} - x^{n+1}) + x^3(-\frac{6}{(n+4)(n-2)} - \frac{x^{n+4}}{n+4} + \frac{x^{n-2}}{n-2})\right] \\
&= \frac{1}{(n+4)(n-2)}\left[x^{n+1} - x^3\right]
\end{aligned}
$$

We restrict $n \neq 2$ for this solution to apply.

When $n = 2$

$$
\begin{aligned}
u &= \frac{1}{6}\left[\frac{1}{6}(x^9 - x^3) + \frac{x^3}{6}(1 - x^6) + x^3 \log x\right] \\
&= \frac{1}{6}x^3 \log x.
\end{aligned}
$$

Note that the boundary conditions cannot be satisfied if $n \leq -1$.

1.11 Find the Green's function for

$$u'' + \omega^2 u = f(x), \quad u(0) = 0, \quad u(\pi) = 0.$$

Examine the special cases $\omega = n, \quad n = 0, 1, \cdots$.

Solution

The Green's function satisfies

$$g'' + \omega^2 g = \delta(x - \xi).$$

$$g_1 = \sin \omega x, \quad g_2 = \sin \omega(\pi - x).$$

$$g = C \begin{cases} g_1(x) g_2(\xi), & x < \xi \\ g_2(x) g_1(\xi), & x > \xi \end{cases}$$

The jump condition gives

$$C[g_2'(\xi) g_1(\xi) - g_1'(\xi) g_2(\xi)] = 1.$$

That is

$$C\omega[-\cos \omega(\pi - \xi) \sin \omega\xi - \sin \omega(\pi - \xi) \cos \omega\xi] = 1.$$

$$C = -\frac{1}{\omega \sin \pi\omega}$$

$$g = \frac{1}{\omega \sin \pi\omega} \begin{cases} \sin \omega x \sin \omega(\xi - \pi), & x < \xi \\ \sin \omega\xi \sin \omega(x - \pi), & x > \xi \end{cases}$$

When $\omega = n$, $g = \sin nx$ satisfies both the boundary conditions and a regular Green's function does not exist. We have to look for a generalized Green's function.

1.12 Express the equation

$$u'' - 2u' + u = f(x), \quad u(0) = 0, \quad u(1) = 0,$$

in the self-adjoint form. Obtain the solution using the Green's function when $f(x) = e^x$.

Solution

Comparing with the Sturm-Liouville form,

$$p'/p = -2, \quad p = e^{-2x}.$$

Our equation becomes

$$(e^{-2x}u')' + e^{-2x}u = e^{-2x}f.$$

Trying a solution of the form $u = e^{\alpha x}$,

$$\alpha^2 - 2\alpha + 1 = 0, \quad \alpha = 1, 1.$$

To find the Green's function, let

$$g_1 = xe^x, \quad g_2 = (x-1)e^x.$$

$$g = C \begin{cases} x(\xi - 1)e^{x+\xi}, & x < \xi \\ \xi(x-1)e^{x+\xi}, & x > \xi \end{cases} .$$

The jump condition gives

$$C[\xi + (\xi - 1)\xi - (\xi - 1) - \xi(\xi - 1)]e^{2\xi} = e^{2\xi}.$$

Then $C = 1$ and

$$g = e^{x+\xi} \begin{cases} x(\xi - 1), & x < \xi \\ \xi(x-1), & x > \xi \end{cases} .$$

When $f = e^x$,

$$\begin{aligned} u &= \int_0^x \xi(x-1)e^{x+\xi-\xi}d\xi + \int_x^1 x(\xi-1)e^{x+\xi-\xi}d\xi \\ &= (x-1)e^x\frac{x^2}{2} - xe^x\frac{(x-1)^2}{2} \\ &= \frac{x-1}{2}e^x[x^2 - x(x-1)] \\ &= \frac{1}{2}x(x-1)e^x. \end{aligned}$$

1.13 Transform the equation

$$xu'' + 2u' = f(x); \quad u'(0) = 0, \quad u(1) = 0,$$

into the self-adjoint form. Find the Green's function, and express the solution in terms of $f(x)$. State the restrictions on $f(x)$ for the solution to exist.

Solution

Comparing with the Sturm-Liouville form,

$$p'/p = 2/x, \quad p = x^2.$$

The self-adjoint form is

$$(x^2u')' = xf.$$

The homogeneous equation is

$$(x^2u')' = 0,$$

which can be integrated to get

$$x^2u' = A, \quad u' = \frac{A}{x^2}, \quad u = -\frac{A}{x} + B.$$

To satisfy $g_1'(0) = 0$, we choose $A = 0$ and $B = 1$. To satisfy $g_2(1) = 0$, we choose $A = 1$ and $B = 1$.

$$g = C \begin{cases} 1 - \frac{1}{\xi}, & x < \xi \\ 1 - \frac{1}{x}, & x > \xi \end{cases} .$$

The jump condition gives

$$C/\xi^2 = 1/\xi^2, \quad C = 1.$$

Then

$$g = \begin{cases} 1 - \frac{1}{\xi}, & x < \xi \\ 1 - \frac{1}{x}, & x > \xi \end{cases} .$$

For a given function f,

$$u = \int_0^x \left(1 - \frac{1}{x}\right) \xi f d\xi + \int_x^1 \left(1 - \frac{1}{xi}\right) \xi f d\xi'$$

Solution exists if ξf is integrable in $(0, x)$ and $(1 - \xi)f$ is integrable in $(x, 1)$

1.14 Find the Green's function for

$$x^2u'' - xu' + u = f(x), \quad u(0) = 0, \quad u(1) = 0.$$

Solution

With the solution $u = x^n$,

$$n(n-1) - n + 1 = 0, \quad n = 1, 1.$$

For this repeated index,

$$u_1 = x, \quad u_2 = x \log x.$$

These satisfy the left and right boundary conditions, respectively.

$$g = C \begin{cases} x\xi \log \xi, & x < \xi \\ \xi x \log x, & x > \xi \end{cases}.$$

The jump condition gives

$$C[\xi(\log \xi + 1) - \xi \log \xi] = 1/\xi^2, \quad C = 1/\xi^3.$$

$$g = \begin{cases} x\xi^{-2} \log \xi, & x < \xi \\ \xi^{-2} x \log x, & x > \xi \end{cases}.$$

1.15 Using the self-adjoint form of the differential equation

$$x^2u'' + 3xu' - 3u = f(x), \quad u(0) = 0, \quad u(1) = 0,$$

find the Green's function and obtain an explicit solution when $f(x) = x$.

Solution

The self-adjoint form for this equation is

$$(x^3u')' - 3xu = xf.$$

Using $u = x^n$, we find

$$n(n-1) + 3n - 3 = 0, \quad n = 1, -3.$$

$$u_1 = x, \quad \text{and} \quad u_2 = x - x^{-3}$$

satisfy the required boundary conditions.

$$g = C \begin{cases} x(\xi - \xi^{-3}), & x < \xi \\ \xi(x - x^{-3}), & x > \xi \end{cases} .$$

The jump condition gives

$$C[\xi(1 + 3\xi^{-4}) - (\xi - \xi^{-3})] = \xi^{-3}, \quad C = \frac{1}{4}.$$

Then,

$$g = \frac{1}{4} \begin{cases} x(\xi - \xi^{-3}), & x < \xi \\ \xi(x - x^{-3}), & x > \xi \end{cases} .$$

For $f = x$,

$$\begin{aligned} u &= \frac{1}{4}\left[\int_0^x (x - x^{-3})\xi^3 d\xi + \int_x^1 x(\xi - \xi^{-3})\xi^2 d\xi\right] \\ &= \frac{1}{4}\left[(x - x^{-3})\frac{x^4}{4} + x\frac{1 - x^4}{4} + x\log x\right] \\ &= \frac{1}{4}x\log x. \end{aligned}$$

1.16 Solve the equation

$$x^2u'' + 3xu' = x^2, \quad u(1) = 1, \quad u(2) = 2,$$

using the Green's function.

Solution

This equation can be written as

$$(x^3u')' = x^3.$$

To obtain homogeneous boundary conditions, let

$$u = v + x,$$

where the function x satisfies the non-homogeneous boundary conditions.

Then,

$$u' = v' + 1$$

and

$$(x^3v')' = x^3 - 3x^2.$$

The homogeneous equation can be integrated to get

$$(x^3v')' = 0, \quad x^3v' = A, \quad v' = \frac{A}{x^3}, \quad v = -\frac{A}{2x^2} + B.$$

To satisfy $v_1(1) = 0$ we choose

$$v_1 = 1 - \frac{1}{x^2}$$

and to satisfy $v_2(2) = 0$ we choose

$$v_2 = 1 - \frac{4}{x^2}.$$

With these the Green's function can be written as

$$g = C \begin{cases} \left(1 - \frac{1}{x^2}\right)\left(1 - \frac{4}{\xi^2}\right), & x < \xi \\ \left(1 - \frac{4}{x^2}\right)\left(1 - \frac{1}{\xi^2}\right), & x > \xi \end{cases}$$

Using the jump condition

$$C\left[\left(1-\frac{1}{\xi^2}\right)\frac{8}{\xi^3}-\left(1-\frac{4}{\xi^2}\right)\frac{2}{\xi^3}\right]=\frac{1}{\xi^3}.$$

$$C=\frac{1}{6}.$$

The solution of the non-homogeneous equation is

$$\begin{aligned} v &= \frac{1}{6}\left[\left(1-\frac{4}{x^2}\right)\int_0^x\left(1-\frac{1}{\xi^2}\right)(\xi^3-3\xi^2)d\xi\right. \\ &\quad \left.+\left(1-\frac{1}{x^2}\right)\int_x^1\left(1-\frac{4}{\xi^2}\right)(\xi^3-3\xi^2)d\xi\right] \\ &= \frac{x^2}{8}-x-\frac{5}{6x^2}+\frac{41}{24}. \end{aligned}$$

and

$$u=\frac{x^2}{8}-\frac{5}{6x^2}+\frac{41}{24}.$$

1.17 For the problem

$$Lu = u'' + u' = 0, \quad u(0) = 0, \quad u'(1) = 0,$$

obtain the adjoint system. Solve the eigenvalue problems,

$$Lu = \lambda u, \quad L^* v = \lambda v,$$

and show that their eigenfunctions are bi-orthogonal.

Solution

To obtain the adjoint system, we use integration by parts.

$$\begin{aligned} \langle v, Lu \rangle &= \int_0^1 v[u'' + u']dx \\ &= (vu' - v'u + vu)\big|_0^1 + \int_0^1 u[v'' - v']dx. \end{aligned}$$

Then

$$L^* v = v'' - v', \quad v(0) = 0, \quad v'(1) - v(1) = 0.$$

The two eigenvalue problems are:

$$\begin{array}{ll} u'' + u' - \lambda u = 0, & v'' - v' - \lambda v = 0 \\ u = e^{\alpha x}, & v = e^{\beta x} \\ \alpha^2 + \alpha - \lambda = 0, & \beta^2 - \beta - \lambda = 0 \\ \alpha = -\frac{1}{2} \pm \sqrt{\frac{1}{4} + \lambda}, & \beta = \frac{1}{2} \pm \sqrt{\frac{1}{4} + \lambda} \end{array}$$

Let $i\mu = \sqrt{\lambda + 1/4}$

$$u = e^{-x/2}[A \sin \mu x + B \cos \mu x], \quad v = e^{x/2}[C \sin \mu x + D \cos \mu x]$$

Using $u(0) = 0$, $u'(1) = 0$ and $v(0) = 0$, $v'(1) - v(1) = 0$ we find

$$\begin{array}{ll} B = 0, \quad \tan \mu = 2\mu, & D = 0, \quad \tan \mu = 2\mu. \\ u_i = e^{-x/2} \sin \mu_i x, & v_j = e^{x/2} \sin \mu_j x, \end{array}$$

where μ_i and μ_j are solutions of $\tan \mu = 2\mu$.

When $\mu_i \neq \mu_j$,

$$
\begin{aligned}
I &= \int_0^1 \sin \mu_i x \sin \mu_j x dx \\
&= \left[\frac{-\cos \mu_i x}{\mu_i} \sin \mu_j x + \frac{\sin \mu_i x}{\mu_i^2} \cos \mu_j x \right]_0^1 \\
&\quad + \frac{\mu_j^2}{\mu_i^2} \int_0^1 \sin \mu_i x \sin \mu_j x dx \\
\left(1 - \frac{\mu_j^2}{\mu_i^2}\right) I &= -\frac{\cos \mu_i}{\mu_i} \sin \mu_j + \frac{\mu_j}{\mu_i^2} \sin \mu_i \cos \mu_j \\
&= \frac{\mu_j}{\mu_i} \cos \mu_i \cos \mu_j \left[-\frac{\tan \mu_j}{\mu_j} + \frac{\tan \mu_i}{\mu_i} \right] \\
&= 0
\end{aligned}
$$

Thus $\langle u_i, v_j \rangle = 0$.

1.18 By solving the nonhomogeneous problem

$$u'' = \delta_\epsilon(x-\xi), \quad u(0)=0, \quad u(1)=0,$$

where

$$\delta_\epsilon(x) = \begin{cases} 0, & |x| > \epsilon, \\ \frac{1}{2\epsilon}, & |x| < \epsilon, \end{cases}$$

in three parts:
a) $0 < x < \xi - \epsilon$,
b) $\xi - \epsilon < x < \xi + \epsilon$, and
c) $\xi + \epsilon < x < 1$,
show that, in the limit $\epsilon \to 0$, we recover the Green's function.

Solution

Let u_1, u_2, and u_3 represent the solutions in the three domains.

$$u_1'' = 0, \quad x < \xi - \epsilon.$$

$$u_1 = A_1 x + B_1, \quad u_1(0) = 0, \quad B = 0.$$

$$u_1 = A_1 x$$

$$u_3'' = 0, \quad x > \xi + \epsilon.$$

$$u_3 = A_3(1-x) + B_3, \quad u_3(1) = 0, \quad B_3 = 0.$$

$$u_2'' = \frac{1}{2\epsilon}, \quad \xi - \epsilon < x < \xi + \epsilon.$$

$$u_2 = \frac{x^2}{4\epsilon} + A_2 x + B_2.$$

Using continuity of the solutions

$$u_1(\xi-\epsilon) = u_2(\xi-\epsilon), \quad u_1'(\xi-\epsilon) = u_2'(\xi-\epsilon),$$

$$A_1 = A_2 + \frac{\xi-\epsilon}{2\epsilon}, \quad A_1(\xi-\epsilon) = A_2(\xi-\epsilon) + B_2 + \frac{(\xi-\epsilon)^2}{4\epsilon}.$$

$$u_3(\xi+\epsilon) = u_2(\xi+\epsilon), \quad u_3'(\xi+\epsilon) = u_2'(\xi+\epsilon),$$

$$-A_3 = A_2 + \frac{(\xi+\epsilon)}{2\epsilon}, \quad A_3(1-\xi-\epsilon) = A_2(\xi+\epsilon) + B_2 + \frac{(\xi+\epsilon)^2}{4\epsilon}.$$

$$B_2 = \frac{(\xi-\epsilon)^2}{4\epsilon},$$

$$A_2(\xi+\epsilon) + \frac{(\xi-\epsilon)^2}{4\epsilon} + \frac{(\xi+\epsilon)^2}{4\epsilon} = -A_2(1-\xi-\epsilon) - \frac{(\xi+\epsilon)^2}{2\epsilon}(1-\xi-\epsilon),$$

$$A_2 = -\frac{\xi+\epsilon}{2\epsilon}(1-\xi-\epsilon) - \frac{(\xi-\epsilon)^2+(\xi+\epsilon)^2}{4\epsilon},$$

$$A_1 = -\frac{\xi+\epsilon}{2\epsilon}(1-\xi-\epsilon) + \frac{\xi-\epsilon}{2\epsilon} - \frac{(\xi-\epsilon)^2+(\xi+\epsilon)^2}{4\epsilon},$$

$$A_3 = -\frac{\xi+\epsilon}{2\epsilon}(1-\xi-\epsilon) - \frac{\xi+\epsilon}{2\epsilon} + \frac{(\xi-\epsilon)^2+(\xi+\epsilon)^2}{4\epsilon}.$$

As $\epsilon \to 0$, $u_2(\xi)$ can be evaluated as

$$\begin{aligned} u_2 &= \lim_{\epsilon\to 0}\left[\frac{x^2}{4\epsilon} + A_2 x + B_2\right] \\ &= \lim_{\epsilon\to 0}\left[\frac{\xi^2}{4\epsilon} + -\frac{\xi^2}{2\epsilon}(1-\xi) - \frac{2\xi^3}{4\epsilon} + \frac{\xi^2}{4\epsilon}\right] + \xi(\xi-1) \\ &= \xi(\xi-1), \end{aligned}$$

where ϵ^2 terms have been neglected from the outset.

$$A_1 = (\xi-1), \quad A_3 = -\xi.$$

$$u_1 = x(\xi-1), \quad u_3 = \xi(x-1).$$

Finally, we have

$$g = \begin{cases} x(\xi-1), & x<\xi \\ \xi(x-1), & x>\xi \end{cases} \quad .$$

1.19 Expanding

$$g(x,\xi) = \begin{cases} x(\xi - 1), & x < \xi, \\ \xi(x-1), & x > \xi, \end{cases}$$

in terms of $u_n = \sqrt{2}\sin n\pi x$ as a Fourier series, show that

$$g(x,\xi) = \sum_{n=1} \frac{u_n(x)u_n(\xi)}{\lambda_n}, \quad \lambda_n = -\pi^2 n^2.$$

Solution

Let

$$g(x,\xi) = \sum_{n=1}^{\infty} A_n \sin n\pi x, \quad A_n = 2\int_0^1 g(x,\xi)\sin n\pi x dx.$$

$$\begin{aligned}
A_n &= 2\left\{(\xi-1)\int_0^{\xi} x\sin x dx + \xi\int_{\xi}^1 (x-1)\sin n\pi x dx\right\} \\
&= 2\left\{(\xi-1)\left[x\frac{-\cos n\pi x}{n\pi} - \frac{-\sin n\pi x}{(n\pi)^2}\right]\Bigg|_0^{\xi}\right. \\
&\quad \left. + \xi\left[(x-1)\frac{-\cos n\pi x}{n\pi} - \frac{-\sin n\pi x}{(n\pi)^2}\right]\Bigg|_{\xi}^1\right\} \\
&= 2\left\{(\xi-1)\left[-\xi\frac{\cos n\pi\xi}{n\pi} + \frac{\sin n\pi\xi}{(n\pi)^2}\right]\right. \\
&\quad \left. + \xi\left[(\xi-1)\frac{\cos n\pi\xi}{n\pi} - \frac{\sin n\pi\xi}{(n\pi)^2}\right]\right\} \\
&= -2\frac{\sin n\pi\xi}{n^2\pi^2}.
\end{aligned}$$

Then

$$g(x,\xi) = -\sum_{n=1}^{\infty}\frac{2}{n^2\pi^2}\sin n\pi x\sin n\pi\xi = \sum_{n=1}^{\infty}\frac{1}{\lambda_n}u_n(x)u_n(\xi).$$

1.20 Obtain the Green's function for

$$\kappa\nabla^2 u(x_1, x_2) = v_1\frac{\partial u}{\partial x_1} + v_2\frac{\partial u}{\partial x_2},$$

where v_1 and v_2 are constants, by transforming the dependent variable.

Solution

$$\frac{\partial^2 u}{\partial x_1^2} - \frac{v_1}{\kappa} + \frac{\partial^2 u}{\partial x_2^2} - \frac{v_2}{\kappa}\frac{\partial u}{\partial x_2} = \frac{f}{\kappa}$$

Let

$$u = e^{\lambda_1 x_1 + \lambda_2 x_2}\phi(x_1, x_2).$$

Then

$$\frac{\partial u}{\partial x_1} = e^{\lambda_1 x_1 + \lambda_2 x_2}[\lambda_1\phi + \frac{\partial \phi}{\partial x_1}],$$

$$\frac{\partial^2 u}{\partial x_1^2} = e^{\lambda_1 x_1 + \lambda_2 x_2}\left[\lambda_1^2\phi + 2\lambda_1\frac{\partial \phi}{\partial x_1} + \frac{\partial^2 \phi}{\partial x_1^2}\right].$$

The given equation becomes

$$\frac{\partial^2 \phi}{\partial x_1^2} + \frac{\partial^2 \phi}{\partial x_2^2} + \left(\lambda_1^2 + \lambda_2^2 - \frac{\lambda_1 v_1}{\kappa} - \frac{\lambda_2 v_2}{\kappa}\right)\phi$$

$$+\left(2\lambda_1 - \frac{v_1}{\kappa}\right)\frac{\partial \phi}{\partial x_1} + \left(2\lambda_2 - \frac{v_2}{\kappa}\right)\frac{\partial \phi}{\partial x_2} = \frac{fe^{-(\lambda_1 x_1 + \lambda_2 x_2)}}{\kappa}.$$

We choose

$$\lambda_1 = \frac{v_1}{2\kappa}, \quad \lambda_2 = \frac{v_2}{2\kappa},$$

to get

$$\nabla^2\phi - k^2\phi = F,$$

where

$$k^2 = \frac{v_1^2 + v_2^2}{4\kappa^2}, \quad F = \frac{fe^{-(\lambda_1 x_1 + \lambda_2 x_2)}}{\kappa}.$$

The solutions of the homogeneous equation are the modified Bessel functions, $K_0(kr)$ and $I_0(kr)$, with

$$r = \sqrt{x_1^2 + x_2^2}.$$

For a bounded solution for $r \to \infty$, we choose

$$\phi = AK_0(kr).$$

As $kr \to 0$, K_0 behaves as $-\log r$ and we choose

$$A = -\frac{1}{2\pi},$$

by comparison to the two dimensional Laplace operator.

For the original equation with f on the right hand side,

$$g = -\frac{1}{2\pi} e^{-[v_1(x_1-\xi_1)+v_2(x_2-\xi_2)]/(2\kappa)} K_0(k\rho)$$

where $\rho = \sqrt{(x_1-\xi_1)^2 +)x_2-\xi_2)^2}$.

1.21 The anisotropic Laplace equation in a two dimensional infinite domain is given by

$$k_1^2 \frac{\partial^2 u}{\partial x_1^2} + k_2^2 \frac{\partial^2 u}{\partial x_2^2} = 0.$$

Find the Green's function for this equation.

SolutionThe Green's function satisfies

$$k_1^2 \frac{\partial^2 u}{\partial x_1^2} + k_2^2 \frac{\partial^2 u}{\partial x_2^2} = \delta(x_1, x_2),$$

if the source is at the origin. Let

$$y_1 = x_1/k_1, \quad y_2 = x_2/k_2.$$

Then, u satisfies the Laplace equation

$$\frac{\partial^2 u}{\partial y_1^2} + \frac{\partial^2 u}{\partial y_2^2} = 0,$$

away from the origin.

$$g(y_1, y_2) = \frac{1}{2\pi} \log \sqrt{y_1^2 + y_2^2}.$$

In terms of the original variables, with the source being at (ξ_1, ξ_2)

$$g(x_1, x_2, \xi_1 \xi_2) = \frac{1}{2\pi} \log \left[\frac{(x_1 - \xi_1)^2}{k_1^2} + \frac{(x_2 - \xi_2)^2}{k_1^2} \right]^{1/2}.$$

Note: When g is used in an integral, the original area element $dy_1 dY_2 = dx_1 dx_2/(k_1 k_2)$.

1.22 In a semi-infinite medium, $-\infty < x < \infty$, $0 < y < \infty$, the pressure fluctuations satisfy the wave equation

$$\nabla^2 p = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2},$$

where c is the wave speed and t is time. If the boundary, $y = 0$, is subjected to a pressure $p = P_0 \delta(x) h(t)$, show that

$$p(x, y, t) = A\frac{y}{r^2}\frac{t}{\sqrt{t^2 - k^2 r^2}} h(t - kr),$$

where $r = \sqrt{x^2 + y^2}$ and $k = 1/c$, is a solution of the wave equation. Evaluate the constant A using equilibrium of the medium in the neighborhood of the applied load. Hint: Use polar coordinates.

Solution

In polar coordinates, the wave equation becomes

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + \frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2} = k^2 \frac{\partial^2 p}{\partial t^2}.$$

Away from $r = 0$, the given solution,

$$p = A\frac{t \sin\theta}{r\beta}, \quad \beta = \sqrt{t^2 - k^2 r^2},$$

has the derivatives

$$
\begin{aligned}
\frac{\partial p}{\partial t} &= A\frac{\sin\theta}{r}\left\{\frac{1}{\beta}-\frac{t^2}{\beta^3}\right\} \\
&= -A\frac{\sin\theta}{r}\frac{k^2r^2}{\beta^3} \\
k^2\frac{\partial^2 p}{\partial t^2} &= A\sin\theta\frac{3k^4rt}{\beta^5} \\
\frac{1}{r^2}\frac{\partial^2 p}{\partial\theta^2} &= -A\frac{\sin\theta}{r^3}\frac{t}{\beta} \\
\frac{\partial p}{\partial r} &= A\sin\theta\left\{\frac{k^2t}{\beta^3}-\frac{t}{\beta r^2}\right\} \\
&= A\frac{t\sin\theta}{r^2\beta^3}\left\{k^2r^2-\beta^2\right\} \\
&= A\frac{t\sin\theta}{r^2\beta^3}\left\{2k^2r^2-t^2\right\} \\
&= A\frac{t\sin\theta}{\beta^3}\left\{2k^2-\frac{t^2}{r^2}\right\} \\
\frac{\partial^2 p}{\partial r^2} &= At\sin\theta\left\{\left(2k^2-\frac{t^2}{r^2}\right)\frac{3k^2r}{\beta^5}+\frac{2t^2}{r^3\beta^3}\right\} \\
&= A\frac{t\sin\theta}{r^3\beta^5}\left\{6k^4r^4-3k^2t^2r^2+2t^2(t^2-k^2r^2)\right\} \\
&= A\frac{t\sin\theta}{r^3\beta^5}\left\{6k^4r^4-5k^2t^2r^2+2t^4\right\}
\end{aligned}
$$

Now, the left hand side of the wave equation becomes

$$
\begin{aligned}
&= At\sin\theta\left\{\frac{1}{r^3\beta^5}(6k^4r^4-5k^2t^2r^2+2t^4)+\frac{1}{r^3\beta^3}(2k^2r^2-t^2)-\frac{1}{r^3\beta}\right\} \\
&= A\frac{t\sin\theta}{r^3\beta^5}\left\{6k^4r^4-5k^2t^2r^2+2t^4+2k^2r^2(t^2-k^2r^2)\right. \\
&\qquad \left.-t^2(t^2-k^2r^2)-(t^4-2k^2r^2t^2+k^4r^4)\right\} \\
&= 3A\frac{t\sin\theta}{\beta^5}k^4r.
\end{aligned}
$$

This is identical to the right hand side quantity.

To find the value of A, we draw a semi-circle under the concentrated load P_0 and balance the vertical force. We let $r \to 0$.

$$\begin{aligned} P_0 &= 2\int_0^{\pi/2} p \sin\theta r d\theta \\ &= 2A\int_0^{\pi/2} \sin^2\theta d\theta = A\pi/2. \end{aligned}$$

Then

$$A = \frac{2P_0}{\pi}$$

1.23 For the two dimensional wave equation

$$\nabla^2 u = \frac{1}{c^2}\frac{\partial^2 u}{\partial t^2},$$

steady state solutions are obtained using $u(x, y, t) = v(x, y)e^{-i\Omega t}$ where v satisfies the Helmholtz equation,

$$Lv = 0, \quad L = \nabla^2 + k^2, \quad k = \Omega/c.$$

Show that the Hankel functions $H_0^{(1)}(kr)$ and $H_0^{(2)}(kr)$ satisfy $Lg = \delta(x, y)$ when $r = \sqrt{x^2 + y^2} \neq 0$. Examine their asymptotic forms for $kr << 1$ and for $kr >> 1$, using the results shown in Abramowitz and Stegun (1965) and select multiplication constants A and B to have $g = AH_0^{(1)}$ or $g = BH_0^{(2)}$ by comparing the asymptotic form with the Green's function for the Laplace operator ($k \to 0$). Show that $H_0^{(1)}$ corresponds to an outgoing wave and $H_0^{(2)}$ to an incoming wave.

Solution

The Helmholtz equation,

$$(\nabla^2 + k^2)v = \delta(x - \xi, y - \eta)$$

with a source at the origin, has the polar form

$$(rv')' + k^2 rv = \delta(r),$$

where we assume axisymmetry.

Solutions are the Hankel functions as

$$v = H_0^{(1)}(kr), H_0^{(2)}(kr).$$

When $kr << 1$,

$$H_0^{(1)}(kr) = J_0 + iY_0 \sim \frac{2}{\pi}\log r,$$

$$H_0^{(2)}(kr) = J_0 - iY_0 \sim -\frac{2}{\pi}\log r,$$

Comparing with the 2D Laplace operator

$$g = \frac{-i}{4} H_0^{(1)}(kr), \quad \text{or} \quad g = \frac{i}{4} H_0^{(2)}(kr).$$

When $kr >> 1$,

$$H_0^{(1)}(kr) \sim \sqrt{\frac{2}{\pi kr}} e^{i(kr-\pi/4)},$$

$$H_0^{(2)}(kr) \sim \sqrt{\frac{2}{\pi kr}} e^{-i(kr-\pi/4)},$$

Combining these with $\exp(-i\Omega t)$, we find terms of the form $(r - ct)$ for $H_0^{(1)}(kr)$ and $(r + ct)$ for $H_0^{(2)}(kr)$, with former being an outgoing wave and the latter an incoming wave.

1.24 Show that

$$g = -\frac{e^{\pm ikr}}{4\pi r},$$

satisfies the Helmholtz equation

$$\nabla^2 g + k^2 g = \delta(\boldsymbol{x} - \boldsymbol{\xi}),$$

in a 3D infinite domain, with $r = |\boldsymbol{x} - \boldsymbol{\xi}|$. Assuming the Helmholtz equation is obtained from the wave equation by separating the time dependence using a factor $e^{-i\Omega t}$, show that $(\pm)$ signs correspond to outgoing and incoming waves, respectively.

Solution

In spherical coordinates the spherically symmetric Green's function satisfies

$$(r^2 g')' + k^2 r^2 g = 0,$$

away from the source point. Substituting the given solution and neglecting the factor $-1/(4\pi)$,

$$\begin{aligned}
&\left\{ r^2 \left[-\frac{1}{r^2} \pm \frac{ik}{r} \right] e^{\pm ikr} \right\}' + k^2 r e^{\pm ikr} \\
=\;& \{(-1 \pm ikr)' + (-1 \pm ikr)(\pm ik)\} + k^2 r \\
=\;& \pm ik \mp ik - k^2 r + k^2 r \\
=\;& 0.
\end{aligned}$$

Integrating the equation for the Green's function over a sphere of radius $r << 1$,

$$\begin{aligned}
4\pi r^2 g'|_r &= 1 \\
\lim_{r \to 0} 4\pi r^2 \left(\frac{1}{4\pi r^2} \mp \frac{ik}{4\pi r} \right) e^{\pm ikr} &= 1
\end{aligned}$$

For the $(+)$ sign, we find the exponential term $(r - ct)$ representing an outgoing wave and for the $(-)$ sign, $(r + ct)$ representing an incoming wave. Here $c = \Omega/k$.

1.25 Consider a volume V enclosed by the surface S in 3D. From

$$\nabla^2 u + k^2 u = f, \quad \nabla^2 g + k^2 g = \delta(\boldsymbol{x} - \boldsymbol{\xi}),$$

obtain the solution

$$u(\boldsymbol{\xi}) = \int_V g(\boldsymbol{\xi}, \boldsymbol{x}) f(\boldsymbol{x}) dV - \int_S \left[g\frac{\partial u}{\partial n} - u\frac{\partial g}{\partial n} \right] dS.$$

Solution

Using the inner products of the first equation with g and the second equation with u and subtracting

$$\langle g, Lu \rangle - \langle u, Lg \rangle = \langle g, f \rangle - \langle u, \delta \rangle.$$

For a self-adjoint operator, the left hand side can be integrated by parts (Gauss theorem) to get

$$u(\boldsymbol{\xi}) = \int_V g(\boldsymbol{\xi}, \boldsymbol{x}) f(\boldsymbol{x}) dV - \int_S \left[g\frac{\partial u}{\partial n} - u\frac{\partial g}{\partial n} \right] dS.$$

1.26 In the previous problem, assuming V is a sphere of radius R centered at $\boldsymbol{x} = \boldsymbol{0}$ and $\boldsymbol{x}$ is a point on its surface, and g is the Green's function for the 3D infinite space, show that

$$u(\boldsymbol{\xi}) = \frac{1}{4\pi} \int_S \left[\frac{e^{ikr}}{r} \frac{\partial u}{\partial n} - u \frac{\partial}{\partial n} \left(\frac{e^{ikr}}{r} \right) \right] dS,$$

if $f = 0$. If the included angle between $\boldsymbol{x}$ and $\boldsymbol{x} - \boldsymbol{\xi}$ is ψ, show that $dr/dn = \cos\psi$. Also show that as $R \to \infty$, $r \to R$ and $R^2(1 - \cos\psi)$ is finite and for the surface integral to exist

$$r \left(\frac{\partial u}{\partial r} - iku \right) \to 0, \quad \text{and} \quad u \to 0.$$

These are known as the Sommerfeld radiation conditions.

Solution

For an infinite 3D space the Green's function for the Helmholtz operator is

$$g = -\frac{1}{4\pi r} e^{ikr}.$$

When there is no forcing function

$$u(\boldsymbol{\xi}) = \frac{1}{4\pi} \int_S \left[\frac{e^{ikr}}{r} \frac{\partial u}{\partial n} - u \frac{\partial}{\partial n} \left(\frac{e^{ikr}}{r} \right) \right] dS$$

From Fig. 1.1,

$$\frac{dr}{dn} = \cos\psi$$

Then

$$\begin{aligned} u(\boldsymbol{\xi}) &= \frac{1}{4\pi} \int_S \left[\frac{1}{r} \frac{\partial u}{\partial n} - \frac{ik}{r} u \cos\psi + \frac{u}{r^2} \cos\psi \right] e^{ikr} dS \\ &= \frac{1}{4\pi} \int_S \left[\frac{1}{r} \frac{\partial u}{\partial n} - \frac{iku}{r} + (1 - \cos\psi) \frac{ik}{r} + \frac{u}{r^2} \cos\psi \right] e^{ikr} dS \end{aligned}$$

From the triangle in Fig. 1.1, with $R = |\boldsymbol{x}|$, we find

$$\xi^2 = R^2 + r^2 - 2Rr\cos\psi.$$

As $R \to \infty$ and $r \to \infty$, $dS \sim 4\pi R^2$ and

$$2R^2(1 - \cos\psi) = \xi^2$$

is finite and the condition for the existence of the integral is

$$r\left[\frac{\partial u}{\partial n} - iku\right] \to 0, \quad \text{and} \quad u \to 0.$$

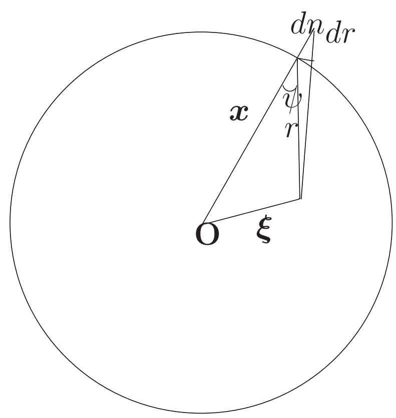

Figure 1.1: A source inside a spherical domain

1.27 By reconsidering the previous problem, show that the surface integral exists if the less restricted condition

$$r\left[\frac{\partial u}{\partial r} - iku + \frac{u}{r}\right] \to 0.$$

is satisfied.

Solution

In the previous problem, if we set

$$\frac{dr}{dn} = \cos\psi,$$

$$u(\boldsymbol{\xi}) = \frac{1}{4\pi}\int_S \left[\frac{1}{r}\frac{\partial u}{\partial r} - \frac{iku}{r} + \frac{u}{r^2}\right] e^{ikr}\cos\psi dS$$

As $dS \sim 4\pi r^2$ and R and r go to infinity, the integral exists if

$$r\left[\frac{\partial u}{\partial r} - iku\right] + u \to 0.$$

1.28 A wedge-shaped 2D domain has the boundaries: $y = 0$ and $y = x$. Obtain the Green's function for the Poisson's equation, $\nabla^2 u = f(x, y)$, for this domain if $u(x, 0) = 0$ and $u(x, x) = 0$. Use the method of images for your solution.

Solution

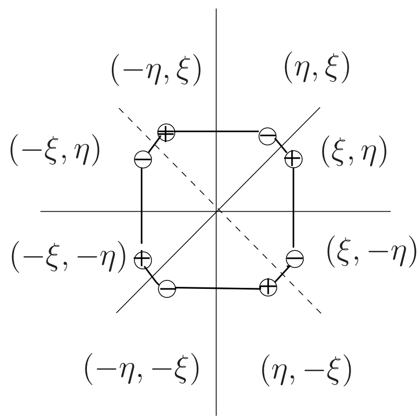

Figure 1.2: Sources and sinks for the wedge bounded by $y = 0$ and $y = x$

As shown in Fig. 1.2, there are 4 sources and 4 sinks to satisfy the boundary conditions. Using the notation

$$r_1 = \sqrt{(x-\xi)^2 + (y-\eta)^2}, \quad r_2 = \sqrt{(x+\eta)^2 + (y-\xi)^2},$$

$$r_3 = \sqrt{(x+\xi)^2 + (y+\eta)^2}, \quad r_4 = \sqrt{(x-\eta)^2 + (y+\xi)^2},$$

$$r_5 = \sqrt{(x-\xi)^2 + (y+\eta)^2}, \quad r_6 = \sqrt{(x+\eta)^2 + (y+\xi)^2},$$

$$r_7 = \sqrt{(x+\xi)^2 + (y-\eta)^2}, \quad r_8 = \sqrt{(x-\eta)^2 + (y-\xi)^2},$$

the Green's function is

$$g = \frac{1}{2\pi} \log \frac{r_1 r_2 r_3 r_4}{r_5 r_6 r_7 r_8}.$$

1.29 A unit circle is mapped into a *cardioid* by the transformation

$$w = c(1+z)^2,$$

where c is a real number. Obtain the Green's function, g, for a domain in the shape of a cardioid with $g = 0$ on the boundary.

Solution

We have

$$w = c(1+z)^2.$$

For a unit circle,

$$g = \frac{1}{4\pi} \log \frac{r^2 + \rho^2 - 2r\rho\cos(\theta - \phi)}{r^2\rho^2 + 1 - 2r\rho\cos(\theta - \phi)},$$

where $r = |z|$, and $\rho = |\zeta|$, where ζ is the source location.

Let ω be the image of the source point ζ. Then

$$z = \left(\frac{w}{c}\right)^{1/2} - 1, \quad \zeta = \left(\frac{\omega}{c}\right)^{1/2} - 1.$$

Let

$$w/c = pe^{i\alpha}, \quad \omega/c = qe^{i\beta},$$

where p and q are real.

$$z = \sqrt{p}e^{i\alpha/2}, \quad \zeta = \sqrt{q}e^{i\beta/2}.$$

From the Green's function for the unit circle,

$$g = \frac{1}{2\pi} \log \frac{z - \zeta}{1 - z\bar{\zeta}},$$

the mapping gives

$$\begin{aligned}
g &= \frac{1}{2\pi} \log \frac{\sqrt{p}e^{i\alpha/2} - \sqrt{q}e^{i\beta/2}}{1 - \sqrt{pq}e^{i(\alpha-\beta)/2}} \\
&= \frac{1}{2\pi} \log \frac{\sqrt{p} - \sqrt{q}e^{i\psi}}{1 - \sqrt{pq}e^{i\psi}}, \quad \psi = \frac{\alpha - \beta}{2} \\
&= \frac{1}{4\pi} \log \frac{(\sqrt{p} - \sqrt{q}e^{i\psi})(\sqrt{p} - \sqrt{q}e^{-i\psi})}{(1 - \sqrt{pq}e^{i\psi})(1 - \sqrt{pq}e^{-i\psi})} \\
&= \frac{1}{4\pi} \log \frac{p + q - 2\sqrt{pq}\cos\psi}{1 + pq - 2\sqrt{pq}\cos\psi}.
\end{aligned}$$

1.30 The steady-state temperature in a semi-infinite plate satisfies

$$\nabla^2 u(x, y) = 0, \quad -\infty < x < \infty, \quad 0 < y < \infty,$$

On the boundary, $y = 0$, the temperature is given as $u = f(x)$. Obtain the temperature distribution in the domain using the Green's function.

Solution

From

$$\nabla^2 u = 0, \quad \nabla^2 g = \delta(\boldsymbol{x} - \boldsymbol{\xi}),$$
$$u(\boldsymbol{\xi}) = \int_A [u\nabla^2 g - g\nabla^2 u] dA.$$

Using a semi-circle of large radius R, the area integral can be transformed into a line integral by using Gauss theorem, to get

$$u(\boldsymbol{\xi}) = \oint_C \left[u\frac{\partial g}{\partial n} - g\frac{\partial u}{\partial n} \right] ds.$$

The Green's function for this proble is

$$g = \frac{1}{4\pi} \left\{ \log[(x-\xi)^2 + (y-\eta)^2] - \log[(x-\xi)^2 + (y+\eta)^2] \right\}.$$

On the semi-circle, u and $\partial u/\partial n$ vanish, and the contour integral becomes

$$u(\boldsymbol{\xi}) = \int_{-\infty}^{\infty} \left[f\frac{\partial g}{\partial n} - g\frac{\partial u}{\partial n} \right] dx.$$

On the x-axis, $u = 0$ and

$$\begin{aligned} \frac{\partial g}{\partial n} &= -\frac{\partial g}{\partial y} = -\frac{1}{4\pi}\frac{-2\eta - 2\eta}{(x-\xi)^2 + \eta^2} \\ &= \frac{1}{\pi}\frac{\eta}{(x-\xi)^2 + \eta^2} \end{aligned}$$

The solution is

$$u(x, y) = \frac{y}{\pi} \int_{-\infty}^{\infty} \frac{f(\xi) d\xi}{(\xi - x)^2 + \eta^2}.$$

1.31 Show by substitution that the solutions of

$$[(1-x^2)u']' - \frac{h^2}{1-x^2}u = 0,$$

are

$$u = \left(\frac{1-x}{1+x}\right)^{\pm h/2}.$$

Solution

Let

$$\begin{aligned}
u &= (1-x)^{\alpha}(1+x)^{-\alpha}, \\
u' &= \alpha[(1-x)^{\alpha-1}(1+x)^{-\alpha} + (1-x)^{\alpha}(1+x)^{-\alpha-1}], \\
(1-x^2)u' &= -\alpha[(1-x)^{\alpha}(1+x)^{-\alpha+1} + (1-x)^{\alpha+1}(1+x)^{-\alpha}], \\
((1-x^2)u')' &= -\alpha[-\alpha(1-x)^{\alpha-1}(1+x)^{-\alpha+1} - (\alpha-1)(1-x)^{\alpha}(1+x)^{-\alpha} \\
&\quad -(\alpha+1)(1-x)^{\alpha}(1+x)^{-\alpha} - \alpha(1-x)^{\alpha+1}(1+x)^{-\alpha-1}] \\
&= \alpha(1-x)^{\alpha}(1+x)^{-\alpha}\left[\alpha\frac{1+x}{1-x} + \alpha - +\alpha + 1 + \alpha\frac{1-x}{1+x}\right] \\
&= \alpha^2(1-x)^{\alpha}(1+x)^{-\alpha}(1-x^2) \\
&= \frac{4\alpha^2 u}{1-x^2}.
\end{aligned}$$

Comparing with the given differential equation

$$\alpha = \pm h/2.$$

1.32 Obtain the Green's function for the above operator when $h \neq 0$ if the boundary conditions are

$$u(-1) = \text{finite}, \quad u(1) = \text{finite}.$$

Solution

For $u(-1)$ to be finite, we take

$$u_1 = \left(\frac{1+x}{1-x}\right)^{h/2},$$

and for $u(1)$ to be finite, we take

$$u_2 = \left(\frac{1-x}{1+x}\right)^{h/2}.$$

Then

$$g = C \begin{cases} (1+x)^{h/2}(1-x)^{-h/2}(1-\xi)^{h/2}(1+\xi)^{-h/2}, & x < \xi \\ (1-x)^{h/2}(1+x)^{-h/2}(1+\xi)^{h/2}(1-\xi)^{-h/2}, & x > \xi \end{cases}.$$

The jump condition

$$[[g']] = 1/p,$$

gives

$$\frac{Ch}{2}\left\{\left(\frac{1+\xi}{1-\xi}\right)^{h/2}[-(1-\xi)^{h/2-1}(1+\xi)^{-h/2} - (1-\xi)^{h/2}(1+\xi)^{-h/2-1}]\right.$$

$$\left. - \left(\frac{1-\xi}{1+\xi}\right)^{h/2}[(1+\xi)^{h/2-1}(1-\xi)^{-h/2} + (1+\xi)^{h/2}(1-\xi)^{-h/2-1}]\right\} = \frac{1}{1-\xi^2}$$

$$C\left\{-\frac{1}{1-\xi} - \frac{1}{1+\xi} - \frac{1}{1+\xi} - \frac{1}{1-\xi}\right\} = \frac{2}{h^2}\frac{1}{1-\xi^2}$$

$$C[-2(1+\xi) - 2(1-\xi)] = 2/h$$

$$C = -\frac{1}{2h}.$$

$$g = -\frac{1}{2h} \begin{cases} (1+x)^{h/2}(1-x)^{-h/2}(1-\xi)^{h/2}(1+\xi)^{-h/2}, & x < \xi \\ (1-x)^{h/2}(1+x)^{-h/2}(1+\xi)^{h/2}(1-\xi)^{-h/2}, & x > \xi \end{cases}.$$

1.33 If $h = 0$ in the preceding problem, show that

$$U(x) = \frac{1}{\sqrt{2}}$$

is a normalized solution of the above homogeneous equation, which satisfies both the boundary conditions. Obtain the generalized Green's function for this case.

Solution

When $h = 0$, the equation is

$$((1 - x^2)u')' = 0.$$

Integrating

$$u' = \frac{A}{1 - x^2} = \frac{A}{2}\left[\frac{1}{1 - x} + \frac{1}{1 + x}\right],$$

$$u = \frac{A}{2}\left[\log|1 - x| + \log|1 + x|\right] + B.$$

For $u(-1)$ and $u(1)$ to be finite, $u = B$. Normalizing this in $(-1, 1)$, we get

$$U = \frac{1}{\sqrt{2}}.$$

The generalized Green's function satisfies

$$[(1 - x^2)g']' \quad = \quad \delta(x - \xi) - U(x)U(\xi),$$

which can be written as

$$\begin{aligned} [(1 - x^2)g_1']' &= -\frac{1}{2}, \quad x < \xi, \\ [(1 - x^2)g_2']' &= -\frac{1}{2}, \quad x > \xi, \end{aligned}$$

Integrating these, we get

$$
\begin{aligned}
(1-x^2)g_1' &= -\frac{x}{2} + C_1, \\
(1-x^2)g_2' &= -\frac{x}{2} + C_2, \\
g_1' &= -\frac{1}{2}\frac{x}{1-x^2} + \frac{C_1}{2}\left\{\frac{1}{1-x} + \frac{1}{1-x}\right\}, \\
g_2' &= -\frac{1}{2}\frac{x}{1-x^2} + \frac{C_2}{2}\left\{\frac{1}{1-x} + \frac{1}{1-x}\right\}, \\
g_1 &= \frac{1}{4}\log(1-x^2) + \frac{C_1}{2}[\log(1-x) + \log(1+x)] + C_3, \\
g_2 &= \frac{1}{4}\log(1-x^2) + \frac{C_2}{2}[\log(1-x) + \log(1+x)] + C_4,
\end{aligned}
$$

For $g_1(-1)$ to be finite, $C_1 = -1/2$ and for $g_2(1)$ to be finite, $C_2 = -1/2$. Then

.

$$
\begin{aligned}
g_1 &= \frac{1}{2}\log(1-x) + C_3, \\
g_2 &= \frac{1}{2}\log(1+x) + C_4,
\end{aligned}
$$

For continuity at $x = \xi$, we take

$$
\begin{aligned}
C_3 &= D + \frac{1}{2}\log(1+\xi), \\
C_4 &= D + \frac{1}{2}\log(1-\xi),
\end{aligned}
$$

where D is a new constant. It is found using $\langle U, g\rangle = 0$.

$$
2D + \frac{1+\xi}{2}\log(1+\xi) + \frac{1}{2}\int_{-1}^{\xi}\log(1-x)dx + \frac{1-\xi}{2}\log(1-\xi) + \frac{1}{2}\int_{\xi}^{1}\log(1+x)dx = 0.
$$

$$
\begin{gathered}
2D + \frac{1+\xi}{2}\log(1+\xi) + \frac{1-\xi}{2}\log(1-\xi) \\
+ \frac{1}{2}x\log(1-x)\Big|_{-1}^{\xi} + \frac{1}{2}\int_{-1}^{\xi}\frac{xdx}{1-x} \\
+ \frac{1}{2}x\log(1+x)\Big|_{\xi}^{1} - \frac{1}{2}\int_{\xi}^{1}\frac{xdx}{1+x} = 0.
\end{gathered}
$$

$$2D + \frac{1+\xi}{2}\log(1+\xi) + \frac{1-\xi}{2}\log(1-\xi)$$
$$+\frac{1}{2}\xi\log(1-\xi) - \frac{1}{2}(-1)\log 2 + \frac{1}{2}\log 2 - \frac{1}{2}\xi\log(1+\xi)$$
$$-\frac{1}{2}(1+\xi) - \frac{1}{2}(1-\xi) - \frac{1}{2}\log(1-x)\Big|_{-1}^{\xi} + \frac{1}{2}\log(1+x)\Big|_{\xi}^{1} = 0.$$
$$2D + \frac{1+\xi}{2}\log(1+\xi) + \frac{1-\xi}{2}\log(1-\xi)$$
$$+\frac{1}{2}\xi\log(1-\xi) + \log 2 - \frac{1}{2}\xi\log(1+\xi)$$
$$-1 - \frac{1}{2}\log(1-\xi) + \log 2 - \frac{1}{2}\log(1+\xi) = 0.$$
$$D = \frac{1}{2} - \log 2$$
$$g = \frac{1}{2} - \log 2 + \frac{1}{2}\begin{cases}\log[(1-x)(1+\xi)], & x < \xi,\\ \log[(1+x)(1-\xi)], & x > \xi.\end{cases}$$

1.34 Find the generalized Green's function for

$$u'' = f(x), \quad u(1) = u(-1), \quad u'(1) = u'(-1).$$

Solution

$$u'' = 0, \quad u = Ax + B.$$

$$u'(1) = u'(-1), \quad A = A.$$

$$u(1) = u(-1), \quad A + B = -A + B, \quad A = 0.$$

Then $u = B$ and normalizing this in $(-1, 1)$, we find

$$U = \frac{1}{\sqrt{2}}.$$

The generalized Green's function satisfies

$$g'' = \delta(x - \xi) - \frac{1}{2}.$$

$$\begin{array}{ll} g_1'' = -\frac{1}{2}, & g_2'' = -\frac{1}{2}, \\ g_1' = -\frac{1}{2}x + C_1, & g_2' = -\frac{1}{2}x + C_2, \\ g_1 = -\frac{1}{4}x^2 + C_1 x + D_1, & g_2 = -\frac{1}{4}x^2 + C_2 x + D_2. \end{array}$$

The condition, $g_1(-1) = g_2(1)$, gives

$$\frac{1}{2} + C_1 = -\frac{1}{2} + C_2.$$

This satisfied by using

$$C_1 = C - \frac{1}{2}, \quad C_2 = C + \frac{1}{2}.$$

The condition, $g_1(-1) = g_2(1)$, gives

$$D_1 + \frac{1}{2} - C = D - 2 + \frac{1}{2} + C.$$

We choose

$$D_1 = D + C, \quad D_2 = D - C$$

to satisfy this. So far

$$\begin{aligned} g_1 &= -\tfrac{1}{4}x^2 + Cx - \tfrac{1}{2}x + D + C, \quad & g_2 &= -\tfrac{1}{4}x^2 + Cx + \tfrac{1}{2}x + D - C, \\ g_1 &= -\tfrac{1}{4}x^2 + C(x+1) - \tfrac{1}{2}x + D, \quad & g_2 &= -\tfrac{1}{4}x^2 + C(x-1) + \tfrac{1}{2}x + D. \end{aligned}$$

The continuity condition, $g_1(\xi) = g_2(\xi)$, gives

$$C(\xi+1) - \frac{\xi}{2} = C(\xi-1) + \frac{\xi}{2}, \quad C = \frac{\xi}{2}.$$

Then

$$g_1 = -\frac{1}{4}x^2 + \frac{1}{2}\xi(x+1) - \frac{1}{2}x + D,$$

$$g_2 = -\frac{1}{4}x^2 + \frac{1}{2}\xi(x-1) + \frac{1}{2}x + D.$$

The remaining constant D is found from $\langle U, g \rangle = 0$.

$$2D - \int_{-1}^{1} \frac{x^2}{4}dx + \frac{1}{2}(\xi-1)\int_{-1}^{\xi} xdx + \frac{1}{2}\xi(\xi+1)$$

$$\frac{1}{2}(\xi+1)\int_{\xi}^{1} xdx - \frac{1}{2}\xi(1-\xi) = 0.$$

$$2D - \frac{1}{6} + \xi^2 + \frac{1}{4}(\xi-1)(\xi+1)[\xi - 1 - (\xi+1)] = 0.$$

$$D = -\frac{1}{6} - \frac{1}{4}\xi^2.$$

$$g = -\frac{1}{6} - \frac{x^2+\xi^2}{4}\frac{1}{2}x\xi + \begin{cases} \xi - x, & x < \xi, \\ x - \xi, & x > \xi. \end{cases} \quad .$$

1.35 To solve the self-adjoint problem

$$Lu = f(x),$$

with homogeneous boundary conditions at $x = a$ and $x = b$, we use a generalized Green's function g which satisfies

$$Lg = \delta(x - \xi) - U(x)U(\xi),$$

with homogeneous boundary conditions, where U is the normalized solution of the homogeneous equation satisfying the same boundary conditions. The solution is written as

$$u(x) = AU(x) + \int_a^b g(x, \xi) f(\xi)\, d\xi.$$

By operating on this equation using L, show that u is the required solution.

Solution

$$\begin{aligned} Lu &= \int_a^b Lgf d\xi + ALU \\ &= \int_a^b [\delta(x - \xi) - U(x)U(\xi)] f d\xi + 0 \\ &= f(x), \end{aligned}$$

where we have used

$$LU = 0, \quad \text{and the existence condition} \quad \langle U, f \rangle = 0.$$

1.36 For the equation

$$Lu = u'' + u' = 0, \quad u'(0) = u'(1) = 0,$$

obtain the adjoint equation and its boundary conditions. Obtain normalized homogeneous solutions of $LU = 0$ and $L^*U^* = 0$. Construct generalized Green's functions g and g^*.

Solution

$$Lu = u'' + u' = 0, \quad u'(0) = u'(1) = 0.$$

The adjoint problem is found by integrating by parts.

$$\begin{aligned} \langle v, Lu \rangle &= \int_0^1 v[u'' + u']dx \\ &= (vu' - v'u + vu]\!\!\big|_0^1 + \int_0^1 u[v'' - v']dx. \end{aligned}$$

Thus

$$L * v = v'' - v'.$$

To make the bilinear concomitant 0 at the boundary points,

$$v'(0) - v(0) = 0, \quad v'(1) - v(1) = 0.$$

Next, we find the solutions of the homogeneous equations.

$$u'' + u' = 0, \quad u' + u = A, \quad u = Be^{-x} + A.$$

The solution $U = 1$ satisfies both the boundary conditions.

$$v'' - v' = 0, \quad v' - v = C, \quad v = De^x - C.$$

$$v'(0) - v(0) = 0, \quad C = 0.$$

$$v'(1) - v(1) = 0, \quad v = De^x.$$

The function U^* is the normalized form of v, with normalization condition

$$\int_0^1 UU^* = 1, \quad D = \frac{e^x}{e-1}, \quad \text{Let} \quad \beta = \frac{1}{e-1}.$$

Then, g and $g*$ satisfy

$$Lg = \delta(x-\xi) - U^*(\xi)U(x), \quad L^*g^* = \delta(x-\xi) - U(\xi)U^*(x).$$

Using the subscripts, 1 and 2, for $x < \xi$ and $x > \xi$,

$$\begin{array}{ll}
g_1'' + g_1' = -\beta e^{\xi}, & g_2'' + g_2' = -\beta e^{\xi}, \\
g_1' + g_1 = -\beta e^{\xi} x + C_1, & g_2' + g_2 = -\beta e^{\xi} x + C_2, \\
g_1 = g_{1h} + g_{1p}, & g_2 = g_{2h} + g_{2p}, \\
g_{1h} = B_1 e^{-x} + C_1, \quad g_{1p} = \beta e^{\xi}(1-x), & g_{2h} = B_2 e^{-x} + C_2, \quad g_{2p} = \beta e^{\xi}(1-x), \\
g_1 = B_1 e^{-x} + \beta e^{\xi}(1-x) + C_1, & g_2 = B_2 e^{-x} + \beta e^{\xi}(1-x) + C_2.
\end{array}$$

The condition $g_1'(0) = 0$ gives

$$-B_1 - \beta e^{\xi} = 0, \quad B_1 = -\beta e^{\xi}.$$

The condition $g_2(1) = 0$ gives

$$B_2 e^{-1} - \beta e^{\xi} = 0, \quad B_2 = -\beta e^{\xi+1}.$$

Then

$$g_1 = \beta e^{\xi}[-e^{-x} - x + 1] + C_1, \quad g_2 = \beta e^{\xi}[-e^{1-x} - x + 1] + C_2.$$

From $g_1(\xi) = g_2(\xi)$,

$$C_1 = D + \beta e^{\xi}[-e^{1-\xi} - \xi + 1], \quad C_2 = D + \beta e^{\xi}[-e^{-\xi} - \xi + 1]$$

and with these

$$g_1 = D + \beta e^{\xi}[-e^{-x} - e^{1-\xi} - x + 1 - \xi + 1], \quad g_2 = D + \beta e^{\xi}[-e^{1-x} - e^{-\xi} - x + 1 - \xi + 1].$$

The remaining constant D is found form $\langle g, U^* \rangle = 0$.

$$D\beta \int_0^1 e^x dx + \beta^2 e^{\xi}[(2 - \xi - e^{1-\xi}(e^{\xi} - 1) - \xi - \xi e^{\xi} + e^{\xi} - 1]$$

$$\beta^2 e^{\xi}[(2 - \xi - e^{-\xi})(e - e^{\xi}) - e(1-\xi) - e + \xi e^{\xi} + e - e^{\xi}] = 0.$$

$$D + \beta^2 e^{\xi}[(2-\xi)(e-1) - e + e^{1-\xi} - e^{1-\xi} + 1$$
$$-\xi - \xi e^{\xi} + e^{\xi} - 1 - e + e\xi + \xi e^{\xi} - e^{\xi}] = 0.$$

$$D + \beta^2 e^{\xi}[2e - 2 - e\xi + \xi - e - \xi + e\xi - e] = 0.$$

$$D = 2\beta^2 e^{\xi}.$$

$$g = \beta e^{\xi}[2\beta + 2 - x - \xi - e^{\xi - x}] - \begin{cases} e, & x < \xi, \\ 1, & x > \xi, \end{cases} \quad .$$

From symmetry

$$g^* = \beta e^{x}[2\beta + 2 - \xi - x - e^{x - \xi}] - \begin{cases} e, & x > \xi, \\ 1, & x < \xi, \end{cases} \quad .$$

Chapter 2

INTEGRAL EQUATIONS

2.1 Using differentiation, convert the integral equation

$$u(x) = \frac{1}{2}\int_0^1 |x-\xi|u(\xi)d\xi + \frac{x^2}{2}$$

into a differential equation. Obtain the needed boundary conditions and solve the differential equation.

Solution

$$\begin{aligned}
u(x) &= \frac{1}{2}\int_0^1 |x-\xi|u(\xi)d\xi + \frac{x^2}{2}\\
&= \frac{1}{2}\left\{\int_0^x (x-\xi)u(\xi)d\xi + \int_x^1 (\xi - x)u(\xi)d\xi + x^2\right\},\\
u' &= \frac{1}{2}\left\{\int_0^x u d\xi - \int_x^1 u d\xi + 2x\right\}\\
u'' &= u+1.
\end{aligned}$$

Boundary conditions

$$u(0) = \frac{1}{2}\int_0^1 \xi u d\xi, \quad u'(0) = -\frac{1}{2}\int_0^1 u d\xi.$$

General solution is

$$
\begin{aligned}
u &= -1 + A\cosh(x) + B\sinh(x), \\
u' &= A\sinh(x) + B\cosh(x), \\
\int_0^1 u d\xi &= A\sinh(1) + B(\cosh(1) - 1) \\
\int_0^1 \xi u d\xi &= -\frac{1}{2} + \{\xi[A\sinh(\xi) + B\cosh(\xi)]\}\,|_0^1 - [A\cosh(\xi) + B\sinh(\xi)]|_0^1, \\
&= -\frac{1}{2} + A[\sinh(1) - \cosh(1) + 1] + B[\cosh(1) - \sinh(1)].
\end{aligned}
$$

Using these in the boundary conditions

$$
\begin{aligned}
-2 + 2A &= -\frac{1}{2} + A[\sinh(1) - \cosh(1) + 1] + B[\cosh(1) - \sinh(1)], \\
2B &= 1 - A\sinh(1) - B(\cosh 1 - 1).
\end{aligned}
$$

Solving for B

$$B = \frac{1}{1 + \cosh 1} - A\frac{\sinh(1)}{1 + \cosh(1)}.$$

Substituting for A,

$$A = \frac{3}{2} + A\frac{\sinh(1) - \cosh(1) - 1}{1 + \cosh(1)}. \tag{2.1}$$

Then

$$A = \frac{3}{2}\frac{1 + \cosh(1)}{2 + 2\cosh(1) - \sinh(1)}, \quad B = -\frac{3}{2}\frac{\sinh(1)}{2 + 2\cosh(1) - \sinh(1)} + \frac{1}{1 + \cosh(1)}. \tag{2.2}$$

Finally

$$u(x) = -1 + \frac{\sinh x}{1 + \cosh(1)} + \frac{3}{2}\frac{\cosh(x) + \cosh(1 - x)}{2 + 2\cosh(1) - \sinh(1)}.$$

2.2 Convert

$$u(x) = \lambda \int_{-1}^{1} e^{-|x-\xi|} u(\xi) d\xi$$

into a differential equation. Obtain the required number of boundary conditions.

Solution

$$\begin{aligned}
u(x) &= \lambda \int_{-1}^{1} e^{-|x-\xi|} u(\xi) d\xi, \\
u' &= \lambda \left\{ \int_{-1}^{x} e^{\xi - x} u d\xi + \int_{x}^{1} e^{x-\xi} u d\xi \right\} \\
u'' &= \lambda \left\{ -u + \int_{-1}^{x} e^{\xi - x} u d\xi - \int_{x}^{1} e^{x-\xi} u d\xi \right\} \\
u'' &= (1 - 2\lambda) u, \\
u(1) &= \lambda \int_{-1}^{1} e^{\xi - 1} u d\xi, \quad u(-1) = \lambda \int_{-1}^{1} e^{1-\xi} u d\xi.
\end{aligned}$$

2.3 Solve the integral equation

$$u(x) = e^x + \lambda \int_{-1}^{1} e^{(x-\xi)} u(\xi) d\xi,$$

and discuss the conditions on λ for a unique solution.

Solution

$$u(x) = e^x + \lambda \int_{-1}^{1} e^{(x-\xi)} u(\xi) d\xi,$$

Let

$$C = \int_{-1}^{1} e^{-\xi} u(\xi) d\xi,$$

Then

$$\begin{aligned} u &= e^x + \lambda C e^x = (1 + \lambda C) e^x, \\ C &= (1 + \lambda C) 2, \quad C = \frac{2}{1 - 2\lambda} \\ u &= \frac{e^x}{1 - 2\lambda} \end{aligned}$$

For a unique solution $\lambda \neq 1/2$.

2.4 Solve

$$u(x) = x + \int_0^1 (1 - x\xi)u(\xi)d\xi.$$

Solution

Let $C_1 = \int_0^1 ud\xi$ and $C_2 = \int_0^1 \xi ud\xi$

$$\begin{aligned}
u &= x + C_1 - C_2 x, \\
C_1 &= \frac{1}{2} + C_1 - \frac{1}{2}C_2, \quad C_2 = \frac{1}{3} + \frac{1}{2}C_1 - \frac{1}{3}C_2, \\
C_1 &= 2, \quad C_2 = 1, \\
u(x) &= 2.
\end{aligned}$$

2.5 Solve

$$u(x) = \sin x + \int_0^{\pi} \cos(x-\xi)u(\xi)d\xi.$$

Solution

$$u(x) = \sin x + \int_0^{\pi} [\cos x \cos\xi + \sin x \sin\xi]u(\xi)d\xi.$$

Let $C_1 = \langle u, \cos\rangle$, and $C_2 = \langle u, \sin\rangle$.

$$\begin{aligned}
u &= \sin x + C_1 \cos x + C_2 \sin x, \\
C_1 &= (1+C_2)\int_0^{\pi} \sin x \cos x dx + C_1 \int_0^{\pi} \cos^2 x dx = \frac{\pi}{2}C_1 \\
C_1 &= 0 \\
C_2 &= (1+C_2)\frac{\pi}{2}, \\
C_2 &= \frac{\pi}{2-\pi}, \\
u(x) &= \frac{2}{2-\pi}\sin x
\end{aligned}$$

2.6 Obtain the eigenvalues and eigenfunctions of the equation

$$u(x) = \lambda \int_0^{2\pi} \cos(x - \xi) u(\xi) d\xi.$$

Solution

$$u = \lambda \int_0^{2\pi} [\cos x \cos \xi + \sin x \sin \xi] u(\xi) d\xi,$$

Let $C_1 = \langle u, \cos \rangle$, and $C_2 = \langle u, \sin \rangle$.

$$\begin{aligned} u &= \lambda [C_1 \cos x + C_2 \sin x] \\ C_1 &= \lambda \pi C_1, \quad C_2 = \lambda \pi C_2, \end{aligned}$$

When $\lambda = 1/\pi$, we may obtain two solutions corresponding to $C_1 = 1, C_2 = 0$ and $C_1 = 0, C_2 = 1$

$$u_1 = \frac{1}{\sqrt{\pi}} \cos x, \quad u_2 = \frac{1}{\sqrt{\pi}} \sin x$$

are the normalized eigenfunctions for the eigenvalue $\lambda = 1/\pi$.

2.7 Find the eigenvalues and eigenfunctions of

$$u(x) = \lambda \int_0^1 (1 - x\xi) u(\xi) d\xi.$$

Solution

Let

$$\begin{aligned} C_1 &= \langle 1, u \rangle, \quad C_2 = \langle \xi, u \rangle \\ u &= \lambda[C_1 - C_2 x] \\ C_1 &= \lambda[C_1 - \frac{1}{2}C_2], \\ C_2 &= \lambda[\frac{1}{2}C_1 - \frac{1}{3}C_2], \end{aligned}$$

The determinant of this homogeneous system has to be zero.

$$\begin{vmatrix} 1-\lambda & \frac{1}{2}\lambda \\ -\frac{1}{2}\lambda & 1+\frac{1}{3}\lambda \end{vmatrix} = 0$$

Expanding,

$$\begin{aligned} 1 - \frac{2}{3}\lambda - \frac{1}{3}\lambda^2 + \frac{1}{4}\lambda^2 &= 0 \\ -\frac{1}{12}\lambda^2 - \frac{2}{3}\lambda + 1 &= 0 \\ \lambda^2 + 8\lambda - 12 &= 0 \end{aligned}$$

This has the solutions

$$\lambda = -4 \pm 2\sqrt{7}$$

When $\lambda = -4 + 2\sqrt{7}$

$$C_2 = \frac{5 - 2\sqrt{7}}{2 - \sqrt{7}} C_1, \quad u_1 = 1 - \frac{5 - 2\sqrt{7}}{2 - \sqrt{7}} x.$$

When $\lambda = -4 - 2\sqrt{7}$

$$C_2 = \frac{5 + 2\sqrt{7}}{2 + \sqrt{7}} C_1, \quad u_2 = 1 - \frac{5 + 2\sqrt{7}}{2 + \sqrt{7}} x.$$

2.8 From the differential equation and the boundary conditions obtained for Exercise 2.2, find the values of λ for the existence of non-trivial solutions.

Solution

The differential equation is

$$u'' - \mu^2 u = 0, \quad \mu^2 \equiv 1 - 2\lambda$$

The solution is given by

$$u = Ae^{\mu x} + Be^{-\mu x},$$

From the given integral equation

$$\begin{aligned}
u(x) &= \lambda \left\{ \int_{-1}^{x} e^{\xi - x} u d\xi + \int_{x}^{1} e^{x-\xi} u d\xi \right\} \\
u(0) &= \lambda \left\{ \int_{-1}^{0} e^{\xi} u d\xi + \int_{0}^{1} e^{-\xi} u d\xi \right\} = \lambda \int_{0}^{1} e^{-\xi}[u(\xi) + u(-\xi)] d\xi \\
u'(x) &= \lambda \left\{ u - u - \int_{-1}^{x} e^{\xi - x} u d\xi + \int_{x}^{1} e^{x-\xi} u d\xi \right\} \\
u'(0) &= \lambda \left\{ \int_{0}^{1} e^{-\xi} u d\xi - \int_{-1}^{0} e^{\xi} d\xi \right\} = \lambda \int_{0}^{1} e^{-\xi}[u(\xi) - u(-\xi)] d\xi \\
u(0) + u'(0) &= (1 - \mu^2) \int_{0}^{1} e^{-\xi} u(\xi) d\xi \\
u(0) - u'(0) &= (1 - \mu^2) \int_{0}^{1} e^{-\xi} u(-\xi) d\xi
\end{aligned}$$

Using the solution

$$\begin{aligned}
(1+\mu)A + (1-\mu)B &= (1-\mu^2) \left\{ A \frac{e^{\mu-1} - 1}{\mu - 1} + B \frac{e^{-\mu-1} - 1}{-\mu - 1} \right\} \\
(1-\mu)A + (1+\mu)B &= (1-\mu^2) \left\{ A \frac{e^{-\mu-1} - 1}{-\mu - 1} + B \frac{e^{\mu-1} - 1}{\mu - 1} \right\}
\end{aligned}$$

This can be simplified to get

$$\begin{aligned}
A \frac{e^{\mu}}{\mu - 1} - B \frac{e^{-\mu}}{\mu + 1} &= 0 \\
A \frac{e^{-\mu}}{\mu + 1} - B \frac{e^{\mu}}{\mu - 1} &= 0
\end{aligned}$$

The determinant of this system gives

$$(\mu + 1)e^{\mu} = (\mu - 1)e^{-\mu}$$

or

$$\frac{e^{\mu} - e^{-\mu}}{e^{\mu} + e^{-\mu}} = -\frac{1}{\mu} \tag{2.3}$$

Let $\mu = i\nu$ to obtain the characteristic equation

$$\tan \nu = 1/\nu,$$

which allows an infinite number of eigenvalues.

2.9 Show that the equation

$$u(x) = \lambda \int_0^1 |x - \xi| u(\xi) d\xi$$

has an infinite number of eigenvalues and eigenfunctions.

Solution

$$\begin{aligned} u(x) &= \lambda \left\{ \int_0^x (x - \xi) u d\xi + \int_x^1 (\xi - x) u d\xi \right\} \\ u'(x) &= \lambda \left\{ \int_0^x u d\xi - \int_x^1 u d\xi \right\} \\ u''(x) &= 2\lambda u(x). \end{aligned}$$

The boundary conditions may be obtained as

$$\begin{aligned} u(0) &= \lambda \int_0^1 \xi u d\xi, \quad u(1) = \lambda \int_0^1 (1 - \xi) u d\xi, \\ u'(0) &= -\lambda \int_0^1 u d\xi, \quad u'(1) = \lambda \int_0^1 u d\xi. \end{aligned}$$

From these we get

$$u(0) + u(1) + u'(0) = 0, \quad u'(0) + u'(1) = 0.$$

Using $\mu^2 = -2\lambda$, the solution of the differential equation is

$$u(x) = A \cos \mu x + B \sin \mu x. \tag{2.4}$$

$$u'(x) = -\mu A \sin \mu x + \mu B \cos \mu x.$$

Then,

$$u(0) = A, \quad u'(0) = \mu B,$$

$$u(1) = A \cos \mu + B \sin \mu, \quad u'(1) = -\mu A \sin \mu + \mu B \cos \mu.$$

The boundary conditions become

$$\begin{aligned} A + A \cos \mu + B \sin \mu + \mu B &= 0, \\ \mu B - -\mu A \sin \mu + \mu B \cos \mu &= 0. \end{aligned}$$

The determinant of this system gives

$$\mu[(1+\cos\mu)^2 + \sin\mu(\mu+\sin\mu)] = 0.$$

This can be simplified to get

$$\mu = -2\frac{1+\cos\mu}{\sin\mu}.$$

This is equivalent to

$$\tan\mu/2 = 2/\mu. \tag{2.5}$$

This characteristic equation has an infinite number of solutions.

2.10 Obtain the values of λ and a for the equation

$$u(x) = \lambda \int_0^1 (x - \xi)u(\xi)d\xi + a + x^2$$

to have (a) a unique solution, (b) a nonunique solution, and (c) no solution.

Solution

Let

$$C_1 = \int_0^1 u d\xi, \quad C_2 = \int_0^1 \xi u d\xi,$$

Then

$$\begin{aligned} u(x) &= \lambda[C_1 x - C_2] + a + x^2, \\ C_1 &= \lambda[\frac{1}{2}C_1 - C_2] + a + \frac{1}{3}, \\ C_2 &= \lambda[\frac{1}{3}C_1 - \frac{1}{2}C_2] + \frac{1}{2}a + \frac{1}{4}, \end{aligned}$$

This system of equations can be written as

$$\begin{aligned} (1 - \frac{\lambda}{2})C_1 + \lambda C_2 &= a + \frac{1}{3}, \\ -\frac{\lambda}{3}C_1 + (1 + \frac{\lambda}{2})C_2 &= \frac{a}{2} + \frac{1}{4}, \end{aligned}$$

The determinant of this system is

$$1 - \frac{\lambda^2}{4} + \frac{\lambda^2}{3} = 0, \quad \lambda^2 = -12.$$

We have unique solutions for C_1 and C_2 if $\lambda^2 \neq -12$.

Eliminating C_2,

$$(1 + \frac{\lambda^2}{12})C_1 = a + \frac{1}{3} - \frac{5}{12}\lambda.$$

When $\lambda^2 = -12$, the left hand side goes to zero. For the right hand side to go to zero

$$a = -\frac{1}{3} + \frac{5}{12}\lambda.$$

In this case C_1 is arbitrary and we have multiple solutions.

For other values of a, there is no solution.

2.11 Obtain the resolvent kernel for

$$u(x) = \lambda \int_0^{2\pi} e^{in(x-\xi)} u(\xi) d\xi + f(x).$$

Solution

$$\begin{aligned}
u(x) &= \lambda \int_0^{2\pi} e^{in(x-\xi)} u(\xi) d\xi + f(x). \\
C &= \int_0^{2\pi} e^{-in\pi\xi} u d\xi, \\
u &= \lambda C e^{inx} + f, \\
C &= 2\pi\lambda C + \langle f, e^{-in\xi} \rangle, \\
C &= \frac{1}{1-2\pi\lambda} \langle f, e^{-in\xi} \rangle, \\
u &= f + \frac{\lambda}{1-2\pi\lambda} \langle f, e^{-in\xi} \rangle, \\
g(x,\xi) &= \frac{1}{1-2\pi\lambda} e^{in(x-\xi)}.
\end{aligned}$$

2.12 Find the resolvent kernel for

$$u(x) = \lambda \int_0^\pi \cos(x-\xi)u(\xi)d\xi + f(x).$$

Solution

$$\begin{aligned}
u &= \lambda \int_0^\pi [\cos x \cos\xi + \sin x \sin\xi] u d\xi + f, \\
C_1 &= \langle u, \cos\rangle, \quad C_2 = \langle u, \sin\rangle, \\
u &= f + \lambda[C - 1\cos x + C - 2\sin x], \\
C_1 &= \langle f, \cos\rangle + \frac{\lambda\pi}{2} C_1, \\
C_2 &= \langle f, \sin\rangle + \frac{\lambda\pi}{2} C_2,
\end{aligned}$$

Solving for the constants

$$\begin{aligned}
C_1 &= \frac{1}{1-\pi\lambda/2}\langle f, \cos\rangle, \\
C_2 &= \frac{1}{1-\pi\lambda/2}\langle f, \sin\rangle, \\
u &= f + \frac{\lambda}{1-\pi\lambda/2}[\langle f, \cos\rangle \cos x + \langle f, \sin\rangle \sin x], \\
u &= f + \lambda\langle g, f\rangle, \\
g &= \frac{1}{1-\pi\lambda/2}\cos(x-\xi).
\end{aligned}$$

2.13 Show that for a Fredholm equation

$$u(x) = \lambda \int_a^b k(x,\xi)u(\xi)d\xi + f(x),$$

the resolvent kernel $g(x,\xi)$ satisfies

$$g(x,\xi) = k(x,\xi) + \lambda \int_a^b k(x,\eta)g(\eta,\xi)d\eta.$$

Thus, the resolvent kernel satisfies the integral equation for u when the forcing function f is replaced by k with ξ and λ kept as parameters.

Solution

$$\begin{aligned} u(x) &= \lambda \int_a^b k(x,\xi)u(\xi)d\xi + f(x), \\ u(x) &= f(x) + \lambda \int_a^b g(x,\xi)f(\xi)d\xi. \end{aligned}$$

Using the second equation in the first

$$\begin{aligned} f(x) + \lambda \int_a^b g(x,\xi)f(\xi)d\xi &= \lambda \int_a^b k(x,\xi)[f(\xi) + \lambda \int_a^b g(\xi,\eta)f(\eta)d\eta] + f(x), \\ \int_a^b g(x,\xi)f(\xi)d\xi &= \int_a^b k(x,\xi)f(\xi)d\xi + \lambda \int_a^b \int_a^b k(x,\eta)g(\eta,\xi)f(\xi)d\eta d\xi. \end{aligned}$$

As $f(x)$ is arbitrary,

$$g(x,\xi) = k(x,\xi) + \lambda \int_a^b k(x,\eta)g(\eta,\xi)d\eta.$$

2.14 Demonstrate the preceding result when $k = 1 - 2x\xi$ in the domain $0 < x, \xi < 1$.

Solution

$$u(x) = \lambda \int_0^1 (1 - 2x\xi) u(\xi) d\xi + f(x),$$

Let $C_1 = \int_0^1 u d\xi$ and $C_2 = \int_0^1 \xi u d\xi$

$$\begin{aligned}
u(x) &= \lambda[C_1 - 2C_2 x] + f, \\
C_1 &= \lambda[C_1 - C_2] + \langle f, 1\rangle, \\
C_2 &= \lambda[\frac{1}{2}C_1 - \frac{2}{3}C_2] + \langle f, \xi\rangle, \\
(1-\lambda)C_1 + \lambda C_2 &= \langle f, 1\rangle, \\
-\frac{\lambda}{2}C_1 + (1 + \frac{2}{3}\lambda)C_2 &= \langle f, \xi\rangle,
\end{aligned}$$

The determinant of the system is $1 - \lambda/3 - \lambda^2/6$

$$\begin{aligned}
C_1 &= [(1 + 2\lambda/3)\langle f, 1\rangle - \lambda\langle f, \xi\rangle]/(1 - \lambda/3 - \lambda^2/6), \\
C_2 &= [(\lambda/2)\langle f, 1\rangle + (1 - \lambda)\langle f, \xi\rangle]/(1 - \lambda/3 - \lambda^2/6), \\
g(x, \xi) &= \frac{(1 + 2\lambda/3) - \lambda(x + \xi) - 2(1 - \lambda)x\xi}{1 - \lambda/3 - \lambda^2/6}.
\end{aligned}$$

Let $A = \lambda/(1 - \lambda/3 - \lambda^2/6)$.

$$\lambda \int_0^1 (1 - 2x\eta) g(\eta, \xi) d\eta = A \int_0^1 (1 - 2x\eta)[(1 + 2\lambda/3) - \lambda(\eta + \xi)$$
$$-2(1 - \lambda)\eta\xi] d\eta,$$
$$= A\left\{(1 + 2\lambda/3) - \lambda(1/2 + \xi) - (1 - \lambda)\xi\right.$$
$$\left. - 2x[(1 + 2\lambda/3)/2 - \lambda(1/3 + \xi/2) - 2(1 - \lambda)\xi/3]\right\},$$
$$= A\left\{1 + \lambda/6 - \xi - x + x\xi(4/3 - \lambda/6)\right\}.$$
$$g(x, \xi) - k(x, \xi) = (A/\lambda)\left\{(1 + 2\lambda/3) - \lambda(x + \xi) - (1 - \lambda)2x\xi\right.$$

$$
\begin{aligned}
& -(1-\lambda/3-\lambda^2/6)(1-2x\xi)\big\}\,, \\
&= (A/\lambda)\left\{\lambda+\lambda^2/6-(x+\xi)\lambda+x\xi\lambda(4/3+\lambda/6)\right\} \\
&= A\left\{1+\lambda/6-x-\xi+(4/3+\lambda/6)x\xi\right\}.
\end{aligned}
$$

Thus,

$$
g(x,\xi)=k(x,\xi)+\lambda\int_0^1 k(x,\eta)g(\eta,\xi)d\eta.
$$

2.15 For the Volterra equation,

$$u(x) = \int_0^x (1 - x\xi)u(\xi)d\xi + 1,$$

starting with $u^{(0)} = 1$, obtain iteratively, $u^{(2)}$.

Solution

$$\begin{aligned}
u(x) &= \int_0^x (1 - x\xi)u(\xi)d\xi + 1. \\
u^{(0)} &= 1, \\
u^{(1)} &= \int_0^x (1 - 2x\xi)d\xi + 1 = 1 + x - x^2/2, \\
u^{(2)} &= \int_0^x (1 - x\xi)(1 + \xi - \xi^2/2)d\xi \\
&= 1 + x + x^2/2 - x^4/8 - x(x^2/2 + x^3/3 - x^5/10) \\
&= 1 + x + x^2/2 - x^3/2 - (11/24)x^4 + x^6/10.
\end{aligned}$$

2.16 For the integral equation of the first kind

$$\int_0^{2\pi} \sin(x+\xi)u(\xi)d\xi = \sin x,$$

discuss the consequence of assuming

$$u(x) = \sum_{n=0} a_n \cos nx + \sum_{n=1} b_n \sin nx,$$

where the constants, a_n and b_n, are unknown.

Solution

We have

$$\int_0^{2\pi} [\sin x \cos\xi + \cos x \sin\xi]u(\xi)d\xi = \sin x.$$

Let

$$u(x) = \sum_{n=0}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx.$$

Then

$$\begin{aligned}
\int_0^{2\pi} u(\xi)\cos\xi d\xi &= \frac{1}{2}\sum_{n=0}^{\infty} a_n \int_0^{2\pi} [\cos(n-1)\xi + \cos(n+1)\xi]d\xi \\
&\quad + \frac{1}{2}\sum_{n=1}^{\infty} b_n \int_0^{2\pi} [\sin(n+1)\xi + \cos(n-1)\xi]d\xi \\
&= \pi a_1. \\
\int_0^{2\pi} u(\xi)\sin\xi d\xi &= \frac{1}{2}\sum_{n=0}^{\infty} a_n \int_0^{2\pi} [\sin(n+1)\xi - \sin(n-1)\xi]d\xi \\
&\quad + \frac{1}{2}\sum_{n=1}^{\infty} b_n \int_0^{2\pi} [\cos(n-1)\xi - \cos(n+1)\xi]d\xi \\
&= \pi b_1.
\end{aligned}$$

Comparing with the term on the right hand side of the given equation

$$b_1 = \frac{1}{\pi}, \quad a_1 = 0.$$

All other a_n and b_n are arbitrary.

2.17 With the quadratic forms,

$$J_1[u] = \int_a^b \int_a^b k(x,\xi)u(x)u(\xi)d\xi dx, \quad J_2[u] = \int_a^b u^2 dx,$$

where u is a smooth function in (a, b), show that the functions u, which extremize

$$J[u] = \frac{J_1[u]}{J_2[u]},$$

are the eigenfunctions of $[k(x,\xi) + k(\xi,x)]/2$. Also show that the extrema of J correspond to the reciprocal of the eigenvalues of $k(x,\xi)$.

Solution

With $J = J_1/J_2$, $\delta J = 0$ gives

$$J_2 \delta J_1 - J_1 \delta J_2 = 0, \quad \delta J_1 - \frac{J_1}{J_2}\delta J_2 = 0.$$

From

$$J_1 = \int_a^b \int_a^b k(x,\xi)u(x)u(\xi)d\xi dx,$$

$$\delta J_1 = \int_a^b \int_a^b k(x,\xi)[u(x)\delta u(\xi) + u(\xi)\delta u(x)]d\xi dx,$$

which can be written as

$$\int_a^b \int_a^b [k(x,\xi) + k(\xi,x)]u(\xi)\delta u(x)d\xi dx.$$

Similarly,

$$J_2 = \int_a^b u^2 dx,$$

gives

$$\delta J_2 = 2\int_a^b u(x)\delta u(x)dx.$$

Using these in the first variation of J, and considering $\delta u(x)$ as arbitrary,

$$\frac{1}{2}[k(x,\xi) + k(\xi,x)]u(\xi)d\xi = \frac{J_1}{J_2}u(x).$$

Let $J_1/J_2 = 1/\lambda$. Then λ is the eigenvalue of the kernel $[k(x,\xi) + k(\xi,x)]/2$.

As $\delta J = 0$, the eigenvalues correspond to the extremum value of J.

2.18 In the preceding problem, assuming k is symmetric, compute $J[v]$ if

$$v(x) = \sum_n a_n u_n(x),$$

where u_n are normalized eigenfunctions.

Solution

We have

$$v = \sum a_n u_n.$$

Then

$$\begin{aligned}
J_2 &= \|v\|^2 = \int_a^b \sum a_n u_n \sum a_m u_m dx = \sum a_n^2. \\
J_1 &= \int_a^b \int_a^b k(x,\xi) \sum a_n u_n(x) \sum a_m u_m(\xi) d\xi dx \\
&= \int_a^b \sum \frac{a_m}{\lambda_m} u_m(x) \sum a_n u_n(x) dx \\
&= \sum \frac{a_n^2}{\lambda_n}. \\
J &= \frac{\sum a_n^2/\lambda_n}{\sum a_n^2}.
\end{aligned}$$

2.19 The generalized Abel equation,

$$\int_0^x \frac{u(\xi)d\xi}{(x-\xi)^\alpha} = f(x),$$

is solved by multiplying both sides by $(\eta - x)^{\alpha-1}$ and integrating with respect to x from 0 to η. Implement this procedure, and discuss the allowable range for the index, α.

Solution

Multiplying the equation by $1/(\eta-x)^{1-\alpha}$ on both sides and integrating form $x = 0$ to $x = \eta$,

$$\int_0^\eta \int_0^x \frac{u(\xi)d\xi}{(\eta - x)^\beta (x-\xi)^\alpha} \quad = \quad \int_0^\eta \frac{f(x)dx}{(\eta - x)^\beta}.$$

where $\beta = 1 - \alpha$.

Interchanging the order of integration,

$$\int_0^\eta \int_\xi^\eta \frac{dx}{(\eta - x)^\beta (x-\xi)^\alpha} u(\xi)d\xi \quad = \quad \int_0^\eta \frac{f(x)dx}{(\eta - x)^\beta}.$$

Let

$$I = \int_\xi^\eta \frac{dx}{(\eta - x)^\beta (x-\xi)^\alpha}.$$

Using

$$x = (\eta - \xi)t + \xi,$$

$$I = \int_0^1 \frac{dt}{t^\alpha (1-t)^\beta},$$

which is a constant. Then

$$u(x) = \frac{1}{I}\frac{d}{dx}\int_0^x \frac{f(\xi)d\xi}{(x-\xi)^\beta}.$$

To evaluate the integral I, consider the complex valued function

$$f(z) = \frac{1}{z^\alpha (z-1)^\beta}$$

1

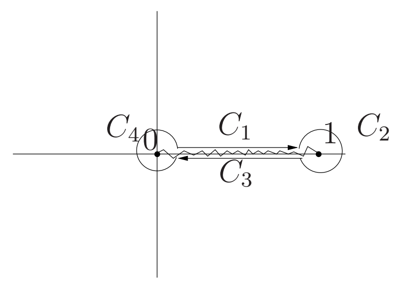

Figure 2.1: Contour for evaluating the integral I

where $\beta = 1 - \alpha$. As shown in Fig. 2.1, we integrate f along the contour $C_1 + C_2 + C_3 + C4$. Let $z = re^{i\theta}$ and $z - 1 = r_1 e^{i\theta_1}$ with the restriction

$$0 < \theta < 2\pi, \quad 0 < \theta_1 < 2\pi.$$

This creates a branch cut from 0 to 1. On C_1:

$$\theta = 0, \quad \theta_1 = \pi, \quad dz = dr$$

and

$$\int_{C_1} f dz = \int_0^1 \frac{dr}{r^\alpha (1-r)^\beta} e^{-i\beta\pi} = e^{-i\beta\pi} I.$$

On C_3:

$$\theta = 2\pi, \quad \theta_1 = \pi, \quad dz = dr$$

and

$$\int_{C_3} f dz = \int_1^0 \frac{dr}{r^\alpha (1-r)^\beta} e^{-2\alpha\pi i - \beta\pi i} = e^{-i\alpha\pi} I.$$

The integrals over C_2 and C_4 vanish when their radii shrink to zero, if $\alpha > 0$ and $\beta > 0$, respectively. This is equivalent to $0 < \alpha < 1$. Thus, the sum of the four integrals is

$$(e^{-\alpha\pi i} + e^{-\beta\pi i}) I.$$

Our next step is to stretch the contour to a circle of radius $R >> 1$. On this circle $r = r_1 = R$ and $\theta = \theta_1$. Then, as $R \to \infty$,

$$\oint f(z) dz = -2\pi i.$$

Thus,

$$\frac{e^{\alpha\pi i} - e^{-\alpha\pi i}}{2i} I = \pi.$$

$$I = \frac{\pi}{\sin \alpha\pi}.$$

Our solution for the integral equation becomes

$$u(x) = \frac{\sin \alpha\pi}{\pi} \frac{d}{dx} \int_0^x \frac{f(\xi) d\xi}{(x - \xi)^{1-\alpha}}.$$

2.20 Solve the singular equation

$$\int_0^x \frac{u(\xi)d\xi}{(x^2 - \xi^2)^{1/2}} = f(x),$$

using a change of the independent variable.

Solution

Let $x^2 = t$ and $\xi^2 = \tau$.

$$d\xi = \frac{d\tau}{2\sqrt{\tau}}.$$

The integral equation becomes

$$\int_0^t \frac{u(\sqrt{\tau})d\tau}{2\sqrt{\tau}\sqrt{t-\tau}} = f(\sqrt{t}).$$

Let

$$v(\tau) = \frac{u(\sqrt{\tau})}{2\sqrt{\tau}}, \quad g(t) = f(\sqrt{t}).$$

Then, we get the Abel equation

$$\int_0^t \frac{v(\tau)d\tau}{\sqrt{t-\tau}} = g(t).$$

The solution of this equation is

$$v(t) = \frac{1}{\pi}\frac{d}{dt}\int_0^t \frac{g(\tau)d\tau}{\sqrt{t-\tau}}.$$

Converting to the original variables, we get

$$\begin{aligned} \frac{u(x)}{2x} &= \frac{1}{2\pi x}\frac{d1}{dx}\int_0^x \frac{f(\xi)2\xi d\xi}{\sqrt{x^2-\xi^2}}, \\ u(x) &= \frac{2}{\pi}\frac{d}{dx}\int_0^x \frac{f(\xi)\xi d\xi}{\sqrt{x^2-\xi^2}}. \end{aligned}$$

2.21 If an elastic, 3D half space ($z > 0$) is subjected to an axi-symmetric z-displacement, $w(r)$, where $r = \sqrt{x^2 + y^2}$, we need to solve the integral equation for the distributed pressure on the horizontal surface, $z = 0$,

$$\int_0^r \frac{g(\rho)d\rho}{\sqrt{r^2 - \rho^2}} = \frac{\mu}{1-\nu} w(r), \quad 0 < r < a,$$

where μ is the shear modulus, ν is the Poisson's ratio, and a is the contact radius. Obtain the pressure distribution $g(r)$ (see Barber, (2002)).

Solution

From the solution of Exercise 2.20, we have

$$g(r) = \frac{2\mu}{\pi(1-\nu)} \frac{d}{dr} \int_0^r \frac{w(\rho)\rho d\rho}{\sqrt{r^2 - \rho^2}}.$$

2.22 In the preceding problem, if the contact displacement is due to a rigid sphere of radius R pressing against the elastic half space, we assume

$$w(r) = d - \frac{r^2}{2R},$$

where d is the maximum indentation. Obtain the pressure distribution, the value of the contact radius a, and the total vertical force.

Solution

From the solution of Exercise 2.21, we have

$$g(r) = \frac{1\mu}{\pi(1-\nu)R}\frac{d}{dr}\int_0^r \frac{(2Rd-\rho^2)\rho d\rho}{\sqrt{r^2-\rho^2}}.$$

Substituting for w,

$$\begin{aligned} g(r) &= \frac{2\mu}{\pi(1-\nu)}\frac{d}{dr}\int_0^r \frac{w(\rho)\rho d\rho}{\sqrt{r^2-\rho^2}} \\ &= \frac{2\mu}{\pi(1-\nu)}\frac{d}{dr}\left\{(2Rd-\rho^2)\sqrt{r^2-\rho^2}\Big|_r^0 - \frac{1}{3}(r^2-\rho^2)^{3/2}\Big|_0^r\right\} \\ &= \frac{\mu}{\pi(1-\nu)R}\frac{d}{dr}\left\{(2Rd-r^2)r + \frac{1}{3}r^3\right\} \\ &= \frac{\mu}{\pi(1-\nu)R}\left\{2Rd - 2r^2\right\}. \end{aligned}$$

The extent of the contact radius a is found using the circle on which the pressure g is zero.

$$a = \sqrt{Rd}.$$

The total vertical force can be calculated by integrating the pressure.

$$\begin{aligned} P &= 2\pi\int_0^a g r dr \\ &= \frac{2\mu}{(1-\nu)R}\left\{Rda^2 - \frac{1}{2}a^4\right\} \\ &= \frac{\mu R d^2}{(1-\nu)}. \end{aligned}$$

2.23 Consider the integral equation

$$u(x) = \int_0^1 k(x,\xi)u(\xi)d\xi + x^2,$$

where

$$k(x,\xi) = \begin{cases} x(\xi - 1), & x < \xi, \\ \xi(x-1), & x > \xi. \end{cases}$$

Obtain an exact solution to this equation. Using the approximate kernel $Ax(1-x)$, find A using the method of least square error. Obtain an approximate solution for u. Compare the values of the exact and approximate solutions at $x = 0.5$.

Solution

The given k is the Green's function for the operator $L = d^2/dx^2$. Then

$$u'' = \int_0^1 Lku(\xi)d\xi + Lx^2 = u(x) + 2.$$

$$u'' - u = 2, \quad u(0) = 0, \quad u(1) = 1.$$

This equation can be solved in the form

$$u = A\cosh x + B\sinh x - 2.$$

The condition $u(0) = 0$ gives

$$A = 2,$$

and the condition $u(1) = 1$ gives

$$B = \frac{3 - 2\cosh 1}{\sinh 1}.$$

$$u_{exact} = \frac{2\sinh(1-x) + 3\sinh x}{\sinh 1} - 2.$$

To obtain an approximate solution, let

$$Ax(1-x) = k(x,\xi), \quad A = \frac{\int_0^1 x(1-x)k(x,\xi)dx}{\int_0^1 x^2(1-x)^2 dx}.$$

$$\int_0^1 x^2(1-x)^2 dx = \frac{1}{30}.$$

$$A = -\frac{5}{2}\xi(1-2\xi^2+\xi^3).$$

The approximate integral equation is

$$u(x) = -\frac{5}{2}x(1-x)\int_0^1 \xi(1-2\xi^2+\xi^3)u d\xi + x^2.$$

Let

$$C = \int_0^1 \xi(1-2\xi^2+\xi^3)u d\xi.$$

$$u = -\frac{5C}{2}x(1-x).$$

$$C = \frac{2}{37}.$$

The approximate solution is

$$u = x^2 - \frac{5}{37}x(1-x).$$

$$u_{exact}(x=0.5) = 0.217047, \quad u_{approx}(x=0.5) = 0.216216.$$

2.24 Consider the differential equation

$$u'' + u = x, \quad u(0) = 0, \quad u(1) = 0.$$

Find an exact solution. Obtain a finite difference solution by dividing the domain into four equal intervals.

Convert the differential equation into an integral equation using the Green's function for the operator $L = d^2/dx^2$. Obtain a numerical solution of the integral equation by dividing the domain into four intervals and using the trapezoidal rule and the collocation method. Compare the results with the exact solution.

Solution

The exact solution is

$$u = A \cos x + B \sin x + x.$$

The condition $u(0) = 0$ gives $A = 0$ and $u(1) = 0$ gives

$$B = -\frac{1}{\sin 1}.$$

Thus

$$u = x - \frac{\sin x}{\sin 1}.$$

With step size $h = 0.25$, the difference equation is

$$u_{n+2} - 2u_n + u_{n+1} + h^2 u_n = x_n.$$

With $u_0 = u_4 = 0$, the remaining unknowns satisfy

$$\begin{bmatrix} -2 + \frac{1}{16} & 1 & 0 \\ 1 & -2 + \frac{1}{16} & 1 \\ 0 & 1 & -2 + \frac{1}{16} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \end{Bmatrix} = \frac{1}{16} \begin{Bmatrix} 1/4 \\ 1/2 \\ 3/4 \end{Bmatrix}$$

Using the Green's function, the differential equation can be converted to

$$u + \int_0^1 g(x, \xi) u(\xi) d\xi = \int_0^1 g(x, \xi) \xi d\xi.$$

After evaluating the integral on the right hand side, we have

$$u + \int_0^1 g(x,\xi)u(\xi)d\xi = \frac{1}{6}x(x^2-1).$$

With $g_{ij} = g(x_i, \xi_j)$, using the trapezoidal rule, we get the system of equations

$$u_i + \frac{1}{4}\sum_{j=1}^{3} g_{ij}u_j = \frac{1}{6}x_i(x_i^2-1), \quad i = 1,2,3.$$

Solving the systems of equations

	Exact	Finite Diff.	Intgral Eq.
u_1	-0.044014	-0.044274	-0.044274
u_2	-0.069747	-0.070156	-0.070156
u_3	-0.060056	-0.060403	-0.060403

2.25 Consider the nonlinear integral equation

$$u(x) - \lambda \int_0^1 u^2(\xi) d\xi = 1.$$

Show that for $\lambda < 1/4$ this equation has two solutions and for $\lambda > 1/4$ there are no real solutions. Also show that one of these solutions is singular at $\lambda = 0$ and two solutions coalesce at $\lambda = 1/4$. Sketch the solutions as functions of λ. (Based on Tricomi (1957).)

Solution

We have

$$u(x) - \lambda \int_0^1 u^2(\xi) d\xi = 1.$$

Let

$$A = \int_0^1 u^2 d\xi.$$

Then

$$u = 1 + \lambda A, \quad A = (1 + \lambda A)^2.$$

This gives a quadratic equation for A:

$$A^2 + \frac{2\lambda - 1}{\lambda^2} A + \frac{1}{\lambda^2} = 0.$$

$$A = \frac{1/2 - \lambda}{\lambda^2} \pm \sqrt{\frac{(1/2 - \lambda)^2}{\lambda^4} - \frac{1}{\lambda^2}}.$$

Corresponding to ($\pm$), there are two solutions:

$$u_1 = 1 + \frac{1}{2\lambda} \left[1 - 2\lambda + \sqrt{1 - 2\lambda}\right],$$

$$u_2 = 1 + \frac{1}{2\lambda} \left[1 - 2\lambda - \sqrt{1 - 2\lambda}\right].$$

Note:

$$\lambda \to 0, \quad u_1 \to -\infty, \quad u_2 \to 1.$$

$$\lambda \to 1/4, \quad u_1 \to 2, \quad u_2 \to 2.$$

For $\lambda > 1/4$, there is no real solution.

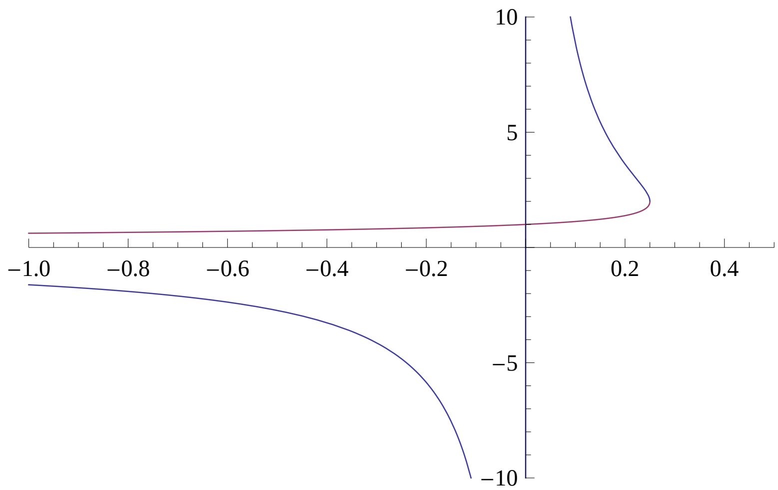

Figure 2.2: The functions u_1 and u_2 versus λ.

Chapter 3

FOURIER TRANSFORMS

3.1 From the Fourier transform of

$$f(x) = h(1 - |x|),$$

show that

$$\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

Solution

$$\begin{aligned}
\mathcal{F}[h(1 - |x|)] &= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{ix\xi} dx \\
&= \frac{1}{\sqrt{2\pi}} \frac{1}{i\xi} \left[e^{i\xi} - e^{-i\xi} \right] = \sqrt{\frac{2}{\pi}} \frac{\sin \xi}{\xi}.
\end{aligned}$$

Inverting this

$$\begin{aligned}
\frac{1}{\sqrt{2\pi}} \sqrt{\frac{2}{\pi}} \int_{-\infty}^{\infty} \frac{\sin \xi}{\xi} e^{-ix\xi} d\xi &= h(1 - |x|) \\
\int_{-\infty}^{\infty} \frac{\sin \xi}{\xi} e^{-ix\xi} d\xi &= \pi h(1 - |x|).
\end{aligned}$$

When $x = 0$,

$$\int_0^\infty \frac{\sin \xi}{\xi} d\xi = \frac{\pi}{2}.$$

3.2 Obtain the cosine and sine transforms of

$$f = e^{-ax}\cos bx,$$

where a and b are positive constants.

Solution

$$\begin{aligned}
\mathcal{F}_c[e^{-ax}] &= \sqrt{\frac{2}{\pi}}\frac{a}{a^2+\xi^2} \\
\mathcal{F}_c[e^{-ax}\cos bx] &= \sqrt{\frac{2}{\pi}}\int_0^\infty \frac{1}{2}\left[\cos(\xi+b)+\cos(\xi-b)\right]dx \\
&= \frac{1}{\sqrt{2\pi}}\left[\frac{a}{a^2+(\xi+b)^2}+\frac{a}{a^2+(\xi-b)^2}\right]. \\
\mathcal{F}_s[e^{-ax}] &= \sqrt{\frac{2}{\pi}}\frac{\xi}{a^2+\xi^2} \\
\mathcal{F}_s[e^{-ax}\cos bx] &= \sqrt{\frac{2}{\pi}}\int_0^\infty \frac{1}{2}\left[\sin(\xi+b)+\sin(\xi-b)\right]dx \\
&= \frac{1}{\sqrt{2\pi}}\left[\frac{\xi}{a^2+(\xi+b)^2}+\frac{\xi}{a^2+(\xi-b)^2}\right].
\end{aligned}$$

3.3 From the Fourier transform of

$$f(x) = (1 - |x|)h(1 - |x|),$$

compute the integrals

$$I_n = \int_0^\infty \left(\frac{\sin x}{x}\right)^n dx,$$

for $n = 2, 3, 4$.

Solution

The given function f is even.

$$\begin{aligned} F(\xi) &= \frac{2}{\sqrt{2\pi}} \int_0^1 (1-x)\cos x\xi dx \\ &= \sqrt{\frac{2}{\pi}} \left[(1-x)\frac{\sin \xi x}{\xi}\bigg|_0^1 - (-1)\frac{(-\cos \xi x)}{\xi^2}\bigg|_0^1 \right] \\ &= \sqrt{\frac{2}{\pi}} \frac{1-\cos\xi}{\xi^2} = 2\sqrt{\frac{2}{\pi}} \frac{\sin^2 \xi/2}{\xi^2}. \end{aligned}$$

Inversion of this gives

$$\begin{aligned} \frac{1}{\sqrt{2\pi}} \frac{2\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^\infty \frac{\sin^2 \xi/2}{\xi^2} \cos \xi x d\xi &= f(x) \\ \frac{4}{\pi} \int_0^\infty \frac{\sin^2 \xi/2}{2(\xi/2)^2} \cos \xi x d(\xi/2) &= f(x) \\ \frac{2}{\pi} \int_0^\infty \frac{\sin^2 \xi}{\xi^2} \cos 2\xi x d\xi &= 1 - x, \quad 0 \le x < 1 \end{aligned}$$

When $x = 0$,

$$I_2 = \frac{\pi}{2}.$$

Integrating the expression for $(1 - x)$ with respect to x,

$$\begin{aligned}
\frac{2}{\pi}\int_0^\infty \frac{\sin^2\xi}{\xi^2}\int_0^x \cos 2\xi x dx d\xi &= x - \frac{x^2}{2} \\
\frac{2}{\pi}\int_0^\infty \frac{\sin^2\xi}{\xi^2}\frac{\sin 2\xi x}{2\xi} d\xi &= x - \frac{x^2}{2}
\end{aligned}$$

When $x = 1/2$, we get

$$I_3 = \frac{3\pi}{8}.$$

Integrating with x one more time,

$$\begin{aligned}
\frac{1}{\pi}\int_0^\infty \frac{\sin^2\xi}{\xi^3}\int_0^x \sin 2\xi x dx d\xi &= \frac{x^2}{2} - \frac{x^3}{6} \\
\frac{1}{\pi}\int_0^\infty \frac{\sin^2\xi}{\xi^3}\frac{1-\cos 2\xi}{2\xi} d\xi &= \frac{x^2}{2} - \frac{x^3}{6}
\end{aligned}$$

When $x = 1$, we get

$$I_4 = \frac{\pi}{3}.$$

3.4 Obtain the Fourier transform of

$$f(x) = e^{-ax^2} \cos bx,$$

where a and b are positive constants.

Solution

$$\begin{aligned}
\mathcal{F}[e^{-ax^2}] &= \mathcal{F}_c[e^{-ax^2}] = \frac{1}{\sqrt{2a}} e^{-\xi^2/(4a)} \\
\sqrt{\frac{2}{\pi}} \int_0^\infty e^{-ax^2} \cos \xi x dx &= \frac{1}{\sqrt{2a}} e^{-\xi^2/(4a)} \\
\mathcal{F}[e^{-ax^2} \cos bx] &= \sqrt{\frac{2}{\pi}} \int_0^\infty \frac{1}{2} e^{-ax^2} [\cos(\xi + b)x + \cos(\xi - b)x] dx \\
&= \frac{1}{2\sqrt{2a}} \left[e^{-(\xi+b)^2/(4a)} + e^{-(\xi-b)^2/(4a)} \right].
\end{aligned}$$

3.5 Compute the sine transform of

$$f = \frac{e^{-ax}}{x},$$

and find its limit as $a \to 0$.

Solution

$$\begin{aligned}
F_s(\xi) &= \sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-ax} \frac{\sin \xi x}{x} dx \\
&= \sqrt{\frac{2}{\pi}} \int_0^{\infty} \int_0^{\xi} e^{-ax} \cos \xi x d\xi dx \\
&= \sqrt{\frac{2}{\pi}} \int_0^{\xi} \frac{a}{a^2+\xi^2} d\xi \\
&= \sqrt{\frac{2}{\pi}} \int_0^{\xi} \frac{d\xi/a}{1+(\xi/a)^2} = \sqrt{\frac{2}{\pi}} \tan^{-1}(\xi/a).
\end{aligned}$$

As $a \to 0$, $\tan^{-1}(\infty) = \pi/2$,

$$\mathcal{F}_s[1/x] = \sqrt{\frac{\pi}{2}}.$$

3.6 Using $f = 1/\sqrt{x}$ in the cosine and sine forms of the Fourier integral theorem, show that

$$\int_0^\infty \frac{\cos x}{\sqrt{x}} dx = \int_0^\infty \frac{\sin x}{\sqrt{x}} dx = \sqrt{\frac{\pi}{2}},$$

and

$$\mathcal{F}_c[1/\sqrt{x}] = \mathcal{F}_s[1/\sqrt{x}] = 1/\sqrt{\xi}.$$

Solution

Fourier integral theorem gives

$$\begin{aligned}
\frac{2}{\pi}\int_0^\infty \cos\xi x \int_0^\infty \frac{\cos\xi t dt}{\sqrt{t}} d\xi &= \frac{1}{\sqrt{x}} \\
\frac{2}{\pi}\int_0^\infty \cos\xi x \int_0^\infty \frac{\cos\tau d\tau}{\sqrt{\tau}} \frac{d\xi}{\sqrt{\xi}} &= \frac{1}{\sqrt{x}}, \quad \tau = \xi t \\
\frac{2}{\pi}\int_0^\infty \frac{\cos\xi x}{\sqrt{\xi}} d\xi \int_0^\infty \frac{\cos\tau}{\sqrt{\tau}} d\tau &= \frac{1}{\sqrt{x}}
\end{aligned}$$

When $x = 1$,

$$\int_0^\infty \frac{\cos x}{\sqrt{x}} dx = \sqrt{\frac{\pi}{2}}$$

Similarly, using the Sine transform,

$$\begin{aligned}
\frac{2}{\pi}\int_0^\infty \sin\xi x \int_0^\infty \frac{\sin\xi t dt}{\sqrt{t}} d\xi &= \frac{1}{\sqrt{x}} \\
\int_0^\infty \frac{\sin x}{\sqrt{x}} dx &= \sqrt{\frac{\pi}{2}}
\end{aligned}$$

Using these

$$\begin{aligned}
\mathcal{F}_c\left[\frac{1}{\sqrt{x}}\right] &= \frac{1}{\sqrt{\xi}}, \\
\mathcal{F}_s\left[\frac{1}{\sqrt{x}}\right] &= \frac{1}{\sqrt{\xi}}.
\end{aligned}$$

3.7 Invert the following

(a)

$$F(\xi) = \frac{1}{\xi^2 + 4\xi + 13},$$

(b)

$$F(\xi) = \frac{1}{(\xi^2 + 4)^2}.$$

Solution

(a)

$$\begin{aligned}
\mathcal{F}^{-1}\left[\frac{1}{\xi^2 + 4\xi + 13}\right] &= \mathcal{F}^{-1}\left[\frac{1}{(\xi + 2)^2 + 9}\right] \\
&= e^{2ix}\mathcal{F}^{-1}\left[\frac{1}{\xi^2 + 9}\right] \\
&= \frac{1}{3}e^{-3|x|+2ix}.
\end{aligned}$$

(b) We have

$$\mathcal{F}^{-1}\left[\frac{a}{a^2 + \xi^2}\right] = \sqrt{\frac{\pi}{2}}e^{-a|x|}.$$

Differentiating this with respect to a,

$$\mathcal{F}^{-1}\left[\frac{1}{a^2 + \xi^2} - \frac{2a^2}{(a^2 + \xi^2)^2}\right] = -\sqrt{\frac{\pi}{2}}|x|e^{-a|x|}.$$

When $a = 2$, we have

$$\begin{aligned}
\mathcal{F}^{-1}\left[\frac{8}{(4 + \xi^2)^2}\right] &= \sqrt{\frac{\pi}{2}}\left[|x|e^{-2|x|} - \frac{1}{2}e^{-2|x|}\right] \\
\mathcal{F}^{-1}\left[\frac{1}{(4 + \xi^2)^2}\right] &= \frac{1}{8}\sqrt{\frac{\pi}{2}}\left[|x| - 4\right]e^{-2|x|}
\end{aligned}$$

3.8 Invert

$$F(\xi) = \frac{e^{-\xi^2}}{\xi^2 + 4},$$

in terms of complementary error functions.

Solution

Let $f(x)$ be the inverse. We have

$$\begin{aligned}
\mathcal{F}^{-1}[e^{-\xi^2}] &= \frac{1}{\sqrt{2}} e^{-x^2/4}, \\
\mathcal{F}^{-1}\left[\frac{1}{\xi^2 + 4}\right] &= \frac{1}{2}\sqrt{\frac{\pi}{2}} e^{-2|x|}.
\end{aligned}$$

Using convolution integral, we have

$$\begin{aligned}
f(x) &= \frac{1}{\sqrt{2\pi}} \frac{1}{2} \frac{1}{\sqrt{2}} \sqrt{\frac{\pi}{2}} \int_{-\infty}^{\infty} e^{-2|x-t|-t^2/4} dt \\
&= \frac{1}{4\sqrt{2}} \left[\int_{-\infty}^{x} e^{-2x+2t-t^2/4} + \int_{x}^{\infty} e^{2x-2t-t^2/4} dt\right] \\
&= \frac{1}{4\sqrt{2}} \left[e^{-2x} \int_{-\infty}^{x} e^{-t^2/4+2t} dt + e^{2x} \int_{x}^{\infty} e^{-t^2/4-2t} dt\right] \\
&= \frac{1}{4\sqrt{2}} \left[e^{-2x} \int_{-x}^{\infty} e^{-(t/2+2)^2+4} dt + e^{2x} \int_{x}^{\infty} e^{-(t/2+2)^2+4} dt\right] \\
&= \frac{\sqrt{\pi}}{4\sqrt{2}} \left[e^{4-2x} \operatorname{erfc}(2 - x/2) + e^{4+2x} \operatorname{erfc}(2 + x/2)\right]
\end{aligned}$$

where

$$\operatorname{erfc}(x) = \frac{2}{\sqrt{\pi}} \int_{x}^{\infty} e^{-y^2} dy.$$

3.9 Invert

$$F(\xi) = \frac{e^{-ia\xi}}{(\xi-1)^2+4}.$$

Solution

$$\begin{aligned}
f(x) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-ia\xi-ix\xi}d\xi}{(\xi-1)^2+4} \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-iy\xi}d\xi}{(\xi-1)^2+4}, \quad y = x+a, \\
&= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\frac{e^{-iy(\eta+1)}d\eta}{\eta^2+4}, \quad \eta = \xi - 1, \\
&= \frac{1}{\sqrt{2\pi}}\sqrt{\frac{\pi}{2}}\frac{e^{-iy}}{2}e^{-2|y|} \\
&= \frac{1}{4}e^{-i(x+a)-2|x+a|}.
\end{aligned}$$

3.10 For $0 < a < \pi$, compute

$$f(x) = \mathcal{F}^{-1}\left[\frac{\cosh a\xi}{\sinh \pi\xi}\right], \quad g(x) = \mathcal{F}^{-1}\left[\frac{\sinh a\xi}{\sinh \pi\xi}\right].$$

Solution

Let

$$I = \int_{-\infty}^{\infty} \frac{\cosh a\xi}{\sinh \pi\xi} e^{-i\xi x} d\xi, \quad J = \int_{-\infty}^{\infty} \frac{\sinh a\xi}{\sinh \pi\xi} e^{-i\xi x} d\xi.$$

Consider the complex valued functions

$$F_1(z) = \frac{\cosh az}{\sinh \pi z} e^{-izx}, \quad F_2(z) = \frac{\sinh az}{\sinh \pi z} e^{-izx}.$$

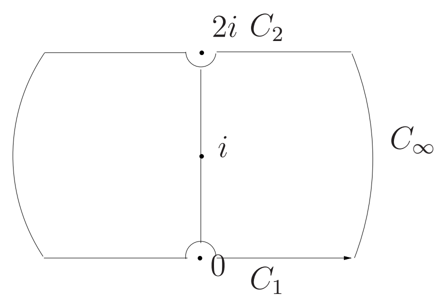

Figure 3.1: Contour for integration

There are two poles on the contour at $z = 0$ and at $z = 2i$. and one inside at $z = i$. Using Cauchy principal value, the line integrals of F_1 can be evaluated as follows:

$$\int_{C_1} F_1(z)dz = I.$$

The line integrals on C_∞ go to zero.

$$\begin{aligned}
\int_{C_2} F_1 dz &= \int_{\infty}^{-\infty} \frac{\cosh a(\xi + 2i)}{\sinh \pi(\xi + 2i)} e^{-ix(\xi+2i)} d\xi \\
&= -e^{2x} \int_{-\infty}^{\infty} \frac{\cosh a\xi \cos 2a + i \sinh a\xi \sin 2a}{\sinh \pi\xi} e^{-ix\xi} d\xi \\
&= -e^{2x}[\cos 2aI + i \sin 2aJ]
\end{aligned}$$

The residues are:

$$
\begin{aligned}
\mathrm{Res}(0) &= \frac{1}{\pi} \\
\mathrm{Res}(i) &= -\frac{1}{\pi}\cos a e^{x} \\
\mathrm{Res}(0) &= \frac{1}{\pi}\cos 2a e^{2x}
\end{aligned}
$$

From the closed contour, we get

$$
(1-e^{2x}\cos 2a)I - ie^{2x}\sin 2aJ = i\left[1+\cos 2ae^{2x} - 2\cos ae^{x}\right]
$$

Next, we work with F_2,

$$
\begin{aligned}
\int_{C_1} F_2(z)dz &= J, \\
\int_{C_2} F_2(z)dz &= \int_{\infty}^{-\infty} \frac{\sinh a(\xi+2i)}{\sinh \pi(\xi+2i)} e^{-ix(\xi+2i)} d\xi \\
&= -e^{2x}\int_{-\infty}^{\infty} \frac{\sinh a\xi\cos 2a + i\cosh a\xi\sin 2a}{\sinh \pi\xi} e^{-ix\xi} d\xi \\
&= -e^{2x}[\cos 2aJ + i\sin 2aI]
\end{aligned}
$$

The residues are:

$$
\begin{aligned}
\mathrm{Res}(i) &= -\frac{i}{\pi}\sin a e^{x} \\
\mathrm{Res}(0) &= \frac{i}{\pi}\sin 2a e^{2x}
\end{aligned}
$$

From the closed contour, we get

$$
(1-e^{2x}\cos 2a)J - ie^{2x}\sin 2aI = -\left[\sin 2ae^{2x} - 2\sin ae^{x}\right]
$$

The two simultaneous equations for I and J can be solved to get

$$
I = 2i\frac{\cos a - \cosh x}{\cosh 2x - \cos 2a}\sinh x, \quad J = 2\frac{\cosh x - \cos a}{\cosh 2x - \cos 2a}\sin a.
$$

These can be simplified to get

$$
I = -i\frac{\sinh x}{\cosh x + \cos a}, \quad J = \frac{\sin a}{\cosh x + \cos a}.
$$

Then

$$\begin{aligned}
\mathcal{F}^{-1}\left[\frac{\cosh a\xi}{\sinh \pi\xi}\right] &= -i\frac{1}{\sqrt{2\pi}}\frac{\sinh x}{\cosh x + \cos a}, \\
\mathcal{F}^{-1}\left[\frac{\sinh a\xi}{\sinh \pi\xi}\right] &= \frac{1}{\sqrt{2\pi}}\frac{\sin a}{\cosh x + \cos a}.
\end{aligned}$$

3.11 Evaluate the integral

$$I = \int_{-\infty}^{\infty} \frac{dx}{(x^2+a^2)(x^2+b^2)},$$

by (a) using partial fractions and (b) using the convolution theorem.

Solution

(a)

$$\frac{1}{(x^2+a^2)(x^2+b^2)} = \frac{1}{b^2-a^2}\left\{\frac{1}{x^2+a^2} - \frac{1}{x^2+b^2}\right\}.$$

$$I = \frac{1}{b^2-a^2}\left\{\frac{1}{a}\tan^{-1}\frac{x}{a} - \frac{1}{b}\tan^{-1}\frac{x}{b}\right\}\Bigg|_{-\infty}^{\infty} = \frac{\pi}{ab(a+b)}.$$

(b) To use convolution, let

$$J(x) = \int_{-\infty}^{\infty} \frac{dt}{(t^2+a^2)[(x-t)^2+b^2]}.$$

$$\begin{aligned}
\mathcal{F}[J] &= \sqrt{2\pi}\mathcal{F}\left[\frac{1}{x^2+a^2}\right]\mathcal{F}\left[\frac{1}{x^2+b^2}\right] \\
&= \sqrt{2\pi}\frac{1}{ab}\sqrt{\frac{\pi}{2}}\sqrt{\frac{\pi}{2}}e^{-a|\xi|}e^{-b|\xi|} = \sqrt{\frac{\pi}{2}}\frac{\pi}{ab}e^{-(a+b)|\xi|}, \\
J(x) &= \frac{\pi}{ab}\frac{a+b}{x^2+(a+b)^2}, \\
I &= J(0) = \frac{\pi}{ab(a+b)}.
\end{aligned}$$

3.12 For

$$f(x) = h(1 - |x|),$$

show that $\|f\| = \|F\|$.

Solution

$$
\begin{aligned}
\|f\|^2 &= \int_{-1}^{1} 1 dx = 2, \quad \|f\| = \sqrt{2}. \\
\mathcal{F}[f] &= F = \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{i\xi x} dx = \frac{1}{\sqrt{2\pi}} \frac{e^{i\xi} - e^{-i\xi}}{i\xi}, \\
F &= \sqrt{\frac{2}{\pi}} \frac{\sin \xi}{\xi}. \\
\|F\|^2 &= \frac{2}{\pi} \int_{-\infty}^{\infty} \frac{\sin^2 \xi}{\xi^2} d\xi = 2. \\
\|F\| &= \|f\|.
\end{aligned}
$$

3.13 Obtain the cosine transform of

$$f = \cos\left(\frac{x^2}{2} - \frac{\pi}{8}\right),$$

and show that it is self-reciprocal.

Solution

$$\begin{aligned}
f &= \cos\left(\frac{x^2}{2} - \frac{\pi}{8}\right) \\
&= \cos\frac{\pi}{8}\cos\frac{x^2}{2} + \sin\frac{\pi}{8}\sin\frac{x^2}{2} \\
\mathcal{F}_c\left[\cos\frac{x^2}{2}\right] &= \frac{1}{\sqrt{2}}\left[\cos\frac{\xi^2}{2} + \sin\frac{\xi^2}{2}\right], \\
\mathcal{F}_c\left[\sin\frac{x^2}{2}\right] &= \frac{1}{\sqrt{2}}\left[\cos\frac{\xi^2}{2} - \sin\frac{\xi^2}{2}\right], \\
\mathcal{F}_c[f] &= \frac{1}{\sqrt{2}}\cos\frac{\xi^2}{2}\left[\cos\frac{\pi}{8} + \sin\frac{\pi}{8}\right] \\
&= \quad +\frac{1}{\sqrt{2}}\sin\frac{\xi^2}{2}\left[\cos\frac{\pi}{8} - \sin\frac{\pi}{8}\right].
\end{aligned}$$

Using

$$\begin{aligned}
\frac{1}{\sqrt{2}}\left[\cos\frac{\pi}{8} + \sin\frac{\pi}{8}\right] &= \cos\frac{\pi}{8}, \\
\frac{1}{\sqrt{2}}\left[\cos\frac{\pi}{8} - \sin\frac{\pi}{8}\right] &= \sin\frac{\pi}{8}, \\
\mathcal{F}_c[f] &= \cos\left(\frac{\xi^2}{2} - \frac{\pi}{8}\right),
\end{aligned}$$

which shows self-reciprocity.

3.14 Show that

$$\frac{1}{2} + \sum_{n=1}^{N} \cos n\theta = \frac{\sin(2N+1)\theta/2}{2\sin\theta/2}.$$

Solution

Let

$$S = \frac{1}{2} + \sum_{n=1}^{N} \cos n\theta = \mathrm{Re}\left[\sum_{n=1}^{N} e^{in\theta}\right] - \frac{1}{2}.$$

Using

$$\begin{aligned}
\sum_{n=1}^{N} r^n &= \frac{1-r^{n+1}}{1-r}, \\
S &= \mathrm{Re}\left[\frac{1-e^{i(N+1)\theta}}{1-e^{i\theta}}\right] - \frac{1}{2} \\
&= \mathrm{Re}\left[\frac{e^{i(N+1)\theta/2} - e^{i(N+1)\theta/2}}{e^{-i\theta/2} - e^{i\theta/2}} \frac{e^{i(N+1)\theta/2}}{e^{i\theta/2}}\right] - \frac{1}{2} \\
&= \frac{\sin(N+1)\theta/2}{\sin\theta/2} \cos\frac{N\theta}{2} - \frac{1}{2} \\
&= \frac{1}{2}\frac{\sin(2N+1)\theta/2}{\sin\theta/2}.
\end{aligned}$$

3.15 Using the definition

$$F_c(\xi) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos x\xi dx,$$

show that

$$\frac{1}{2}F_c(0) + \sum_{n=1}^{N} F_c(n\xi) = \frac{1}{\sqrt{2\pi}} \int_0^\infty f(x) \frac{\sin(2N+1)x\xi/2}{\sin x\xi/2} dx.$$

Solution

$$F_c(\xi) = \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \cos x\xi dx,$$

$$\begin{aligned}
\frac{1}{2}F_c(0) + \sum_{n=1}^{N} F_c(n\xi) &= \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \left[\frac{1}{2} + \sum_{n=1}^{N} \cos n\pi\xi\right] dx \\
&= \sqrt{\frac{2}{\pi}} \int_0^\infty f(x) \frac{\sin(N+1)x\xi/2}{2\sin x\xi/2} dx \\
&= \frac{1}{\sqrt{2\pi}} \int_0^\infty f(x) \frac{\sin(N+1)x\xi/2}{\sin x\xi/2} dx.
\end{aligned}$$

3.16 Using the Riemann-Lebesgue localization lemma, show that the preceding relation yields the Poisson sum formula

$$\sqrt{\xi}\left[\frac{1}{2}F_c(0)+\sum_{n=1}^{\infty}F_c(n\xi)\right]=\sqrt{x}\left[\frac{1}{2}f(0)+\sum_{n}^{\infty}f(nx)\right],$$

where $x\xi = 2\pi$.

Solution

Let

$$\begin{aligned} S &= \frac{1}{2}F_c(0)+\sum_{n=1}^{\infty}F_c(n\xi) \\ &= \lim_{N\to\infty}\frac{1}{\sqrt{2\pi}}\int_0^{\infty} f(x)\frac{\sin(N+1)x\xi/2}{\sin x\xi/2}dx. \end{aligned}$$

Using $t = x\xi/2$,

$$S = \lim_{N\to\infty}\frac{1}{\sqrt{2\pi}}\int_0^{\infty} f(2t/\xi)\frac{\sin(2N+1)t}{\sin t}\frac{2dt}{\xi}$$

From

$$\begin{aligned} \frac{\sin(2N+1)t}{\sin t} &= \frac{\sin 2Nt\cos t}{\sin t}+\cos 2Nt \\ S &= \lim_{N\to\infty}\frac{2}{\xi}\frac{1}{\sqrt{2\pi}}\int_0^{\infty} f(2t/\xi)\frac{\sin 2Nt\cos t}{\sin t}dt \\ &= \lim_{N\to\infty}\frac{2}{\xi}\frac{1}{\sqrt{2\pi}}\left[\int_0^{\epsilon} f(2t/\xi)\frac{\sin 2Nt\cos t}{t}dt\right. \\ &\quad \int_{\epsilon}^{\pi-\epsilon} f(2t/\xi)\frac{\sin 2Nt\cos t}{\sin t}dt \\ &= \left.\int_{\pi-\epsilon}^{\pi+\epsilon} f(2t/\xi)\frac{\sin 2Nt\cos t}{\sin t}dt+\cdots\right] \\ &= \frac{\sqrt{2\pi}}{\xi}\left[\frac{1}{2}f(0)+\sum_{n=1}^{\infty}f(2n\pi/\xi)\right]. \end{aligned}$$

With $2\pi/\xi = x$,

$$\sqrt{\xi}S = \sqrt{x}[\frac{1}{2}f(0) + \sum_{n=1}^{\infty} f(nx)].$$

3.17 From the preceding result, show that the Theta function given by the sum

$$\theta(x) = \sum_{-\infty}^{\infty} e^{-n^2x^2/2},$$

satisfies

$$\theta(x) = \frac{\sqrt{2\pi}}{x}\theta(2\pi/x).$$

Solution

We have

$$f(x) = e{-}x^2/2, \quad F_c(\xi) = e{-}\xi^2/2.$$

$$S(x) = 2\left[\frac{1}{2}f(0) + \sum_{n=1}^{\infty} f(nx)\right].$$

By the Poisson sum formula, using $x\xi = 2\pi$,

$$2[F_C(0) + \sum_{n=-\infty}^{\infty} F_c(n\xi)] = \sqrt{\frac{x}{\xi}}S(x).$$

$$2\left[1 + \sum_{n=-\infty}^{\infty} e^{-n^2\xi^2/2}\right] = S(2\pi/x).$$

Thus

$$\theta(x) = \frac{\sqrt{2\pi}}{x}\theta(2\pi/x).$$

3.18 Solve the wave equation for an infinite string

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2},$$

with the initial conditions

$$u(x,0) = U_0 e^{-x^2}, \quad \frac{\partial u(x,0}{\partial t} = 0.$$

Solution

Taking the Fourier transform

$$-\xi^2 U(\xi) - \ddot{U},$$

which has the general solution

$$U(\xi,t) = Ae^{i\xi t} + Be^{-i\xi t}.$$

The initial conditions give

$$U(\xi,0) = \frac{U_0}{\sqrt{2}} e^{-\xi^2/2}, \quad \dot{U}(\xi,0) = 0.$$

Then

$$A + B = \frac{U_0}{\sqrt{2}} e^{-\xi^2/2}, \quad A - B = 0.$$

Solving

$$A = B = \frac{U_0}{2\sqrt{2}} e^{-\xi^2/2}.$$

$$U = \frac{U_0}{2\sqrt{2}} \left[e^{-\xi^2/2+i\xi t} + e^{-\xi^2/2-i\xi t} \right].$$

Inverting

$$u(x,t) = \frac{U_0}{2} \left[e^{-(x-t)^2} + e^{-(x+t)^2} \right].$$

3.19 Solve the wave equation for an infinite string

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2},$$

with the initial conditions

$$u(x,0) = \frac{U_0}{x^2+1}, \quad \frac{\partial u(x,0}{\partial t} = 0.$$

Solution

Fourier transform gives

$$\ddot{U} + \xi^2 U = 0, \quad U(\xi,0) = \sqrt{\frac{\pi}{2}} U_0 e^{|\xi|}, \quad \dot{U}(\xi,0) = 0.$$

Using the solution

$$U(\xi,t) = Ae^{i\xi t} + Be^{-i\xi t},$$

in the initial conditions, we find

$$A = \sqrt{\frac{\pi}{2}} \frac{U_0}{2} e^{-|\xi|+i\xi t}, \quad B = \sqrt{\frac{\pi}{2}} \frac{U_0}{2} e^{-|\xi|-i\xi t}.$$

Inverting this, we get

$$u(x,t) = \frac{U_0}{2}\left[\frac{1}{(x+t)^2+1} + \frac{1}{(x-t)^2+1}\right].$$

3.20 Lateral motion of an infinite beam satisfies

$$EI\frac{\partial^4 v}{\partial x^4} + m\frac{\partial^2 v}{\partial t^2} = 0,$$

where EI is the bending stiffness and m is the mass per unit length. If the initial conditions are

$$v(x,0) = V_0 e^{-x^2}, \quad \frac{\partial v(x,0)}{\partial t} = 0,$$

and $v \to 0$ as $|x| \to \infty$, obtain $v(x,t)$.

Solution

The given equation

$$a^2 v'''' + \ddot{v} = 0, \quad a^2 = EI/m,$$

under the Fourier transform, becomes

$$\ddot{V} + a^2\xi^4 V = 0.$$

This has the solution

$$V(\xi,t) = A\cos a\xi^2 t + B\sin a\xi^2 t.$$

At $t = 0$,

$$V(\xi,0) = F(\xi), \dot{V}(\xi,0) = 0.$$

Then

$$A = F(\xi), \quad B = 0$$

and

$$V(\xi,t) = F(\xi)\cos a\xi^2 t.$$

We have

$$\mathcal{F}^{-1}[\cos a\xi^2 t] = \frac{1}{2\sqrt{at}}\left[\cos\frac{x^2}{4at} + \sin\frac{x^2}{4at}\right].$$

Using convolution,

$$v(x,t) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} f(x-y)\frac{1}{2\sqrt{at}}\left[\cos\frac{y^2}{4at} + \sin\frac{y^2}{4at}\right] dy.$$

Using

$$z^2 = \frac{y^2}{4at},$$

$$v(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x - \sqrt{4at}z) \left[\cos z^2 + \sin z^2\right] dz.$$

For the special case

$$f(x) = V_0 e^{-x^2}, \quad F(\xi) = \frac{1}{\sqrt{2}} e^{-\xi^2/4}.$$

$$\begin{aligned} v(x,t) &= \sqrt{\frac{2}{\pi}} V_0 \int_0^{\infty} \frac{1}{\sqrt{2}} e^{-\xi^2/4} \cos at\xi^2 \cos \xi x d\xi \\ &= \frac{V_0}{2\sqrt{\pi}} \int_0^{\infty} \left[e^{-\xi^2/4 + iat\xi} + e^{-\xi^2/4 - iat\xi}\right] \cos \xi x d\xi \end{aligned}$$

Using

$$a_1^2 = \frac{1}{4} - iat, \quad a_2^2 = \frac{1}{4} - iat,$$

and

$$\phi = \tan^{-1} 4at, \quad r = [1 + 16a^2t^2]^{1/4},$$

$$a_1 = \frac{1}{2} r e^{-i\phi/2}, \quad a_2 = \frac{1}{2} r e^{i\phi/2}.$$

$$\begin{aligned} v(x,t) &= \frac{V_0}{2\sqrt{\pi}} \int_0^{\infty} \left[e^{-a_1^2\xi^2} + e^{-a_2^2\xi^2}\right] \cos \xi x d\xi \\ &= \frac{V_0}{2\sqrt{2}} \mathcal{F}^{-1} \left[e^{-a_1^2\xi^2} + e^{-a_2^2\xi^2}\right] \\ &= \frac{V_0}{4} \left[\frac{1}{a_1} e^{-x^2/(4a_1^2)} + \frac{1}{a_2} e^{-x^2/(4a_2^2)}\right] \\ &= \frac{V_0}{2r} \left[e^{i\phi/2} e^{-(x^2/r^2)\exp(i\phi)} + e^{-i\phi/2} e^{-(x^2/r^2)\exp(-i\phi)}\right] \\ &= \frac{V_0}{2r} e^{-(x^2/r^2)\cos\phi} \left[e^{i\phi/2} e^{-i(x^2/r^2)\sin\phi} + e^{-i\phi/2} e^{i(x^2/r^2)\sin\phi}\right] \\ &= \frac{V_0}{r} e^{-(x^2/r^2)\cos\phi} \cos\left[\frac{x^2}{r^2}\sin\phi - \frac{\phi}{2}\right]. \end{aligned}$$

3.21 Consider the motion of a concentrated force on an elastically supported taut string. The deflection u satisfies

$$\frac{\partial^2 u}{\partial x^2} + u = \frac{\partial^2 u}{\partial t^2} - \delta(x - ct),$$

with $u \to 0$ as $|x| \to \infty$. Assume the speed of the force, $c > 1$. We are interested in the steady state distribution of the deflection. Introducing the new variables:

$$\xi = x - ct, \quad \tau = t,$$

rewrite the governing equation. For the steady state, let $\partial u/\partial \tau$ and $\partial^2 u/\partial \tau^2$ go to zero. Obtain the solution which is continuous at $\xi = 0$.

Solution

With

$$\xi = x - ct, \quad \tau = t,$$

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi}, \quad \frac{\partial}{\partial t} = \frac{\partial}{\partial \tau} - c\frac{\partial}{\partial \xi}.$$

The equation becomes

$$\frac{\partial^2 u}{\partial \xi^2} + u = \frac{\partial^u}{\partial \tau^2} - 2c\frac{\partial^2 u}{\partial \xi \partial \tau} + c^2\frac{\partial^2 u}{\partial \xi^2} - \delta(\xi).$$

For steady-state we drop all time derivatives to get

$$(c^2 - 1)\frac{\partial^2 u}{\partial \xi^2} - u = \delta(\xi).$$

The solution for positive and negative values of ξ can be written as

$$u = \begin{cases} Ae^{-\beta\xi}, & \xi > 0 \\ Be^{\beta\xi}, & \xi < 0. \end{cases},$$

where

$$\beta = \sqrt{c^2 - 1}.$$

Continuity at $\xi = 0$ requires $A = B$. The jump condition gives

$$u'(a_+) - u'(a_-) = 1/\beta^2, \quad -2\beta A = 1/\beta^2.$$

Then

$$A = -\frac{1}{2\beta^3},$$

and

$$u(x - ct) = -\frac{1}{2\beta^3} e^{-\beta|x-ct||}.$$

3.22 The transient heat conduction in a 1D infinite bar satisfies

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t},$$

with the initial condition

$$u(x,0) = T_0 e^{-|x|}.$$

Obtain the temperature $u(x,t)$.

Solution

Fourier transform gives

$$-\xi^2 U = \frac{\partial U}{\partial t}, \quad U(\xi,0) = T_0\sqrt{\frac{2}{\pi}}\frac{1}{\xi^2+1}.$$

The solution of this equation is

$$U = Ae^{-\xi^2 t}, \quad A = T_0\sqrt{\frac{2}{\pi}}\frac{1}{\xi^2+1}.$$

Then

$$U(\xi,t) = T_0\sqrt{\frac{2}{\pi}}\frac{e^{-\xi^2 t}}{\xi^2+1}.$$

We invert this using the convolution theorem.

$$\begin{aligned}
u(x,t) &= \frac{T_0}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{|x-y|}\frac{1}{\sqrt{2t}}e^{-y^2/(4t)}dy \\
&= \frac{T_0}{\sqrt{2\pi}}\left[\int_{-\infty}^{x} e^{-x+y-y^2/(4t)}\frac{dy}{\sqrt{2t}} + \int_{x}^{\infty} e^{x-y-y^2/(4t)}\frac{dy}{\sqrt{2t}}\right] \\
&= \frac{T_0}{\sqrt{2\pi}}\left[e^{-x}\int_{-x}^{\infty} e^{-y-y^2/(4t)}\frac{dy}{\sqrt{2t}} + e^{x}\int_{x}^{\infty} e^{-y-y^2/(4t)}\frac{dy}{\sqrt{2t}}\right] \\
&= \frac{T_0}{\sqrt{2\pi}}\left[e^{-x}\int_{-x/(2\sqrt{t})}^{\infty} e^{-z^2-2\sqrt{t}z}\sqrt{2}dz + e^{x}\int_{x/(2\sqrt{t})}^{\infty} e^{-z^2-2\sqrt{t}z}\sqrt{2}dz\right] \\
&= \frac{T_0}{\sqrt{\pi}}\left[e^{-(x-t)}\int_{-x/(2\sqrt{t})}^{\infty} e^{-(z-\sqrt{t})^2}dz + e^{x+t}\int_{x/(2\sqrt{t})}^{\infty} e^{-(z-\sqrt{t})^2}dz\right] \\
&= \frac{T_0}{2}\left\{e^{-(x-t)}\mathrm{erfc}\left[-\left(\frac{x}{2\sqrt{t}}+\sqrt{t}\right)\right] + e^{x+t}\mathrm{erfc}\left[\left(\frac{x}{2\sqrt{t}}+\sqrt{t}\right)\right]\right\}.
\end{aligned}$$

3.23 The transient heat conduction in a 1D infinite bar with heat generation satisfies

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t} + \frac{1}{x^2+1},$$

with the initial condition $u(x,0) = T_0$. Obtain $u(x,t)$ in the form of a convolution integral.

Solution

For the Fourier transform to exist the function should be integrable. Let

$$u(x,t) = v(x,t) + T_0.$$

The v satisfies

$$\frac{\partial^2 v}{\partial x^2} = \frac{\partial v}{\partial t} + \frac{1}{x^2+1},$$

with $v(x,0) = 0$. Taking the Fourier transform, we have

$$-\xi^2 V = \frac{\partial V}{\partial t} + \sqrt{\frac{\pi}{2}} e^{-|\xi|}.$$

The solution

$$V(\xi,t) = Ae^{-\xi^2 t} - \sqrt{\frac{\pi}{2}} \frac{e^{-|\xi|}}{\xi^2},$$

with the condition $V(\xi,0) = 0$, gives

$$A = \sqrt{\frac{\pi}{2}} \frac{e^{-|\xi|}}{\xi^2}$$

and

$$V(\xi,t) = \sqrt{\frac{\pi}{2}} \frac{e^{-|\xi|}}{\xi^2} \left[e^{-\xi^2 t} - 1 \right] = -\sqrt{\frac{\pi}{2}} \int_0^t e^{-|\xi|} e^{-\xi^2 t} dt.$$

$$\begin{aligned}
v(x,t) &= -\frac{1}{2} \int_0^t \int_{-\infty}^{\infty} e^{-\xi^2 t - |\xi| - ix\xi} d\xi dt \\
&= -\frac{1}{2} \int_0^t \left\{ \int_{-\infty}^0 e^{-\xi^2 t + \xi - ix\xi} d\xi + \int_0^\infty e^{-\xi^2 t - \xi - ix\xi} d\xi \right\} dt \\
&= -\frac{1}{2} \int_0^t \left\{ \int_0^\infty e^{-\xi^2 t - \xi + ix\xi} d\xi + \int_0^\infty e^{-\xi^2 t - \xi - ix\xi} d\xi \right\} dt
\end{aligned}$$

The terms of the form $-\xi^2 t - (1 - ix)\xi$ can be transformed by completing the square. Let $a_1 = 1 - ix$ and $a_2 = 1 + ix$.

$$\begin{aligned} v(x,t) &= -\frac{1}{2}\int_0^t \left\{ e^{a_1^2/(4t)} \int_0^\infty e^{-t(\xi+a_1/(2t))^2} d\xi + e^{a_2^2/(4t)} \int_0^\infty e^{-t(\xi+a_2/(2t))^2} d\xi \right\} dt \\ &= -\frac{1}{2}\int_0^t \left\{ e^{a_1^2/(4t)} \mathrm{erfc}\left(\frac{a_1}{2\sqrt{t}}\right) + e^{a_2^2/(4t)} \mathrm{erfc}\left(\frac{a_2}{2\sqrt{t}}\right) \right\} \frac{dt}{\sqrt{t}}. \end{aligned}$$

3.24 The transient heat conduction in a 2D infinite plate satisfies

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial t},$$

with

$$u(x, y, 0) = U_0 e^{-(x^2+y^2)}.$$

Obtain $u(x, y, t)$.

Solution

Taking the Fourier transform with $x \to \xi$ and $y \to \eta$,

$$-(\xi^2 + \eta^2)U = \frac{\partial U}{\partial t},$$

which has the solution

$$U = Ae^{-(\xi^2+\eta^2)t}.$$

The initial condition gives

$$A = \frac{1}{2} U_0 e^{-(\xi^2+\eta^2)/4}.$$

Then

$$U = \frac{1}{2} U_0 e^{-(\xi^2+\eta^2)(1/4+t)}.$$

Inverting this, we get

$$u(x, y, t) = \frac{U_0}{1+4t} e^{-(x^2+y^2)/(1+4t)}.$$

3.25 The transient heat conduction in a 2D infinite plate satisfies

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{\partial u}{\partial t},$$

with

$$u(x, y, 0) = U_0 h(1 - |x|) h(1 - |y|).$$

Obtain $u(x, y, t)$.

Solution

We have

$$-(\xi^2 + \eta^2) U = \frac{\partial U}{\partial t}.$$

The initial condition is

$$U(\xi, \eta, 0) = U_0 \mathcal{F}[h(1 - |x|)] \mathcal{F}[h(1 - |y|)].$$

$$\begin{aligned}
\mathcal{F}[h(1-|x|)] &= \sqrt{\frac{2}{\pi}} int_0^1 (1-x)\cos\xi x dx \\
&= \sqrt{\frac{2}{\pi}} \left\{ (1-x)\frac{\sin\xi x}{\xi}\bigg|_0^1 - (-1)\frac{(-\cos\xi x)}{\xi^2}\bigg|_0^1 \right\} \\
&= \sqrt{\frac{2}{\pi}} \frac{1-\cos\xi}{\xi^2} = 2\sqrt{\frac{2}{\pi}} \frac{\sin^2 \xi/2}{\xi^2}.
\end{aligned}$$

The solution

$$U = A e^{-(\xi^+ \eta^2)t},$$

with the initial condition gives

$$A = \frac{8}{\pi} \frac{\sin^2 \xi/2}{\xi^2} \frac{\sin^2 \eta/2}{\eta^2}.$$

Then

$$u(x, y, t) = v(x, t) v(y, t),$$

where

$$
\begin{aligned}
v(x,t) &= \mathcal{F}^{-1}\left[2\sqrt{\frac{2}{\pi}}\frac{\sin^2\xi/2}{\xi^2}e^{-\xi^2 t}\right] \\
&= \frac{1}{\sqrt{4\pi t}}\int_{-1}^{1} e^{-(x-z)^2/(4t)}(1-|z|)dz \\
&= \frac{1}{\sqrt{4\pi t}}\left\{\int_{-1}^{1} e^{-(x-z)^2/(4t)}dz + \int_{-1}^{0} e^{-(x-z)^2/(4t)}zdz - \int_{0}^{1} e^{-(x-z)^2/(4t)}zdz\right\}
\end{aligned}
$$

Using $x-z=2\sqrt{t}r$, $dz=-2\sqrt{t}dr$, we find

$$
\begin{aligned}
v(x,t) &= \frac{1}{\sqrt{\pi}}\left\{\int_{(x-1)/(2\sqrt{t})}^{(x+1)/(2\sqrt{t})} e^{-r^2}dr - \int_{(x+1)/(2\sqrt{t})}^{x/(2\sqrt{t})} e^{-r^2}(x-2\sqrt{t}r)dr\right. \\
&\quad \left. + \int_{x/(2\sqrt{t})}^{(x-1)/(2\sqrt{t})} e^{-r^2}(x-2\sqrt{t}r)dr\right\} \\
&= \frac{1}{2}\left\{\mathrm{erf}\frac{x+1}{2\sqrt{t}} - \mathrm{erf}\frac{x-1}{2\sqrt{t}} - 2x\mathrm{erf}\frac{x}{2\sqrt{t}}\right. \\
&\quad +x\left[\mathrm{erf}\frac{x+1}{2\sqrt{t}} + \mathrm{erf}\frac{x-1}{2\sqrt{t}} - 2\mathrm{erf}\frac{x}{2\sqrt{t}}\right] \\
&\quad \left. -\sqrt{t}\left[e^{-x^2/(4t)} - e^{-(x+1)^2/(4t)} + e^{-(x-1)^2/(4t)} - e^{-x^2/(4t)}\right]\right\} \\
&= \frac{1}{2}\left\{(x+1)\mathrm{erf}\frac{x+1}{2\sqrt{t}} + (x-1)\mathrm{erf}\frac{x-1}{2\sqrt{t}} - 2x\mathrm{erf}\frac{x}{2\sqrt{t}}\right. \\
&\quad \left. +\sqrt{t}\left[e^{-(x+1)^2/(4t)} - e^{-(x-1)^2/(4t)} + e^{-x^2/(4t)}\right]\right\}
\end{aligned}
$$

3.26 A semi-infinite rod has an initial temperature distribution

$$u(x,0) = T_0 x e^{-x^2},$$

and the transient heat conduction is governed by

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}.$$

If the boundary temperature is $u(0,t) = 0$, obtain the temperature $u(x,t)$.

Solution

Taking the Sine transform

$$-\xi^2 U_s = \frac{\partial U_s}{\partial t}, \quad U_s(\xi,t) = Ae^{-\xi^2 t}.$$

$$U_s(\xi,0) = T_0 \mathcal{F}_s[xe^{-x^2}].$$

Differentiating with respect to ξ, the Cosine transform,

$$\mathcal{F}_c[e^{-a^2x^2}] = \frac{1}{\sqrt{2}a} e^{-\xi^2/4a^2},$$

we get

$$-\mathcal{F}_s[xe^{-a^2x^2}] = -\frac{\xi}{2\sqrt{2}a^3} e^{-\xi^2/4a^2}.$$

$$\mathcal{F}_s[xe^{-x^2}] = \frac{\xi}{2\sqrt{2}} e^{-\xi^2/4}.$$

Then

$$A = \frac{T_0}{2\sqrt{2}} \xi e^{-\xi^2/4}.$$

$$U_s = \frac{T_0}{2\sqrt{2}} \xi e^{-\xi^2(t+1/4)}.$$

$$u(x,t) = \frac{T_0}{(1+4t)^{3/2}} x e^{-x^2/(4t+1)}.$$

3.27 In the preceding problem if we replace the condition at $x = 0$ by

$$\frac{\partial u(0,t)}{\partial x} - hu(0,t) = 0,$$

where h is a constant, obtain the solution using the mixed trigonometric transform.

Solution

Taking the mixed transform

$$\mathcal{F}_R[u] = U = \sqrt{\frac{2}{\pi}} \int_0^\infty [\xi \cos \xi x + h \sin \xi x] u(x) dx,$$

$$-\xi^2 U = \frac{\partial U}{\partial t}, \quad U = Ae^{-\xi^2 t}.$$

From the initial condition

$$\mathcal{F}_R[u(x,0)] = -\mathcal{F}_s \left[\frac{\partial u}{\partial x} - hu \right] dx,$$

$$A = -\sqrt{\frac{2}{\pi}} \int_0^\infty T_0 [1 - 2x^2 - hx] e^{-x^2} \sin \xi x dx$$

and

$$U = -\sqrt{\frac{2}{\pi}} T_0 \int_0^\infty [1 - 2x^2 - hx] e^{-x^2} \sin \xi x e^{-\xi^2 t} dx.$$

For inversion, we use

$$\begin{aligned}
u' - hu &= -\mathcal{F}_S^{-1}[U] \\
&= \frac{2T_0}{\pi} \int_0^\infty \int_0^\infty [1 - 2y^2 - hy] e^{-y^2} e^{-\xi^2 t} \sin \xi y \sin \xi x dy d\xi \\
&= \frac{T_0}{\pi} \int_0^\infty \int_0^\infty [1 - 2y^2 - hy] e^{-y^2} e^{-\xi^2 t} [\cos(x-y) - \cos(x+y)] d\xi dy \\
&= \frac{T_0}{\sqrt{2t}\pi} \int_0^\infty (1 - 2y^2 - hy) e^{-y^2} \left[e^{(x-y)^2/4t} - e^{-(x+y)^2/4t} \right] dy
\end{aligned}$$

Next we solve

$$w' - hw = e^{-(x-u)^2/4t}.$$

Using the homogeneous solution, e^{hx}, with variable coefficient

$$w = Ae^{hx},$$

we find

$$\begin{aligned} A' &= e^{-(x^2-2xa+a^2+4htx)/4t} \\ &= e^{-(x-a+2ht)^2/4t} e^{[(a-2ht)^2-a^2]/4t} \end{aligned}$$

Letting $y = x - a + 2ht)/2\sqrt{t}$

,

$$\begin{aligned} A &= e^{h(ht-a)} 2\sqrt{t} \int_{\infty}^{(x-a+2ht)/4t} e^{-y^2} dy. \\ w(x,a) &= -\sqrt{\pi t} e^{h(x+ht-a)} \operatorname{erfc} \frac{x+2ht-a}{2\sqrt{t}} \\ u(x,t) &= \frac{T_0}{\sqrt{2\pi}} \int_0^{\infty} (1-2y^2-hy) e^{-y^2} [g(x,y) - g(x,-y)] dy, \end{aligned}$$

where

$$g(x,y) = e^{h(x+ht+y)} \operatorname{erfc} \frac{x+2ht+y}{2\sqrt{t}}.$$

3.28 Solve the integral equation

$$u(x) = xe^{-|x|} - \int_{-\infty}^{\infty} u(t)e^{-|x-t|}dt.$$

Solution

We have

$$\mathcal{F}[e^{-|x|}] = \sqrt{\frac{2}{\pi}}\frac{1}{\xi^2+1}.$$

$$\mathcal{F}[xe^{-|x|}] = \sqrt{\frac{2}{\pi}}i\frac{2\xi}{(\xi^2+1)^2}.$$

The Fourier transform of the integral equation is

$$U = \sqrt{\frac{2}{\pi}}i\frac{2\xi}{(\xi^2+1)^2} - \sqrt{2\pi}\sqrt{\frac{2}{\pi}}\frac{U}{\xi^2+1}.$$

$$U = \sqrt{\frac{2}{\pi}}i2\xi\left\{\frac{1}{\xi^2+1} - \frac{1}{\xi^2+3}\right\}$$

$$u = 2x\left\{e^{-|x|} - e^{-\sqrt{3}|x|}\right\}.$$

3.29 Find the solution of

$$u(x) = e^{-2|x|} - \int_{-\infty}^{\infty} u(t)e^{-|x-t|}dt.$$

Solution

$$U = \sqrt{\frac{2}{\pi}}\frac{2}{\xi^2+4} - 2\frac{U}{\xi^2+1}.$$

$$U = \sqrt{\frac{2}{\pi}}\frac{2(\xi^2+1)}{(\xi^2+4)(\xi^2+3)} = 2\sqrt{\frac{2}{\pi}}\left[\frac{3}{\xi^2+4} - \frac{2}{\xi^2+3}\right].$$

$$u = 3e^{-2|x|} - \frac{4}{\sqrt{3}}e^{-\sqrt{3}|x|}.$$

3.30 Find the solution of

$$u(x) = |x|^{-p} - \int_{-\infty}^{\infty} u(t)e^{-|x-t|}dt,$$

where p is a real constant: $0 < p < 1$.

Solution

$$F = \mathcal{F}[|x|^{-p}].$$

$$U = \frac{\xi^2+1}{\xi^2+3}F = F - \frac{2}{\xi^2+3}F.$$

$$u = |x|^{-p} - \sqrt{\frac{2}{3\pi}}\int_{-\infty}^{\infty} |t|^{-p}e^{-\sqrt{3}|x-t|}dt.$$

3.31 Solve

$$u(x) = xe^{-x^2} - \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2} u(t)dt,$$

in the form of an infinite series.

Solution

$$\mathcal{F}[xe^{-a^2x^2}] = \frac{i}{(\sqrt{2}a)^3} \xi e^{-\xi^2/4a^2}.$$

$$U = \frac{i}{(\sqrt{2})^3} \xi e^{-\xi^2/4} + \sqrt{\pi} e^{-\xi^2/4} U.$$

$$U = \frac{i}{(\sqrt{2})^3} \xi e^{-\xi^2/4} \frac{1}{1 - \sqrt{\pi} e^{-\xi^2/4}}.$$

$$U = \frac{i}{(\sqrt{2})^3} \xi e^{-\xi^2/4} \left[1 + \sqrt{\pi} e^{-\xi^2/4} + \pi e^{-2\xi^2/4} + \cdots \right].$$

$$u = \sum_{n=0}^{\infty} \frac{\pi^{n/2}(n+1)^{3/2}}{2^{3/2}} xe^{-x^2/(n+1)}.$$

3.32 Solve

$$u(x) = \frac{x}{x^2+4} - \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(x-t)^2} u(t)dt,$$

in the form of a series of convolution integrals.

Solution

$$\mathcal{F}[e^{-x^2}] = \frac{1}{\sqrt{2}} e^{-\xi^2/4}, \quad \mathcal{F}\left[\frac{x}{x^2+4}\right] = F = \sqrt{\frac{\pi}{2}} \operatorname{sgn}(\xi) e^{-2\xi}.$$

The integral equation becomes

$$U = F - \frac{\sqrt{2\pi}}{\sqrt{\pi}} \frac{1}{\sqrt{2}} e^{-\xi^2/4} U.$$

$$U = \frac{F}{1+e^{-\xi^2/4}} = F[1 - e^{-\xi^2/4} + e^{-2\xi^2/4} - \cdots].$$

$$U = F + \sum_{n=1}^{\infty} (-1)^n F e^{-n\xi^2/4}.$$

$$u = \frac{x}{x^2+4} + \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{\pi n}} \int_{-\infty}^{\infty} \frac{t}{t^2+4} e^{-(x-t)^2/n}.$$

3.33 Solve the system of integral equations

$$u_1(x) = \frac{1}{2}\int_{-\infty}^{\infty} u_2(t)e^{-|x-t|}dt + 2e^{-|x|},$$

$$u_2(x) = \frac{1}{2}\int_{-\infty}^{\infty} u_1(t)e^{-|x-t|}dt - 4e^{-2|x|}.$$

Solution

Taking the Fourier transform

$$\begin{aligned} U_1 &= \frac{1}{2}\sqrt{2\pi}\sqrt{\frac{2}{\pi}}\frac{U_2}{1+\xi^2} + 2\sqrt{\frac{2}{\pi}}\frac{1}{1+\xi^2}, \\ U_2 &= \frac{1}{2}\sqrt{2\pi}\sqrt{\frac{2}{\pi}}\frac{U_1}{1+\xi^2} - 4\sqrt{\frac{2}{\pi}}\frac{2}{1+\xi^2}. \end{aligned}$$

Rearranging these equations, we find

$$\begin{aligned} (1+\xi^2)U_1 - U_2 &= \sqrt{\frac{2}{\pi}}2, \\ -U_1 + (1+\xi^2)U_2 &= \sqrt{\frac{2}{\pi}}(-8)\frac{1+\xi^2}{4+\xi^2}. \end{aligned}$$

Solving these, we get

$$\begin{aligned} U_1 &= \frac{1}{(1+\xi^2)^2-1}\sqrt{\frac{2}{\pi}}\left[2(1+\xi^2) - 8\frac{1+\xi^2}{4+\xi^2}\right], \\ U_2 &= \frac{1}{(1+\xi^2)^2-1}\sqrt{\frac{2}{\pi}}\left[2 - 8\frac{(1+\xi^2)^2}{4+\xi^2}\right]. \end{aligned}$$

These may be simplified to the forms,

$$\begin{aligned} U_1 &= \sqrt{\frac{2}{\pi}}\frac{\xi^2+1}{\xi^2+2}\frac{2}{\xi^2+4} = \sqrt{\frac{2}{\pi}}\left[\frac{3}{\xi^2+4} - \frac{1}{\xi^2+2}\right], \\ U_2 &= \sqrt{\frac{2}{\pi}}\frac{\xi^2+1}{\xi^2+2}\frac{2}{\xi^2+4} = \sqrt{\frac{2}{\pi}}\left[\frac{1}{\xi^2+4} - \frac{9}{\xi^2+2}\right], \end{aligned}$$

Inversion gives

$$\begin{aligned} u_1 &= \frac{3}{2}e^{-2|x|} - \frac{1}{\sqrt{2}}e^{-\sqrt{2}|x|}, \\ u_2 &= \frac{1}{\sqrt{2}}e^{-\sqrt{2}|x|} - \frac{9}{\sqrt{2}}e^{-2|x|}. \end{aligned}$$

3.34 Show that

$$u = 1/2, \quad \text{and} \quad u = c - x^2,$$

are solutions of

$$u(x) \pm \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} u(x-t) dt = 1,$$

corresponding to "+" and "−" signs, respectively. Here, c is a constant.

Solution

Let

$$J_1 = u(x) + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} u(x-t) dt.$$

When $u = 1/2$, we have

$$\begin{aligned} J_1 &= \frac{1}{2} + \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} \frac{dt}{2} \\ &= \frac{1}{2} + \frac{1}{2} = 1. \end{aligned}$$

Let

$$J_2 = u(x) - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} u(x-t) dt.$$

When $u = c - x^2$, we have

$$\begin{aligned} J_2 &= c - x^2 - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} [c - (x-t)^2] dt \\ &= c\left[1 - \frac{1}{\sqrt{2\pi}} 2\sqrt{\frac{\pi}{2}}\right] - x^2 \left[1 - \frac{1}{\sqrt{2\pi}} 2\sqrt{\frac{\pi}{2}}\right] - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} [2x - t] t dt \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} t^2 dt \\ &= \frac{1}{\sqrt{2\pi}} \left[e^{-t^2/2}(-t)\Big|_{-\infty}^{\infty} + \int_{-\infty}^{\infty} e^{-t^2/2} dt \right] \\ &= 1. \end{aligned}$$

3.35 Obtain all the solutions of the integral equation

$$u(x) - \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} u(x-t)dt = 1.$$

Solution

We use the Wiener-Hopf method for this, as 1 does not have a Fourier transform

Let $f(x) = 1$.

$$\begin{aligned} F_+ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{i\zeta x} 1 dx = -\frac{1}{\sqrt{2\pi}} \frac{1}{i\zeta} \\ F_- &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{i\zeta x} 1 dx = \frac{1}{\sqrt{2\pi}} \frac{1}{i\zeta}, \\ k &= e^{-x^2/2}, \quad K = e^{-\zeta^2/2}. \end{aligned}$$

Then

$$U_+ = \frac{F_+ - H}{1 - e^{-\zeta^2/2}}, \quad U_- = \frac{F_- + H}{1 - e^{-\zeta^2/2}},$$

where H is any function, which is analytic between the lines: $C_+ : \eta > 0$ and $C_- : \eta < 0$. The solution is given by

$$\begin{aligned} u(x) &= -\frac{1}{\sqrt{2\pi}} \int_{C_+} \frac{1}{\sqrt{2\pi}} \frac{1}{i\zeta} \frac{e^{-i\zeta x} d\zeta}{1 - e^{-\zeta^2/2}} + \frac{1}{\sqrt{2\pi}} \int_{C_-} \frac{1}{\sqrt{2\pi}} \frac{1}{i\zeta} \frac{e^{-i\zeta x} d\zeta}{1 - e^{-\zeta^2/2}} \\ &\quad + \frac{1}{\sqrt{2\pi}} \oint_{C_- - C_+} \frac{H e^{-i\zeta x} d\zeta}{1 - e^{-\zeta^2/2}}. \end{aligned}$$

Using the fact the denominator has a second order zero at $\zeta = 0$, we find

$$u(x) = -x^2 + c_0 + c_1 x.$$

3.36 Solve

$$u(x) - \frac{1}{2}\int_{-\infty}^{\infty} e^{-2|x-t|}u(t)dt = e^{|x|}.$$

Solution

With

$$f(x) = e^{|x|},$$

$$\begin{aligned}
F_+ &= \frac{1}{\sqrt{2\pi}}\int_0^{\infty} e^{x(1+i\zeta)}dx = \frac{1}{\sqrt{2\pi}}\frac{1}{1+i\zeta}, \\
F_- &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{0} e^{x(-1+i\zeta)}dx = \frac{1}{\sqrt{2\pi}}\frac{1}{-1+i\zeta}, \\
u &= \frac{1}{\sqrt{2\pi}}\left[\int_{C_+} U_+e^{-i\zeta x}d\zeta + \int_{C_-} U_-e^{-i\zeta x}d\zeta\right]. \\
k &= \frac{\sqrt{2\pi}}{2}e^{-2|x|}, \quad K = \frac{\sqrt{2\pi}}{2}\sqrt{\frac{2}{\pi}}\frac{2}{\zeta^2+4} = \frac{2}{\zeta^2+4}. \\
U_+ &= \frac{F_+ - H}{1-2/(\zeta^2+4)}, \quad U_- = \frac{F_- + H}{1-2/(\zeta^2+4)},
\end{aligned}$$

$C_+ : 1 < \eta < 2$ and $C_- : -2 < \eta < -1$

$$\begin{aligned}
u &= \frac{1}{2\pi}\int_{C_+}\frac{\zeta^2+4}{\zeta^2+2}\frac{(-e^{-i\zeta x})d\zeta}{1+i\zeta} + \frac{1}{2\pi}\int_{C_-}\frac{\zeta^2+4}{\zeta^2+2}\frac{(e^{-i\zeta x})d\zeta}{-1+i\zeta} \\
&\quad + \frac{1}{\sqrt{2\pi}}\int_{C_- - C_+}\frac{\zeta^2+4}{\zeta^2+2}H(\zeta)e^{-i\zeta x}d\zeta
\end{aligned}$$

The last integral has poles at $\zeta = \pm i\sqrt{2}$ and using the residue theorem, it reduces to $A\cos\sqrt{2}x + B\sin\sqrt{2}x$. The contributions to the first and second integrals from these poles are already included in A and B. Including the contribution from the pole at $\pm i$, we get

$$u = 3e^{|x|} + A\cos\sqrt{2}x + B\sin\sqrt{2}x.$$

3.37 Obtain all the solutions of the integral equation

$$u(x) - \frac{\lambda}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-t^2/2} u(x-t) dt = x,$$

with λ being real and positive.

Solution

With

$$f(x) = x,$$

$$\begin{aligned} F_+ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} x e^{ix\zeta} dx = -\frac{1}{\sqrt{2\pi}} \frac{1}{\zeta^2}, \\ F_- &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} x e^{ix\zeta} dx = \frac{1}{\sqrt{2\pi}} \frac{1}{\zeta^2}. \\ k &= e^{-x^2/2}, \quad K = e^{-\xi^2/2}. \\ U_+ &= \frac{F_+ - H}{1 - \lambda e^{-\xi^2/2}}, \\ U_- &= \frac{F_- + H}{1 - \lambda e^{-\xi^2/2}}, \end{aligned}$$

The denominator has zeros at

$$e^{-\zeta^2/2} = 1/\lambda, \quad \zeta = \pm\sqrt{2\log\lambda} = \pm a.$$

The solution is written as

$$\begin{aligned} u(x) &= -\frac{1}{2\pi} \int_{C_+} \frac{1}{\zeta^2} \frac{e^{-i\zeta x} d\zeta}{1 - \lambda e^{-\xi^2/2}} + \frac{1}{2\pi} \int_{C_-} \frac{1}{\zeta^2} \frac{e^{-i\zeta x} d\zeta}{1 - \lambda e^{-\xi^2/2}} \\ &\quad \frac{1}{\sqrt{2\pi}} \int_{C_- - C_+} \frac{H(\zeta) e^{-i\zeta x} d\zeta}{1 - \lambda e^{-\xi^2/2}}, \end{aligned}$$

where C_+ is a line above $\eta = 0$ and C_- is a line below $\eta = 0$. Using the residue theorem, we get

$$u(x) = \frac{x}{1 - \lambda} + A \cos ax + B \sin ax.$$

3.38 Solve
$$u(x) - \lambda \int_{-\infty}^{\infty} e^{-|x-t|} u(t) dt = \cos x.$$

Solution

Here,
$$f(x) = \cos x,$$
and
$$\begin{aligned}
F_+ &= \frac{1}{\sqrt{2\pi}} \int_0^{\infty} e^{i\zeta x} \frac{e^{ix} + e^{-ix}}{2} dx = -\frac{1}{\sqrt{2\pi}} \frac{1}{2i} \left[\frac{1}{1+\zeta} + \frac{1}{-1+\zeta} \right], \\
F_- &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{i\zeta x} \frac{e^{ix} + e^{-ix}}{2} dx = \frac{1}{\sqrt{2\pi}} \frac{1}{2i} \left[\frac{1}{1+\zeta} + \frac{1}{-1+\zeta} \right]. \\
k &= \sqrt{2\pi} e^{-|x|}, \quad K = \frac{2}{1+\zeta^2}.
\end{aligned}$$

Note that K has poles at $\zeta = \pm i$, F_+ is analytic above $\eta = 0$, and F_- is analytic below $\eta = 0$. We choose inversion contours, C_+ with $0 < \eta < 1$ and C_- with $-1 < \eta < 0$. The integral equation gives
$$\begin{aligned}
U_+ \left[1 - \frac{2\lambda}{1+\zeta^2} \right] &= F_+ - H, \\
U_- \left[1 - \frac{2\lambda}{1+\zeta^2} \right] &= F_- + H.
\end{aligned}$$

The solution is
$$\begin{aligned}
u(x) &= \frac{1}{\sqrt{2\pi}} \int_{C_+} \frac{1+\zeta^2}{\zeta^2 - a^2} F_+ e^{-i\zeta x} d\zeta + \frac{1}{\sqrt{2\pi}} \int_{C_-} \frac{1+\zeta^2}{\zeta^2 - a^2} F_- e^{-i\zeta x} d\zeta \\
&\quad \frac{1}{\sqrt{2\pi}} \int_{C_- - C_+} \frac{H(1+\zeta^2)}{\zeta^2 - a^2} e^{-i\zeta x} d\zeta,
\end{aligned}$$
where $a^2 = 2\lambda - 1$.

(a) When $\lambda > 1/2$, $\zeta = \pm a$ are real, and the last integral gives $A \cos ax + B \sin ax$. Using the residues
$$u = \frac{2}{1-a^2} \cos x + A \cos ax + B \sin ax.$$

(b) When $\lambda < 1/2$, $\zeta = \pm i|a|$ are imaginary, and the last integral gives no residues.

$$u = \frac{2}{1-a^2}\cos x.$$

(c) When $\lambda = 1/2$, $\zeta = 0, 0$ and

$$u = 2\cos x + A + Bx.$$

3.39 Consider the signal

$$f(t) = 3\cos 3t + 4\sin 2t,$$

which is modulated to get

$$g(t) = f(t)\cos 100t.$$

Show that the analytical signal

$$h(t) = g(t) + i\mathcal{H}[g(\tau), \tau \to t],$$

allows the factoring

$$h(t) = f(t)e^{i100t}.$$

Solution

$$\begin{aligned} f(t) &= 3\cos 3t + 4\sin 2t, \\ g(t) &= 3\cos 3t\cos 100t + 4\sin 2t\cos 100t, \\ &= \frac{3}{2}[\cos 103t + \cos 97t] + 2[\sin 102t - \sin 98t]. \end{aligned}$$

We have

$$\begin{aligned} \mathcal{H}[\cos\omega t] &= \sin\omega t, \\ \mathcal{H}[\sin\omega t] &= -\cos\omega t. \end{aligned}$$

$$\mathcal{H}[g] = \frac{3}{2}[\sin 103t + \sin 97t] - 2[\cos 102t - \cos 98t].$$

$$\begin{aligned} g + i\mathcal{H}[g] &= \frac{3}{2}[e^{i103t} + e^{i97t}] - 2i[e^{i102t} - e^{i98t}] \\ &= e^{i100t}[3\cos 3t + 4\sin 2t] = f(t)e^{i100t} \end{aligned}$$

3.40 Show that the Finite Hilbert transform of $(1-x^2)^{1/2}$ is x and that of $(1-x^2)^{-1/2}$ is 0. Use the change of variable:

$$x = \frac{1-t^2}{1+t^2},$$

to simplify the integrals.

Solution

$$T_1 = \mathcal{T}[\sqrt{1-x^2}] = \frac{1}{\pi}\int_{-1}^{1} \frac{\sqrt{1-\xi^2}}{x-\xi} d\xi.$$

Using

$$\xi = \frac{1-\tau^2}{1+\tau^2}, \quad d\xi = -\frac{4\tau d\tau}{(1+\tau^2)^2},$$

$$1-\xi^2 = \frac{4\tau^2}{(1+\tau^2)^2}, \quad x-\xi = \frac{(x+1)\tau^2-(1-x)}{1+\tau^2}.$$

$$T_1 = \frac{1}{\pi}\int_0^\infty \frac{8\tau^2}{1+\tau^2}\frac{d\tau}{(x+1)\tau^2-(1-x)}.$$

Let

$$a^2 = \frac{1-x}{1+x}, \quad a^2 > 0 \quad \text{for} \quad -1 < x < 1.$$

$$T_1 = \frac{8}{\pi(1+x)}\int_0^\infty \frac{\tau^2 d\tau}{(1+\tau^2)^2(\tau^2-a^2)}.$$

Using partial fractions, we have

$$\frac{\tau^2}{(1+\tau^2)^2(\tau^2-a^2)} = \frac{A}{(1+\tau^2)^2} + \frac{B}{1+\tau^2} + \frac{C}{\tau^2-a^2},$$

where

$$A = \frac{1}{a^2+1}, \quad B = \frac{-1}{a^2+1} + \frac{1}{(a^2+1)^2}, \quad C = \frac{a^2}{(a^2+1)^2}.$$

Using the integrals

$$\begin{aligned}
\int_0^\infty \frac{d\tau}{1+\tau^2} &= \frac{\pi}{2}, \\
\int_0^\infty \frac{d\tau}{(1+\tau^2)^2} &= \frac{\pi}{4}, \\
\int_0^\infty \frac{d\tau}{\tau^2 - a^2} &= 0, \\
T_1 &= \frac{8}{\pi(1+x)} \left[\frac{A}{4} + \frac{B}{2} \right] \\
&= \frac{2}{1+x} \left[\frac{1}{a^2+1} - \frac{2}{a^2+1} + \frac{1}{(a^2+1)^2} \right] \\
&= \frac{2}{1+x} \frac{1-a^2}{(a^2+1)^2} \\
&= \frac{2}{1+x} \frac{(1+x)^2}{4} 2 \frac{1+x-1}{1+x} \\
&= x.
\end{aligned}$$

Let

$$T_2 = \mathcal{T}\left[\frac{1}{\sqrt{1-x^2}} \right].$$

Using the change of variable

$$\begin{aligned}
T_2 &= \frac{1}{\pi} \int_0^\infty \frac{1+\tau^2}{2\tau} \frac{4\tau d\tau}{(1+\tau^2)^2} \frac{1+\tau^2}{(1+x)\tau^2 - (1-x)} \\
&= \frac{2}{\pi(1+x)} \int_0^\infty \frac{d\tau}{\tau^2 - a^2} \\
&= 0.
\end{aligned}$$

3.41 Show that the Finite Hilbert transform of an even function in (-1,1) is odd and that of an odd function is even.

Solution

Let

$$\begin{aligned} v(x) &= \frac{1}{\pi}\int_{-1}^{1}\frac{u(\xi)d\xi}{x-\xi} \\ &= \frac{1}{\pi}\left\{\int_{-1}^{0}\frac{u(\xi)d\xi}{x-\xi}+\int_{0}^{1}\frac{u(\xi)d\xi}{x-\xi}\right\} \\ &= \frac{1}{\pi}\int_{0}^{1}\left\{\frac{u(-\xi)}{x+\xi}+\int_{0}^{1}\frac{u(\xi)}{x-\xi}\right\}d\xi \end{aligned}$$

(a) If u is even, $u(-\xi)=u(\xi)$, and

$$\begin{aligned} v(x) &= \frac{1}{\pi}\int_{0}^{1}\left\{\frac{1}{x+\xi}+\frac{1}{x-\xi}\right\}u(\xi)d\xi \\ &= \frac{2x}{\pi}\int_{0}^{1}\frac{u(\xi)d\xi}{x^2-\xi^2}, \end{aligned}$$

which is an odd function.

(b) If u is odd, $u(-\xi)=-u(\xi)$, and

$$\begin{aligned} v(x) &= \frac{1}{\pi}\int_{0}^{1}\left\{\frac{1}{x-\xi}-\frac{1}{x+\xi}\right\}u(\xi)d\xi \\ &= \frac{2}{\pi}\int_{0}^{1}\frac{\xi u(\xi)d\xi}{x^2-\xi^2}, \end{aligned}$$

which is even.

3.42 Show that

$$\frac{1}{\pi}\int_0^{*\pi}\frac{\cos n\phi d\phi}{\cos\theta-\cos\phi}=-\frac{\sin n\theta}{\sin\theta}.$$

In Eq. (3.295), assuming Fourier series

$$f(\theta)=\sum_{n=0}A_n\cos n\theta,\quad g(\theta)=\sum_{n=1}B_n\sin n\theta,$$

show that $B_n=-A_n$ $(n=1,2,\cdots)$. In airfoil theory B_n are known and A_n are to be found.

Solution

Let

$$\begin{aligned}
I &= \frac{1}{2\pi}\int_0^{2\pi}\frac{\cos n\phi d\phi}{\cos\theta-\cos\phi}=\mathrm{Re}[J],\\
J &= \frac{1}{2\pi}\int_0^{2\pi}\frac{2e^{in\phi}d\phi}{e^{i\theta}+e^{-i\theta}-(e^{i\phi}+e^{-i\phi})},\quad s=e^{i\phi},\quad z=e^{i\theta}\\
&= \frac{1}{\pi}\oint_{|s|=1}\frac{s^n ds}{is[z+z^{-1}-(s+s^{-1})]}\\
&= -\frac{1}{\pi i}\oint\frac{s^n ds}{(s-z)(s-z^{-1})}.
\end{aligned}$$

Using the residue at $s=z$, we get

$$\begin{aligned}
J &= -\frac{1}{i}\frac{\cos n\theta+i\sin n\theta}{\sin\theta}.\\
I &= \mathrm{Re}[J]=-\frac{\sin n\theta}{\sin\theta}.
\end{aligned}$$

For

$$f(\theta)=\sum_{n=1}A_n\cos n\theta,$$

$$\begin{aligned}
g(\theta) &= \sum_{n=1}A_n\int_0^{\pi}\frac{\cos n\phi\sin\theta d\phi}{\cos\theta-\cos\phi}\\
&= -\sum_{n=1}A_n\sin n\theta.
\end{aligned}$$

Then,

$$B_n = A_n.$$

Chapter 4

LAPLACE TRANSFORMS

4.1 Find the Laplace transforms of

$$f(t) = e^{at}\cos bt, \quad g(t) = e^{at}\sin bt.$$

Solution

We have

$$\mathcal{L}[\cos bt] = \frac{p}{p^2+b^2}, \quad \mathcal{L}[\sin bt] = \frac{p}{p^2+b^2}.$$

$$\begin{aligned}
\mathcal{L}[e^{at}\cos bt] &= \frac{p-a}{(p-a)^2+b^2}, \\
\mathcal{L}[eat\sin bt] &= \frac{b}{(p-a)^2+b^2}.
\end{aligned}$$

4.2 Using the relation

$$\int_0^\infty e^{-(a^2x^2+b^2/x^2)}dx = \frac{\sqrt{\pi}}{2a}e^{-2ab}, \quad a, b > 0,$$

obtain the Laplace transforms of

$$f(t) = \frac{1}{\sqrt{t}}e^{-x^2/(4t)}, \quad g(t) = \text{erfc}\left(\frac{x}{2\sqrt{t}}\right).$$

Solution

$$\begin{aligned} f(t) &= \frac{1}{\sqrt{t}}e^{-x^2/(4t)}. \\ \bar{f}(p) &= \int_0^\infty e^{-x^2/(4t)-pt}\frac{dt}{\sqrt{t}}. \end{aligned}$$

Using $t = y^2$, $dt = 2ydy$, we get

$$\bar{f} = 2\int_0^\infty e^{-x^2/(4y^2)-py^2}dy.$$

From

$$\int_0^\infty e^{-(a^2t^2+b^2/t^2)}dt = \sqrt{\pi}e^{-2ab}/(2a),$$

we have

$$\begin{aligned} \bar{f} &= \frac{2\sqrt{\pi}}{2\sqrt{p}}e^{-2\sqrt{p}x/2} = \sqrt{\frac{\pi}{p}}e^{-\sqrt{p}x}. \\ g(t) &= \frac{2}{\sqrt{\pi}}\int_{x/(2\sqrt{t})}^\infty e^{-y^2}dy \\ \bar{g} &= \frac{2}{\sqrt{\pi}}\int_0^\infty \int_{x/(2\sqrt{t})}^\infty e^{-y^2-pt}dydt. \end{aligned}$$

We interchange the order of integration in the (t, y)-plane, to have

$$\begin{aligned}
\bar{g} &= \frac{2}{\sqrt{\pi}} \int_0^\infty e^{-y^2} \int_{x^2/(4y^2)}^\infty e^{-pt} dt dy \\
&= \frac{2}{p\sqrt{\pi}} \int_0^\infty e^{-y^2 - px^2/(4y^2)} dy \\
&= \frac{2\sqrt{\pi}}{2\sqrt{\pi}p} e^{-\sqrt{p}x} \\
&= \frac{1}{p} e^{-\sqrt{p}\,x}.
\end{aligned}$$

4.3 Find the Laplace transforms of

$$f(t) = a^t, \quad g(t) = \frac{\cos at}{\sqrt{t}}.$$

Solution

$$f(t) = a^t = e^{t\log a}, \quad \bar{f} = \frac{1}{p - \log a}.$$

$$\begin{aligned}
g(t) &= \frac{\cos at}{\sqrt{t}}. \\
\bar{g}(p) &= \int_0^\infty \frac{\cos at}{\sqrt{t}} e^{-pt} dt \\
&= \int_0^\infty \cos at^2 e^{-pt^2} dt \\
&= \int_0^\infty \left[e^{-(p-ia)t^2} + e^{-(p+ia)t^2} \right] dt \\
&= \frac{\sqrt{\pi}}{2} \left[\frac{1}{\sqrt{p-ia}} + \frac{1}{\sqrt{p+ia}} \right].
\end{aligned}$$

Let

$$\begin{aligned}
e^{i\theta} &= \frac{p+ia}{\sqrt{p^2+a^2}}, \quad e^{-i\theta} = \frac{p-ia}{\sqrt{p^2+a^2}}. \\
\bar{g}(p) &= \sqrt{\pi} \frac{\cos\theta/2}{(p^2+a^2)^{1/4}} = \frac{\sqrt{\pi}}{(p^2+a^2)^{1/4}} \frac{\sqrt{1+\cos\theta}}{\sqrt{2}} \\
&= \frac{\sqrt{\pi}}{\sqrt{2(p^2+a^2)}} [p + \sqrt{p^2+a^2}]^{1/2}.
\end{aligned}$$

4.4 Invert the transforms

$$\bar{f}(p) = \frac{e^{-2p}}{[(p+1)^2+1]^2}, \quad \bar{g}(p) = \frac{\sqrt{p}}{p-a},$$

Solution

$$\begin{aligned}
\bar{f}(p) &= \frac{e^{-2p}}{[(p+1)^2+1]^2}. \\
f(t) &= \frac{1}{2\pi i}\int_\Gamma \frac{e^{-2p+pt}dp}{[(p+1)^2+1]^2}. \\
&= \frac{1}{2\pi i}\int_\Gamma \frac{e^{p\tau}dp}{[(p+1)^2+1]^2}, \quad \tau = t-2, \\
&= \frac{1}{2\pi i}\int_\Gamma \frac{e^{-\tau+q\tau}dq}{(q^2+1)^2}, \quad q = p+1, \\
&= -\frac{1}{2\pi i}e^{-\tau}\frac{d}{da}\int_\Gamma \frac{e^{q\tau}dq}{(q^2+a)}\bigg|_{a=1} \\
&= -e^{-\tau}\frac{d}{da}\left[\frac{\sin\sqrt{a}\tau}{\sqrt{a}}\right]\bigg|_{a=1} \\
&= -e^{-\tau}\left[-\frac{1}{2a^{3/2}}\sin\sqrt{a}\tau + \frac{\tau}{2a}\cos\sqrt{q}\tau\right]\bigg|_{a=1} \\
&= \frac{e^{-\tau}}{2}(\sin\tau - \tau\cos\tau) \\
&= \frac{1}{2}e^{-(t-2)}[\sin(t-2)-(t-2)\cos(t-2)]h(t-2).
\end{aligned}$$

$$\begin{aligned}
\bar{g} &= \frac{1}{\sqrt{p}}\frac{p}{p-a} = \frac{1}{\sqrt{p}}\left[1+\frac{a}{p-a}\right] \\
&= \sqrt{\frac{t}{\pi}} + a\sqrt{\frac{t}{\pi}} * e^{at} \\
&= \sqrt{\frac{t}{\pi}} + \frac{a}{\sqrt{\pi}}\int_0^t \sqrt{\tau}e^{a(t-\tau)}d\tau \\
&= \sqrt{\frac{t}{\pi}} + \frac{1}{2\sqrt{a}}e^{at}\mathrm{erf}\sqrt{at}.
\end{aligned}$$

4.5 Show that the Laplace transform of

$$f(t) = \int_1^\infty e^{-t\tau}\frac{d\tau}{\tau},$$

is

$$\bar{f}(p) = \frac{\log(p+1)}{p}.$$

Solution

$$\begin{aligned}
f(t) &= \int_1^\infty e^{-t\tau}\frac{d\tau}{\tau}, \\
\bar{f}(p) &= \int_0^\infty \int_1^\infty \frac{1}{\tau}e^{-\tau t-pt}d\tau dt \\
&= \int_1^\infty \left.\frac{-e^{-(p+\tau)t}}{\tau(p+\tau)}\right|_)^\infty d\tau \\
&= \int_1^\infty \frac{d\tau}{\tau(p+\tau)} \\
&= \int_1^\infty \left[\frac{1}{\tau} - \frac{1}{p+\tau}\right]\frac{d\tau}{p} \\
&= \frac{1}{p}\log \left.\frac{\tau}{p+\tau}\right|_1^\infty \\
&= \frac{1}{p}\log(p+1).
\end{aligned}$$

4.6 Invert the transforms

$$\bar{f}(p) = \frac{\cosh ap}{p^3 \cosh p}, \quad \bar{g}(p) = \frac{\sinh ap}{p^2 \sinh p}, \quad 0 < a < 1.$$

Solution

$$f(t) = \frac{1}{2\pi i} \int_\Gamma \frac{\cosh ap}{p^3 \cosh p} e^{pt} dp.$$

In the complex p-plane, there are poles at $p = 0$ and at $p = \pm(n + 1/2)\pi i = \pm \bar{n}\pi i$. For the pole at $p = 0$, we expand the function in the form,

$$\frac{\cosh ap}{p^3 \cosh p} e^{pt} = \frac{1}{p^3}(1+a^2p^2/2+\cdots)(1-p^2/2+\cdots)(1+pt+p^2t^2/2+\cdots).$$

$$\text{Res}[0] = (a^2 + t^2 - 1)/2.$$

The residues at $p = \bar{n}\pi i$ is

$$\begin{aligned}
\text{Res}[\bar{n}\pi i] &= \frac{\cos \bar{n}\pi a e^{\bar{n}\pi t i}}{-(\bar{n}\pi)^3 i i \sin \bar{n}\pi} \\
&= \frac{(-1)^n}{(\bar{n}\pi)^3} \cos \bar{n}\pi a e^{\bar{n}\pi t i}.
\end{aligned}$$

The residue at $p = -\bar{n}\pi i$ is

$$\begin{aligned}
\text{Res}[-\bar{n}\pi i] &= \frac{\cos \bar{n}\pi a e^{-\bar{n}\pi t i}}{-(\bar{n}\pi)^3 i i \sin \bar{n}\pi} \\
&= \frac{(-1)^n}{(\bar{n}\pi)^3} \cos \bar{n}\pi a e^{-\bar{n}\pi t i}.
\end{aligned}$$

Adding the residues

$$\begin{aligned}
f(t) &= \frac{1}{2}(a^2 + t^2 - 1) + \sum_{n=1}^{\infty} \frac{2(-1)^n}{(\bar{n}\pi)^3} \cos \bar{n}\pi a \cos \bar{\pi} t \\
&= \frac{1}{4}[(a+t)^2 + (a-t)^2 - 2] \\
&\quad + \sum_{n=1}^{\infty} \frac{(-1)^n}{\pi^3 \bar{n}^3} [\cos \pi\bar{n}(a+t) + \cos \pi\bar{n}(a-t)].
\end{aligned}$$

$$g(t) = \frac{1}{2\pi i}\int_\Gamma \frac{\sinh ap}{p^2 \sinh p} e^{pt} dp.$$

There are poles at $p = 0$ and at $p = \pm n\pi i$. Expanding the function around $p = 0$,

$$\frac{\sinh ap}{p^2 \sinh p} e^{pt} = \frac{1}{p^2}(1 + a^2p^2/3 + \cdot)(1 - p^2/3 + \cdots)(1 + pt + \cdots).$$

$$\text{Res}[0] = t.$$

$$\text{Res}[n\pi i] = \frac{1}{-n^2\pi^2}\frac{i \sin an\pi}{\cos n\pi} e^{n\pi i t}.$$

$$\text{Res}[-n\pi i] = \frac{1}{-n^2\pi^2}\frac{-i \sin an\pi}{\cos n\pi} e^{-n\pi i t}.$$

Adding the residues

$$\begin{aligned} g(t) &= t + \sum_{n=1}^{\infty} \frac{2(-1)^n}{n^2\pi^2} \sin an\pi \sin n\pi t \\ &= \frac{1}{2}[(a+t) - (a-t)] \\ &\quad - \sum_{n=1}^{\infty} \frac{(-1)^n}{\pi^2 n^2}[\cos \pi n(a+t) - \cos \pi n(a-t)]. \end{aligned}$$

4.7 Show that the Laplace transform of the Theta function,

$$\theta(t) = \sum_{n=-\infty}^{\infty} e^{-n^2\pi^2 t},$$

can be expressed as

$$\bar{\theta}(p) = \frac{\coth\sqrt{p}}{\sqrt{p}}.$$

Solution

$$\begin{aligned}
\theta(t) &= \sum_{n=-\infty}^{\infty} e^{-n^2\pi^2 t}, \\
\bar{\theta}(p) &= \sum_{n=-\infty}^{\infty} \frac{1}{p+n^2\pi^2}.
\end{aligned}$$

We may express this as a sum of residues, with

$$f(z) = \frac{1}{p-z^2}\frac{\cosh z}{\sinh z}.$$

This function, when integrated along $z = x + (N+1/2)\pi i$ with x going from ∞ to $-\infty$ and returning along $z = x - (N+1/2)\pi i$ with two vertical lines at $x = \pm\infty$, has poles at $z = n\pi i$ and at $z = \pm\sqrt{p}$. Here, N will go to infinity. The line integrals are zero. Then

$$\sum_{n=-\infty}^{n=\infty} \mathrm{Res}[n\pi i] + \mathrm{Res}[\sqrt{p}] + \mathrm{Res}[-\sqrt{p}] = 0.$$

$$\begin{aligned}
\mathrm{Res}[n\pi i] &= \frac{1}{p^2+n^2\pi^2}, \\
\mathrm{Res}[\sqrt{p}] &= -\frac{\cosh\sqrt{p}}{\sinh\sqrt{p}}\frac{1}{2\sqrt{p}}, \\
\mathrm{Res}[-\sqrt{p}] &= \frac{\cosh\sqrt{p}}{\sinh\sqrt{p}}\frac{(-1)}{2\sqrt{p}}, \\
\bar{\theta}(p) &= \frac{\coth\sqrt{p}}{\sqrt{p}}.
\end{aligned}$$

4.8 Using the expansion

$$\frac{1}{1-e^{-2\sqrt{p}}} = 1 + e^{-2\sqrt{p}} + e^{-4\sqrt{p}} + \cdots,$$

invert the transform

$$\bar{f}(p) = \frac{\coth\sqrt{p}}{\sqrt{p}}.$$

Solution

$$\begin{aligned}
\bar{f}(p) &= \frac{e^{\sqrt{p}} + e^{-\sqrt{p}}}{\sqrt{p}(e^{\sqrt{p}} + e^{-\sqrt{p}})} = \frac{1}{\sqrt{p}}\frac{1+e^{-2\sqrt{p}}}{1-e^{-2\sqrt{p}}} \\
&= \frac{1}{\sqrt{p}}(1+e^{-2\sqrt{p}})(1+e^{-2\sqrt{p}}+e^{-4\sqrt{p}}+\cdots) \\
&= \sum_{n=0}^{\infty}\left[e^{-2n\sqrt{p}} + e^{-2(n+1)\sqrt{p}}\right] \\
f(t) &= \sum_{n=0}^{\infty}\left[\frac{1}{\sqrt{\pi t}}e^{-4n^2/(4t)} + \frac{1}{\sqrt{\pi t}}e^{-4(n+1)^2/(4t)}\right] \\
&= \frac{1}{\sqrt{\pi t}}\sum_{n=0}^{\infty}\left[e^{-n^2/t} + e^{-(n+1)^2/t}\right] \\
&= \frac{1}{\sqrt{\pi t}}\sum_{n=-\infty}^{\infty} e^{-n^2/t}.
\end{aligned}$$

4.9 Assume $p = a$ is a branch point of the transform $\bar{f}(p)$ and near $p = a$ the transform may be expanded as

$$\bar{f}(p) = c_0(p-a)^{\nu_0} + c_1(p-a)^{\nu_1} + c_2(p-a)^{\nu_2} + \cdots ,$$

where $\nu_0 < \nu_1 < \nu_2 < \cdots$. Show that, as $t \to \infty$, $f(t)$ has the expansion

$$f(t) \sim e^{at}\left[\frac{c_0}{\Gamma(-\nu_0)t^{\nu_0+1}} + \frac{c_1}{\Gamma(-\nu_1)t^{\nu_1+1}} + \frac{c_2}{\Gamma(-\nu_2)t^{\nu_2+1}} + \cdots\right].$$

Use the inversion formula

$$\mathcal{L}^{-1}[p^{\nu}] = \frac{1}{\Gamma(-\nu)t^{\nu+1}},$$

for this purpose.

Solution

We have

$$\mathcal{L}^{-1}[p^{\nu}] = \frac{1}{\Gamma(-\nu)t^{\nu+1}}.$$

$$\begin{aligned}
f(t) &= \frac{1}{2\pi i}\int_{\Gamma} e^{pt}\left[c_0(p-a)^{\nu_0} + c_1(p-a)^{\nu_1} + \cdots\right] dp \\
&= \frac{1}{2\pi i}\int_{\Gamma} e^{(pt+at)}\left[c_0 p^{\nu_0} + c_1 p^{\nu_1} + \cdots\right] dp \\
&= e^{at}\left[\frac{c_0}{\Gamma(-\nu_0)t^{\nu_0+1}} + \frac{c_1}{\Gamma(-\nu_1)t^{\nu_1+1}} + \cdots\right].
\end{aligned}$$

4.10 Solve

$$\frac{d^2u}{dt^2} + 3\frac{du}{dt} + 2u = e^{2t}\cos t.$$

with $u(0) = du/dt(0) = 0$.

Solution

Taking the Laplace transform

$$(p^2 + 3p + 2)\bar{u} = \frac{p-2}{(p^-2)^2 + 1},$$

$$\bar{u} = \frac{p-2}{(p^-2)^2 + 1}\frac{1}{(p+2)(p+1)}.$$

From partial fraction expansion

$$\frac{1}{(p+2)(p+1)} = \frac{1}{p+1} - \frac{1}{p+2}.$$

The solution in convolution form is

$$\begin{aligned} u(t) &= \int_0^t e^{2\tau}\cos\tau[e^{-t+\tau} - e^{-2(t-\tau)}]d\tau \\ &= e^{-t}\int_0^t e^{3\tau}\cos\tau d\tau - e^{-2t}\int_0^t e^{4\tau}\cos\tau d\tau. \end{aligned}$$

To evaluate the integrals we use

$$\begin{aligned} I &= \int_0^t e^{a\tau}\cos\tau d\tau \\ &= \frac{1}{a}e^{a\tau}\cos\tau\Big|_0^t - \frac{1}{a^2}e^{a\tau}(-\sin\tau)\Big|_0^t - \frac{1}{a^2}I \\ \frac{a^2+1}{a^2}I &= \frac{e^{at}\cos t - 1}{a} + \frac{e^{at}\sin t}{a^2} \\ I &= \frac{e^{at}}{a^2+1}\left[a(\cos t - 1) + \sin t\right]. \\ u(t) &= \frac{1}{10}e^{-t}e^{3t}[3(\cos t - 1) + \sin t] - \frac{1}{17}e^{-t}e^{4t}[4(\cos t - 1) + \sin t] \\ &= \frac{1}{10}e^{2t}[3(\cos t - 1) + \sin t] - \frac{1}{17}e^{3t}[4(\cos t - 1) + \sin t]. \end{aligned}$$

4.11 Solve

$$\frac{d^3u}{dt^3} - 6\frac{d^2u}{dt^2} + 11\frac{du}{dt} - 6u = t,$$

with

$$u(0) = \frac{du}{dt}(0) = \frac{d^2u}{dt^2}(0) = 0.$$

Solution

Laplace transform gives

$$(p^3 - 6p^2 + 11p - 6)\bar{u} = \frac{1}{p^2}.$$

Noting $p = 1$ is a root of the characteristic polynomial, the other roots are $p = 2$ and $p = 3$. Then

$$\begin{aligned} \bar{u} &= \frac{1}{p^2}\frac{1}{(p-1)(p-2)(p-3)} \\ &= \frac{A}{p^2} + \frac{B}{p} + \frac{C}{p-1} + \frac{D}{p-2} + \frac{E}{p-3}, \end{aligned}$$

where

$$\begin{aligned} A &= -\frac{1}{6}, \quad C = \frac{1}{2}, \quad D = -\frac{1}{4}, \\ E &= \frac{1}{18}, \quad B = -\frac{11}{24}. \\ u(t) &= -\frac{11}{24} - \frac{1}{6}t + \frac{1}{2}e^t - \frac{1}{4}e^{2t} + \frac{1}{18}e^{3t}. \end{aligned}$$

4.12 Solve

$$\frac{d^2u}{dt^2} + 2\frac{du}{dt} + 2u = e^{-t}\sin t, \quad u(0) = \frac{du}{dt}(0) = 0.$$

Solution

Laplace transform gives

$$(p^2 + 2p + 2)\bar{u} = \frac{1}{(p+1)^2 + 1}.$$

$$\bar{u} = \frac{1}{[(p+1)^2 + 1]^2}.$$

Consider

$$\bar{v} = \frac{a}{(p+1)^2 + a^2}, \quad v = e^{-t}\sin at.$$

$$\frac{\partial \bar{v}}{\partial a} = \frac{1}{(p+1)^2 + a^2} - \frac{2a^2}{[(p+1)^2 + a^2]^2}.$$

$$\frac{\partial v}{\partial a} = ate^{-t}\cos t = \frac{1}{a}e^{-t}\sin at - 2a^2\mathcal{L}^{-1}\left[\frac{1}{[(p+1)^2 + a^2]^2}\right].$$

$$\begin{aligned} u &= \mathcal{L}^{-1}\left[\frac{1}{[(p+1)^2 + 1^2]^2}\right] \\ &= \frac{1}{2}[\sin t - t\cos t]e^{-t}. \end{aligned}$$

4.13 Solve the boundary value problem

$$\frac{d^2u}{dx^2} + \frac{du}{dx} - 2u = e^{-x}, \quad u(0) = 1, \quad \frac{du}{dx}(1) = 0.$$

Solution

We assume $u'(0) = C$ and expect to find C using $u'(1) = 0$. Then

$$(p^2\bar{u} - p - C + p\bar{u} - 1 - 2\bar{u} = \frac{1}{p+1}.$$

$$(p^2 + p - 2)\bar{u} = 1 + p + C + \frac{1}{p+1}.$$

$$\bar{u} = \frac{(1+p+C)(p+1)+1}{(p+1)(p+2)(p-1)} = \frac{A}{p+1} + \frac{B}{p+2} + \frac{D}{p-1}.$$

The coefficients are

$$A = -\frac{1}{2}, \quad B = \frac{(C-1)(-1)+1}{(-1)(-3)} = \frac{2-C}{3},$$

$$D = \frac{(C+2)2+1}{(2)(3)} = \frac{2C+5}{6}.$$

$$u(x) = -\frac{1}{2}e^{-x} + \frac{2-C}{3}e^{-2x} + \frac{2C+5}{6}e^{x}.$$

$$u'(x) = \frac{1}{2}e^{-x} - \frac{4-2C}{3}e^{-2x} + \frac{2C+5}{6}e^{x}.$$

As $u'(1) = 0$,

$$0 = \frac{1}{2}e^{-1} - \frac{4-2C}{3}e^{-2} + \frac{2C+5}{6}e.$$

$$C = \frac{8e^{-2} - 3e^{-1} - 5e}{2e + 4e^{-2}} = \frac{8 - 3e - 5e^3}{4 + 2e^3}.$$

4.14 Solve the system of differential equations

$$\frac{du}{dt} - u + v = 0, \quad \frac{dv}{dt} + v - u = e^t,$$

subject to the initial conditions

$$u(0) = 1, \quad v(0) = 0.$$

Solution

We have

$$\begin{aligned} p\bar{u} - 1 - \bar{u} + \bar{v} &= 1, \\ p\bar{v} + \bar{v} - \bar{u} = \frac{1}{p-1}&. \\ (p-1)\bar{u} + \bar{v} &= 1, \\ -\bar{u} + (p+1)\bar{v} &= \frac{1}{p-1}. \end{aligned}$$

Solving for $\bar{u}$ and $\bar{v}$, we get

$$\begin{aligned} \bar{u} &= \frac{1}{p-1}, \quad \bar{v} = \frac{2}{p^2}. \\ u &= e^t, \quad v = 2t. \end{aligned}$$

4.15 In order to solve

$$\frac{\partial^2\psi}{\partial r^2}+\frac{2}{r}\frac{\partial\psi}{\partial r}=\frac{\partial\psi}{\partial t},\quad \psi(a,t)=T_0,\quad \psi(r,0)=0,$$

we use a substitution

$$\psi=\phi/r.$$

Obtain the equation for ϕ. Using the Laplace transform solve for ψ.

Solution

Let

$$\psi=\frac{\phi}{r},\quad \frac{\partial\psi}{\partial r}=-\frac{\phi}{r^2}+\frac{1}{r}\frac{\partial\phi}{\partial r}.$$

Then

$$\frac{\partial}{\partial r}\left[r^2\frac{\partial\psi}{\partial r}\right]=r^2\frac{\partial\psi}{\partial t}$$

becomes

$$\frac{\partial}{\partial r}\left[-\phi_r\frac{\partial\phi}{\partial r}\right]=r\frac{\partial\phi}{\partial t}.$$

$$\frac{\partial^2\phi}{\partial r^2}=\frac{\partial\phi}{\partial t}.$$

Taking Laplace transform,

$$\frac{\partial^2\bar{\phi}}{\partial r^2}=p\bar{\phi},\quad \bar{\phi}(a,p)=\frac{aT_0}{p}.$$

The solution of this equation vanishing at infinity is

$$\bar{\phi}=Ae^{-\sqrt{p}r},\quad A=\frac{aT_0}{p}e^{\sqrt{p}a}.$$

$$\bar{\phi}=\frac{aT_0}{p}e^{-\sqrt{p}(r-a)}.$$

$$\phi=aT_0\mathrm{erfc}\frac{r-a}{\sqrt{4t}}.$$

$$\psi=\frac{aT_0}{r}\mathrm{erfc}\frac{r-a}{\sqrt{4t}}.$$

4.16 Obtain the solution of

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2}, \quad u(0,t) = U_0 \sin \omega t, \quad u(\ell, t) = 0,$$

$$u(x,0) = \frac{\partial u}{\partial t}(x,0) = 0.$$

Solution

Laplace transform gives

$$\begin{aligned}
\bar{u}''(x,p) &= p^2 \bar{u}(x,p), \\
\bar{u}(x,p) &= A \sinh px + B \sinh p(x-\ell), \\
\bar{u}(0,p) &= -B \sinh p\ell = \frac{U_0 \omega}{p^2+\omega^2}. \\
\bar{u}(\ell,p) &= A \sinh p\ell = 0, \quad A = 0. \\
\bar{u}(x,p) &= \frac{U_0 \omega}{p^2+\omega^2} \frac{\sinh p(\ell - x)}{\sinh p\ell}, \\
u(x,t) &= \frac{U_0 \omega}{2\pi i} \int_\Gamma \frac{\sinh pa}{\sinh p\ell} \frac{e^{pt} dp}{p^2+\omega^2}; \quad a = \ell - x.
\end{aligned}$$

To the left of the vertical line Γ, there are poles at $p = \pm i\omega$ and at $p = \pm in\pi/\ell$. Computing the residues,

$$\begin{aligned}
\mathrm{Res}[i\omega] &= \frac{1}{\omega} \frac{\sin \omega a}{\sin \omega \ell} \frac{e^{i\omega t}}{2i}, \\
\mathrm{Res}[-i\omega] &= -\frac{1}{\omega} \frac{\sin \omega a}{\sin \omega \ell} \frac{e^{-i\omega t}}{2i}, \\
\mathrm{Res}[in\pi/\ell] &= \frac{1}{\omega^2 - n^2\pi^2/ell^2} \frac{1}{\ell} \frac{i \sin n\pi a/\ell}{\cos n\pi} e^{in\pi t/\ell},
\end{aligned}$$

Adding the residues

$$u(x,t) = U_0 \left\{ \frac{\sin \omega a}{\sin \omega \ell} \sin \omega t - \sum_{n=1} \frac{(-1)^n \omega \ell}{\omega^2 \ell^2 - n^2 \pi^2} \sin(n\pi a/\ell) \sin(n\pi t/\ell) \right\}.$$

4.17 The propagation of elastic stress in a bar: $0 < x < \ell$, is governed by

$$\frac{\partial^2 \sigma}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 \sigma}{\partial t^2},$$

where c is the wave speed. At time $t = 0$, the end $x = 0$ is subjected to a stress σ_0, while the end $x = \ell$ is subject to $\partial\sigma/\partial x = 0$. The initial conditions are: $\sigma(x, 0) = 0$ and $\partial\sigma/\partial t(x, 0) = 0$. Obtain $\sigma(x, t)$ by expanding $(1 - e^{-2pl/c})^{-1}$ in a binomial series.

Solution

Laplace transform gives

$$\bar{\sigma}'' = \frac{p^2}{c^2}\bar{\sigma},$$

which has the solutions

$$\bar{\sigma} = Ae^{px/c} + Be^{-px/c}.$$

From the boundary conditions

$$\begin{aligned} A + B &= \frac{\sigma_0}{p}, \\ \frac{p}{c}\left[Ae^{p\ell/c} - Be^{-p\ell/c}\right] &= 0. \\ A &= \frac{\sigma_0}{p}\frac{e^{-2p\ell/c}}{1 - e^{-2p\ell/c}}, \quad B = \frac{\sigma_0}{p}\frac{1}{1 - e^{-2p\ell/c}}. \end{aligned}$$

$$\begin{aligned} \bar{\sigma} &= \frac{\sigma_0}{p}\frac{e^{p(x-2\ell)/c} + e^{-px/c}}{1 - e^{-2p\ell/c}} \\ &= \frac{\sigma_0}{p}\left[e^{p(x-2\ell)/c} + e^{-px/c}\right]\left[1 - e^{-2p\ell/c}\right]^{-1} \\ &= \frac{\sigma_0}{p}\left[e^{p(x-2\ell)/c} + e^{-px/c}\right]\left[1 + e^{-2p\ell/c} + e^{-4p\ell/c} + \cdots\right]. \\ \sigma(x, t) &= \sigma_0\left[h\left(t - \frac{x}{c}\right) + h\left(t - \frac{2\ell - x}{c}\right) + h\left(t - \frac{2\ell + x}{c}\right) + \cdots\right]. \end{aligned}$$

4.18 The viscoelastic motion of a semi-infinite bar: $0 < x < \infty$, is governed by

$$c^2 \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} + \beta \frac{\partial u}{\partial t}.$$

If the boundary and initial conditions are

$$u(0,t) = U_0 \sin \omega t, \quad u(x \to \infty, t) = 0, \quad u(x,0) = \frac{\partial u}{\partial x}(x,0) = 0,$$

show that the solution can be written as

$$u(x,t) = U_0 \omega \int_0^t k(x,\tau) \cos \omega (t-\tau) d\tau,$$

where, using a branch cut in the p-plane between $p = 0$ and $p = -\beta$, an expression for $k(x,t)$ can be obtained as

$$k(x,t) = 1 - \frac{1}{\pi} \int_0^\beta \frac{1}{r} e^{-rt} \sin \sqrt{r(\beta - r)}\, dr.$$

Solution

Laplace transform gives

$$c^2 \bar{u}'' = (p^2 + \beta p)\bar{u}.$$

The solution vanishing at infinity has the form

$$\bar{u} = A e^{-\sqrt{p^2+\beta p} x/c}.$$

From the boundary condition

$$A = \frac{U_0 \omega}{p^2 + \omega^2}.$$

$$\begin{aligned}
\bar{u} &= \frac{U_0 \omega}{p^2 + \omega^2} e^{-\sqrt{p^2+\beta p} x/c}, \\
u(x,t) &= \frac{U_0}{2\pi i} \int_\Gamma \frac{\omega}{p^2 + \omega^2} e^{-\sqrt{p^2+\beta p} x/c + pt} dp, \\
&= \frac{U_0}{2\pi i} \int_\Gamma \frac{p}{p^2 + \omega^2} \frac{\omega}{p} e^{-\sqrt{p^2+\beta p} x/c + pt} dp, \\
u(x,t) &= U_0 \omega \int_0^t k(x,\tau) \cos \omega (t-\tau) d\tau.
\end{aligned}$$

where

$$k(x,t) \quad = \quad \frac{1}{2\pi i}\int_{\Gamma} e^{-\sqrt{p^2+\beta p}x/c+pt}\frac{dp}{p}.$$

For manipulating the integral, we create a branch cut along the negative real axis between $(-\beta)$ and 0 by setting

$$p = re^{i\theta}, \quad 0 < \theta < 2\pi,$$

$$p - \beta = r_1 e^{i\theta_1}, \quad 0 < \theta_1 < 2\pi.$$

The inversion integral along Γ can be deformed to go along counter clockwise on a small circle around the origin, along the upper line along the branch, along a small circle around $p = -\beta$, and, finally, along the lower line. Then

$$k(x,t) = 1 - \frac{1}{\pi}\int_0^{\beta} e^{-rt}\sin[\sqrt{(\beta - r)r}x/c]\frac{dr}{r}.$$

4.19 Unsteady heat conduction in a semi-infinite bar is governed by the equation

$$\kappa\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t},$$

with the conditions

$$u(x,0) = 0, \quad u(0,t) = U_0 te^{-at}, \quad u(x \to \infty, t) = 0.$$

Find $u(x,t)$.

Solution

Laplace transform gives

$$\bar{u}'' = (p/\kappa)\bar{u}.$$

The solution which vanishes at infinity is

$$\bar{u} = Ae^{-\sqrt{p/\kappa}x}.$$

Let

$$f(t) = U_0 te^{-at}.$$

Then

$$A = \bar{f}, \quad \text{and} \quad \bar{u} = \bar{f}e^{-\sqrt{p/\kappa}x}.$$

$$\begin{aligned} u(x,t) &= U_0\int_0^t (t-\tau)e^{-(t-\tau)}e^{-x^2/(4\kappa\tau)}\frac{xd\tau}{\sqrt{4\pi\kappa\tau}} \\ &= \frac{U_0 x}{\sqrt{4\pi\kappa}}\int_0^t (t-\tau)e^{-a(t-\tau)-x^2/(4\kappa\tau)}\frac{d\tau}{\sqrt{\tau}}. \end{aligned}$$

4.20 Two semi-infinite bars, A: $-\infty < x < 0$ and B: $0 < x < \infty$, have thermal diffusivities κ_A and κ_B, respectively. Their conductivities are k_A and k_B and they are at uniform temperatures, T_A^0 and T_B^0 when $t < 0$. At time $t = 0$, their ends are made to contact. Obtain the transient temperatures $T_A(x,t)$ and $T_B(x,t)$ for $t > 0$.

Solution

Taking the Laplace transform with t,

$$\begin{aligned}\kappa_A \bar{T}_A'' &= p\bar{T}_A, \quad \bar{T}_A = Ae^{qx} + \frac{T_A^0}{p}, \\ \kappa_B \bar{T}_B'' &= p\bar{T}_B, \quad \bar{T}_B = Be^{-q'x} + \frac{T_B^0}{p},\end{aligned}$$

where

$$q = \sqrt{p/\kappa_A}, \quad q' = \sqrt{p/\kappa_B}$$

Matching the temperatures and the heat fluxes at $x = 0$,

$$A + \frac{T_A^0}{p} = B + \frac{T_B^0}{p},$$

$$\frac{Ak_A}{\sqrt{\kappa_A}} = \frac{Bk_B}{\sqrt{\kappa_B}}$$

Let

$$\theta = T_B^0 - T_A^0.$$

$$h_A = \frac{\sqrt{\kappa_A}k_B}{\sqrt{\kappa_A}k_B + \sqrt{\kappa_B}k_A}, \quad h_B = \frac{\sqrt{\kappa_B}k_A}{\sqrt{\kappa_A}k_B + \sqrt{\kappa_B}k_A}.$$

Then

$$\bar{T}_A = \frac{\theta}{p}h_A e^{qx} + \frac{T_A^0}{p}, \quad \bar{T}_B = -\frac{\theta}{p}h_B e^{-q'x} + \frac{T_B^0}{p}.$$

Inversion gives

$$T_A = T_A^0 + \theta h_A \mathrm{erfc}\left(-\frac{x}{2\sqrt{\kappa_A t}}\right), \quad T_B = T_B^0 - \theta h_B \mathrm{erfc}\left(\frac{x}{2\sqrt{\kappa_B t}}\right).$$

4.21 A semi-infinite bar is made of two materials: A and B. Material A occupies $0 < x < 1$ and B occupies $1 < x < \infty$. Heat conduction in the two materials is governed by

$$\kappa_A \frac{\partial^2 T_A}{\partial x^2} = \frac{\partial T_A}{\partial t}, \quad \kappa_B \frac{\partial^2 T_B}{\partial x^2} = \frac{\partial T_B}{\partial t}.$$

The boundary and initial conditions are:

$$T_A(0,t) = T_0 h(t), \quad T_A(x,0) = T_B(x,0) = 0, \quad T_B(x \to \infty) = 0,$$

where $h(t)$ is the Heaviside step function. At the interface $x = 1$,

$$\frac{\partial T_A}{\partial x} = \alpha \frac{\partial T_B}{\partial x}, \quad T_A = T_B,$$

where a is a constant. Obtain the temperatures in the two sections of the bar.

Solution

Taking the Laplace transform with t,

$$\begin{aligned} \kappa_A \bar{T}_A'' &= p\bar{T}_A, \quad \bar{T}_A = A \sinh qx + B \sinh q(1-x), \\ \kappa_B \bar{T}_B'' &= p\bar{T}_B, \quad \bar{T}_B = Ce^{-q'(x-1)} + De^{q'(x-1)} \end{aligned}$$

where

$$q = \sqrt{p/\kappa_A}, \quad q' = \sqrt{p/\kappa_B}$$

As $x \to \infty$, $\bar{T}_B = 0$. Then, $D = 0$.

$$\bar{T}_A(0,p) = T_0/p, \quad B = \frac{T_0}{p} \frac{1}{\sinh q}$$

$$\bar{T}_A(1,p) = \bar{T}_B(1,p), \quad A \sinh q = C$$

$$\bar{T}_A'(1,p) = \alpha \bar{T}_B'(1,p), \quad q[A \cosh q - B] = -\alpha C q'$$

Solving for A, B, and C,

$$A = \frac{T_0}{p} \frac{f(q)}{\sinh q}, \quad B = \frac{T_0}{p} \frac{1}{\sinh q}, \quad C = \frac{T_0 f(q)}{p}$$

where

$$f(q) = \frac{1}{\cosh q + \beta \sinh q},$$

where

$$\beta = \alpha\sqrt{\kappa_A/\kappa_B}.$$

Using these

$$\begin{aligned}
\bar{T}_A &= \frac{T_0}{p}\left\{\frac{f(q)\sinh qx}{\sinh q} + \frac{\sinh q(1-x)}{\sinh q}\right\} \\
\bar{T}_B &= \frac{T_0 f(q)}{p}e^{-q'(x-1)} = \frac{T_0 f(q)}{p}e^{-q\bar{x}},
\end{aligned}$$

where

$$\bar{x} = \sqrt{\frac{\kappa_A}{\kappa_B}}(x-1).$$

We may write

$$\begin{aligned}
f(q) &= \frac{2}{e^q + e^{-q} + \beta(e^q - e^{-q})} \\
&= \frac{2e^{-q}}{(1+\beta) + (1-\beta)e^{-2q}}, \quad \gamma = \frac{1-\beta}{1+\beta}, \\
&= \frac{2}{1+\beta}e^{-q}[1 - \gamma e^{-2q} + \gamma^2 e^{-4q} - \cdots], \\
\bar{T}_A &= \frac{T_0}{p}\left[f(q)\frac{e^{qx} - e^{-qx}}{e^q - e^{-q}} + \frac{e^{q(1-x)} - e^{-q(1-x)}}{e^q - e^{-q}}\right] \\
&= \frac{T_0}{p}\left[f(q)(e^{-q(1-x)} - e^{-q(1+x)}) + e^{-qx} - e^{-q(2-x)}\right](1 + e^{-2q+e^{-4q}} + \cdots) \\
&= \frac{T_0}{p}\left[\frac{2}{1+\beta}(e^{-q(2-x)} - e^{-qx})(1 - \gamma e^{-2q} + \cdots) + e^{-qx} - e^{-q(2-x)}\right](1 + e^{-2q} + \cdots) \\
T_A(x,t) &= T_0\left[\frac{2}{1+\beta}\left(\operatorname{erfc}\frac{2-x}{\sqrt{4\kappa_A t}} - \operatorname{erfc}\frac{x}{\sqrt{4\kappa_A}}\right) + \cdots + \operatorname{erfc}\frac{x}{\sqrt{4\kappa_A}} - \operatorname{erfc}\frac{2-x}{\sqrt{4\kappa_A}}\cdots\right], \\
T_B(x,t) &= \frac{2T_0}{1+\beta}\left[\operatorname{erfc}\frac{\bar{x}+1}{\sqrt{\kappa_A t}} - \gamma\operatorname{erfc}\frac{\bar{x}+3}{\sqrt{\kappa_A t}} + \cdots\right].
\end{aligned}$$

4.22 If u is the solution of

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad u(0,t) = \frac{\partial u}{\partial x}(0,t) = 0, \quad u(x,0) = f(x),$$

and v is the solution of

$$\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial t^2}, \quad v(0,t) = \frac{\partial v}{\partial x}(0,t) = 0, \quad v(x,0) = 0, \quad \frac{\partial v}{\partial t}(x,0) = f(x),$$

show that

$$u(x,t) = \frac{1}{\sqrt{4\pi t^3}} \int_0^\infty v(x,\tau) e^{-\tau^2/4t} \tau d\tau.$$

(from Snedddon, 1972).

Solution

$$\begin{aligned} \bar{u}'' - p\bar{u} &= -f(x), \quad \bar{u}(0,p) = \bar{u}'(0,p) = 0, \\ \bar{v}'' - p^2\bar{v} &= -f(x), \quad \bar{v}(0,p) = \bar{v}'(0,p) = 0, \end{aligned}$$

We may write the complementary solution in the form

$$\bar{u} = Ae^{\sqrt{p}x} + Be^{-\sqrt{p}x}.$$

The particular solution can be found by the method of variable constants. Assuming A and B are functions of x,

$$\begin{aligned} A'e^{\sqrt{p}x} + B'e^{-\sqrt{p}x} &= 0, \\ A'e^{\sqrt{p}x} - B'e^{-\sqrt{p}x} &= -f(x)/\sqrt{p}. \end{aligned}$$

$$A = -\int_0^x \frac{f(\xi)}{2\sqrt{p}} e^{-\sqrt{p}\xi} d\xi, \quad B = \int_0^x \frac{f(\xi)}{2\sqrt{p}} e^{\sqrt{p}\xi} d\xi.$$

The general solution is

$$\begin{aligned} \bar{u} &= \int_0^x \frac{f(\xi)}{2\sqrt{p}} \left[e^{\sqrt{p}(\xi - x)} - e^{-\sqrt{p}(\xi - x)} \right] d\xi \\ &= \int_0^x \frac{f(\xi)}{2\sqrt{p}} \sinh \sqrt{p}(\xi - x) d\xi. \end{aligned}$$

Similarly
$$\bar{v}(x,p) = \int_0^x \frac{f(\xi)}{2p} \sinh p(\xi - x) d\xi.$$
Note that
$$\bar{u}(x,p) = \bar{v}(x,\sqrt{p}).$$
We have
$$\begin{aligned}
u(x,t) &= \frac{1}{2\pi i} \int_\Gamma \bar{u}(x,p) e^{pt} dp \\
&= \frac{1}{2\pi i} \int_\Gamma \bar{v}(x,\sqrt{p}) e^{pt} dp \\
&= \frac{1}{2\pi i} \int_\Gamma \int_0^\infty v(x,\tau) e^{-\sqrt{p}\tau} d\tau e^{pt} dp \\
&= \int_0^\infty v(x,\tau) \mathcal{L}^{-1} \left[e^{-\sqrt{p}} \right] d\tau \\
&= \int_0^\infty v(x,\tau) \frac{\tau}{\sqrt{4\pi t^3}} e^{-\tau^2/(4t)} d\tau \\
u(x,t) &= \frac{1}{\sqrt{4\pi t^3}} \int_0^\infty v(x,\tau) e^{-\tau^2/(4t)} \tau d\tau.
\end{aligned}$$

4.23 If u is the solution of

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}, \quad u(0,t) = \frac{\partial u}{\partial x}(0,t) = 0, \quad u(x,0) = f(x),$$

and v is the solution of

$$\frac{\partial^2 v}{\partial x^2} = \frac{\partial^2 v}{\partial t^2}, \quad v(0,t) = \frac{\partial v}{\partial x}(0,t) = 0, \quad v(x,0) = f(x), \quad \frac{\partial v}{\partial t}(x,0) = 0,$$

show that

$$u(x,t) = \frac{1}{\sqrt{\pi t}} \int_0^\infty v(x,\tau) e^{-\tau^2/4t} \tau d\tau.$$

(from Snedddon, 1972).

Solution

$$\begin{aligned} \bar{u}'' - p\bar{u} &= -f(x), \quad \bar{u}(0,p) = \bar{u}'(0,p) = 0, \\ \bar{v}'' - p^2\bar{v} &= -pf(x), \quad \bar{v}(0,p) = \bar{v}'(0,p) = 0, \end{aligned}$$

We may write the complementary solution in the form

$$\bar{u} = Ae^{\sqrt{p}x} + Be^{-\sqrt{p}x}.$$

The particular solution can be found by the method of variable constants. Assuming A and B are functions of x,

$$\begin{aligned} A'e^{\sqrt{p}x} + B'e^{-\sqrt{p}x} &= 0, \\ A'e^{\sqrt{p}x} - B'e^{-\sqrt{p}x} &= -f(x)/\sqrt{p}. \end{aligned}$$

$$A = -\int_0^x \frac{f(\xi)}{2\sqrt{p}} e^{-\sqrt{p}\xi} d\xi, \quad B = \int_0^x \frac{f(\xi)}{2\sqrt{p}} e^{\sqrt{p}\xi} d\xi.$$

The general solution is

$$\begin{aligned} \bar{u} &= \int_0^x \frac{f(\xi)}{2\sqrt{p}} \left[e^{\sqrt{p}(\xi - x)} - e^{-\sqrt{p}(\xi - x)} \right] d\xi \\ &= \int_0^x \frac{f(\xi)}{2\sqrt{p}} \sinh \sqrt{p}(\xi - x) d\xi. \end{aligned}$$

Similarly

$$\bar{v}(x,p) = \int_0^x f(\xi)\sinh p(\xi - x)d\xi.$$

Note that

$$\bar{u}(x,p) = \frac{1}{\sqrt{p}}\bar{v}(x,\sqrt{p}).$$

We have

$$\begin{aligned}
u(x,t) &= \frac{1}{2\pi i}\int_\Gamma \bar{u}(x,p)e^{pt}dp \\
&= \frac{1}{2\pi i}\int_\Gamma \frac{1}{\sqrt{p}}\bar{v}(x,\sqrt{p})e^{pt}dp \\
&= \frac{1}{2\pi i}\int_\Gamma \frac{1}{\sqrt{p}}\int_0^\infty v(x,\tau)e^{-\sqrt{p}\tau}d\tau e^{pt}dp \\
&= \int_0^\infty v(x,\tau)\mathcal{L}^{-1}\left[\frac{e^{-\sqrt{p}}}{\sqrt{p}}\right]d\tau \\
&= \int_0^\infty v(x,\tau)\frac{1}{\sqrt{\pi t}}e^{-\tau^2/(4t)}\tau d\tau \\
u(x,t) &= \frac{1}{\sqrt{\pi t}}\int_0^\infty v(x,\tau)e^{-\tau^2/(4t)}\tau d\tau.
\end{aligned}$$

4.24 Solve the integral equation

$$u(t) - a\int_0^t e^{-a\tau}u(t-\tau)d\tau = e^{-bt},$$

where a and b are constants.

Solution

Laplace transform gives

$$\bar{u} - \frac{a}{p+a}\bar{u} = \frac{1}{p+b}.$$

$$\bar{u} = \frac{1}{p}\frac{p+a}{p+b} = \frac{1}{p} + \frac{a-b}{p+b}.$$

Inverting

$$u(t) = 1 + (a-b)e^{-bt}.$$

4.25 Solve the integral equation

$$\int_0^t \frac{\cos[k\sqrt{t^2-\tau^2}]}{\sqrt{t^2-\tau^2}}f(\tau)d\tau = g(t).$$

Solution

Let

$$x^2 = x', \quad t^2 = t', \quad dt = \frac{dt'}{2\sqrt{t'}}.$$

$$f(t) = F(t'), \quad g(x) = G(x').$$

With these, the integral equation becomes

$$\int_0^{x'} \frac{\cos\sqrt{x'-t'}}{\sqrt{x'-t'}}F(t')\frac{dt'}{2\sqrt{t'}} = G(x').$$

Taking the Laplace transform, we get

$$\mathcal{L}\left[\frac{\cos k\sqrt{t}}{\sqrt{t}}\right]\mathcal{L}\left[\frac{F(t)}{2\sqrt{t}}\right] = \mathcal{L}[G].$$

$$\begin{aligned}
\mathcal{L}\left[\frac{\cos k\sqrt{t}}{\sqrt{t}}\right] &= \int_0^\infty \frac{\cos k\sqrt{t}}{\sqrt{t}} e^{-pt} dt = 2\int_0^\infty \cos kt e^{-pt^2} dt \\
&= 2\sqrt{\frac{\pi}{2}} \frac{e^{-k^2/(4p)}}{\sqrt{p}}. \\
\mathcal{L}\left[\frac{F(t)}{2\sqrt{t}}\right] &= \frac{1}{\sqrt{2\pi}} \sqrt{p} e^{kr/(4p)} \mathcal{L}[G].
\end{aligned}$$

Using $k \to ik$,

$$\begin{aligned}
\mathcal{L}\left[\frac{\cosh k\sqrt{t}}{\sqrt{t}}\right] &= \sqrt{2\pi} \frac{e^{k^2/(4p)}}{\sqrt{p}}. \\
\mathcal{L}\left[\frac{F(t)}{2\sqrt{t}}\right] &= \frac{2}{\pi} p \mathcal{L}\left[\frac{\cosh k\sqrt{t}}{\sqrt{t}}\right] \mathcal{L}[G].
\end{aligned}$$

$$\begin{aligned}
F &= \frac{2}{\pi} \frac{x}{2} \frac{d}{dx'} \int_0^{x'} \frac{\cosh k\sqrt{x'-t'}}{\sqrt{x'-t'}} G(t') dt'. \\
f(x) &= \frac{2}{\pi} \frac{d}{dx} \int_0^x \frac{\cosh k\sqrt{x^2-t^2}}{\sqrt{x^2-t^2}} t g(t) dt.
\end{aligned}$$

4.26 Solve the integral equation

$$\int_0^x \frac{u(\xi)d\xi}{(x-\xi)^{2/3}} = \int_0^x \frac{f(\xi)d\xi}{(x-\xi)^{1/3}}.$$

Solution

Using

$$\mathcal{L}\left[t^{\alpha-1}\right] = \frac{\Gamma(\alpha)}{p^\alpha},$$

the integral equation becomes

$$\bar{u} = \frac{\Gamma(2/3)}{\Gamma(1/3)} \frac{\bar{f}}{p^{1/3}}.$$

Inversion results in

$$u = \frac{\Gamma(2/3)}{[\Gamma(1/3)]^2} \int_0^x \frac{f(\xi)d\xi}{(x-\xi)^{2/3}}.$$

4.27 Solve the integral equation

$$\int_0^x \frac{u(\xi)d\xi}{\sqrt{x^2-\xi^2}} = \frac{1}{\sqrt{x}}.$$

Solution

Let

$$x^2 = y, \quad \xi^2 = \eta, \quad \frac{u(\sqrt{\eta})}{2\sqrt{\eta}} = v(\eta).$$

$$\int_0^y \frac{v(\eta)}{\sqrt{y-\eta}} d\eta = \frac{1}{y^{1/4}}.$$

Taking the Laplace transform, we get

$$\begin{aligned}
\frac{\Gamma(1/2)}{\sqrt{p}}\bar{v} &= \frac{\Gamma(3/4)}{p^{3/4}} \\
\bar{v} &= \frac{\Gamma(3/4)}{\Gamma(1/2)}\frac{1}{p^{1/4}} \\
v(y) &= \frac{\Gamma(3/4)}{\Gamma(1/2)\Gamma(1/4)}\frac{1}{y^{3/4}} \\
u(\sqrt{y}) &= \frac{\Gamma(3/4)}{\Gamma(1/2)\Gamma(1/4)}\frac{2y^{1/2}}{y^{3/4}} \\
u(x) &= \frac{2}{\sqrt{\pi}}\frac{\Gamma(3/4)}{\Gamma(1/4)}\frac{1}{\sqrt{x}}.
\end{aligned}$$

4.28 Find the solution of

$$u_{n+1} - au_n = nb^n, \quad u_0 = 1.$$

Solution

We have

$$\mathcal{Z}[b^n] = 1 + bz + b^2z^2 + \cdots = \frac{1}{1-bz}.$$

$$\mathcal{Z}[nb^n] = b\sum_{n=1} nb^{n-1} = b\frac{d}{db}\frac{1}{1-bz} = \frac{bz}{(1-bz)^2}.$$

$$\sum nb^n z^{n-1} = \frac{b}{(1-bz)^2}.$$

Let

$$Z = \mathcal{Z}[u_n].$$

The given difference equation becomes

$$\begin{aligned}
\frac{1}{z}(Z-1) - aZ &= \frac{bz}{(1-bz)^2}, \\
Z(1-az) &= 1 + \frac{bz^2}{(1-bz)^2}, \\
Z &= \frac{1}{1-az} + \frac{bz^2}{(1-bz)^2(1-az)},
\end{aligned}$$

Using partial fractions,

$$\begin{aligned}
\frac{bz^2}{(1-bz)^2(1-az)} &= \frac{A}{1-az} + \frac{B}{1-bz} + \frac{C}{(1-bz)^2}, \\
A = \frac{b/a^2}{(1-b/a)^2}, \quad C &= \frac{1/b}{1-a/b}, \\
B &= -\frac{b/a^2}{(1-b/a)^2} - \frac{1/b}{1-a/b}.
\end{aligned}$$

$$u_n = a^n + \frac{b}{(a-b)^2}a^n - \frac{b-(a-b)}{(a-b)^2}b^n + \frac{(n+1)}{(b-a)b}b^n,$$

$$u_n = \left[1 + \frac{b}{(a-b)^2}\right]a^n + \left[\frac{a}{(a-b)^2} + \frac{(n+1)}{(b-a)b}\right]b^n.$$

4.29 Find the solution of the difference equation

$$u_{n+2} - 2bu_{n+1} + b^2 u_n = b^n, \quad u_0 = u_1 = 0.$$

Solution

Let

$$\mathcal{Z}[u_n] = Z.$$

$$\begin{aligned}
Z\left[\frac{1}{z^2} - \frac{2b}{z} + b^2\right] &= \frac{1}{1-bz} \\
Z &= \frac{z^2}{(1-bz)^3}, \\
\mathcal{Z}b^n &= \frac{1}{1-bz}, \\
\mathcal{Z}nb^{n-1} &= \frac{z}{(1-bz)^2}, \\
\mathcal{Z}n(n-1)b^{n-2} &= \frac{2z^2}{(1-bz)^3}, \\
u_n &= \frac{1}{2}n(n-1)b^{n-2}.
\end{aligned}$$

4.30 Obtain the frequencies ω corresponding to the periodic solutions of

$$u_{n+2} - (2 - \omega^2)u_{n+1} + u_n = 0,$$

subject to the conditions

$$u_0 = u_4 = 1, \quad u_2 = u_5.$$

Solution

$$\omega = 2 \sin \pi/N.$$

When $N = 4$,

$$\omega = 2 \sin \pi/4 = \sqrt{2}.$$